The working (6,7): a clearer graphical derivation

101 listed moves in 71 independent steps | Clarity before count

READ THE PICTURES AS THREE DIFFERENT ACTIONS. A blue disk substitution changes factors by a named identity. A yellow sphere redraw changes only the drawing of the same curve or arc. A peripheral cleanup is shown on a separate line, after its curve is already peripheral.

What a step means. A numbered step is a set of independent factor-interval edits. Moves a,b,... are listed one after another for printing, but each starts from the SAME step input and they may be performed in either order. Every proposed parallel pair has a literal two-order certificate. A singleton step contains one move. Some independent edits are deliberately shown in separate steps for readability.

These numbers do not claim a shorter proof. This follows the longer 85-operation source, with the simpler early prefix and the delayed petal expansion: the F2 half-twist neighborhood has length 4 rather than 12, and the half-neighborhood total at the matched checkpoint drops from 80 to 72. These are support lengths in this chart; the maximum stays 30. The shorter 79-operation route remains archived. More complicated or more tangled curves are not preferable merely because they use fewer arrows. Future research will prioritize simpler arc/curve supports and their interactions.

Delayed petals. Keep the three-point root intact while the last two half twists are handled. Expand it only after those halves have combined, on its own step. This keeps 17 rows rather than 19 during the gather. The two actual half-twist classes are unchanged; this reduces diagram clutter, not arc complexity.

Framing ledger. The ledger is separate from the main factor list. The represented word is ledger times main factorization. During the five-point rechart, each same-curve redraw records its boundary correction here. The SIX corrections are kept explicitly, redrawn as one peripheral powered curve, and only then cleared. An individual curve rechart can carry a deck sign; the ledger retains it. A single five-point cap would have -I and is never silently used.

Drawing conventions. Read factors downward. All support diagrams keep the same unnumbered six-point grid: 18pt spacing and 1.26pt radius. Braids represent the pictured disk word and remain connected. Individual support pictures are not joint cut-system certificates. Parallel factor edits do not assert a simultaneous ambient isotopy of an entire joint cut system.

Let the final half twists meet lower

The same two arcs; a different meeting place; two more listed moves.

The two remaining half twists already have the same arc. The old route carried the lower half twist upward through all six intervening factors. That made several surrounding curves wind more before the half twists could combine.

The new route. Carry the upper half twist DOWN past the first five intervening factors. Independently, carry the lower half twist UP past the squared pair. Each carried half twist keeps its original arc; the crossed factors change by the displayed Hurwitz moves. These two carries act on disjoint factor intervals, so both start from the same input and either order gives exactly the same result.

The halves are now adjacent near the bottom of that interval. Combine them THERE into their full twist. Only then carry the full twist upward through the five neighbors. This rejoins exactly the previous complete derivation; the retained three-point root still expands later on its own step.

What this improves. While the two half twists remain, the exact closed-neighborhood crossing sum is at most 936 rather than 1094, and the largest pair has 36 crossings rather than 48. Several surrounding curves show less nested winding. The half arcs themselves are unchanged. These are closed-neighborhood counts, not intersections of the arcs or one jointly minimal drawing.

The tradeoff. The support-word maximum in this local route is 18 rather than 17. Four listed moves and eleven adjacent checks replace two moves and six checks. The final full-twist carry restores the old endpoint geometry after the halves disappear. This is a choice about clarity during the half-twist part, not a shorter proof or a reduction in overall geometric complexity.

Contents

Independent steps, their individual moves, and complete redraw discrepancies.

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101 moves; 71 independent steps. No shortening claim.

I. Prepare and apply the disk lantern

Every substitution and sphere redraw is separate. Independent moves share a step number; their individual diagrams remain visible.

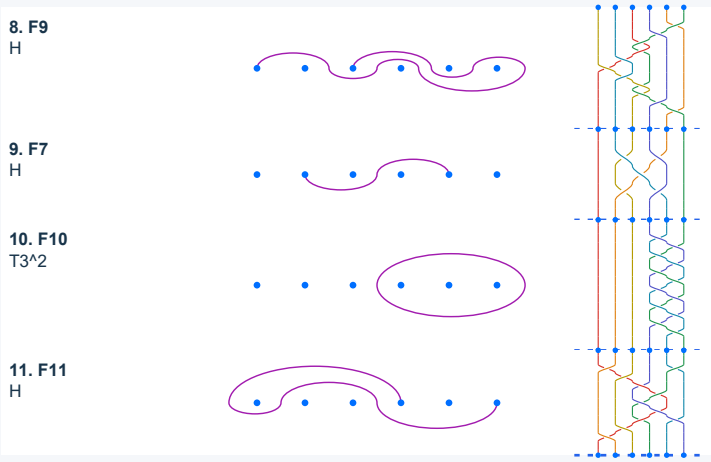
1 - step 001

Change the working ordering by an inverse Hurwitz move

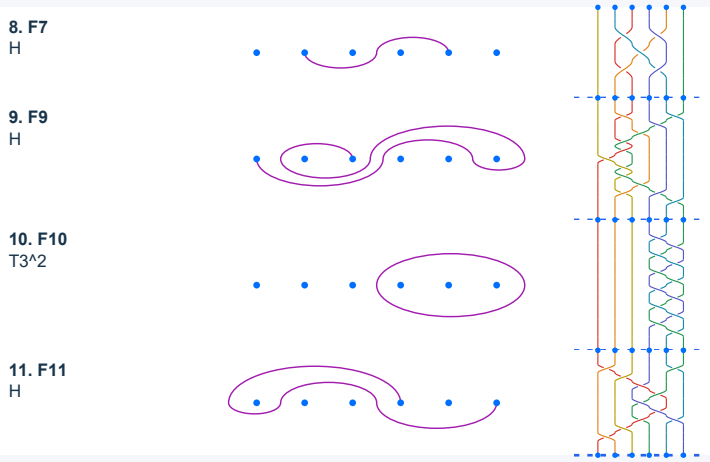
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 2 below. Only the changed interval is shown.

Before



After



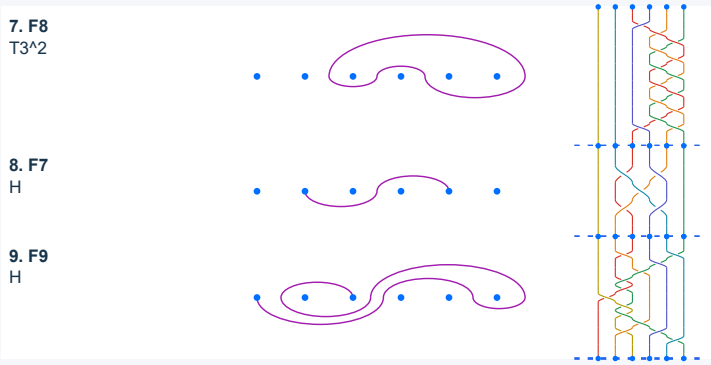
1 - step 002a

Change the working ordering by an inverse Hurwitz move

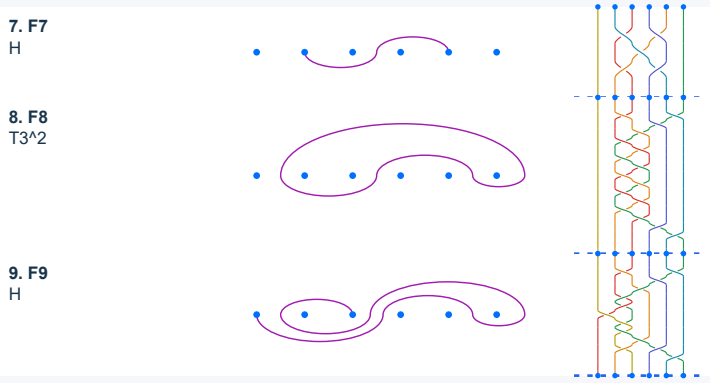
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

6 unchanged displayed factors above; 4 below. Only the changed interval is shown. PARALLEL STEP 2: moves a, b are independent and each starts from the SAME input. Both orders are certified.

Before



After



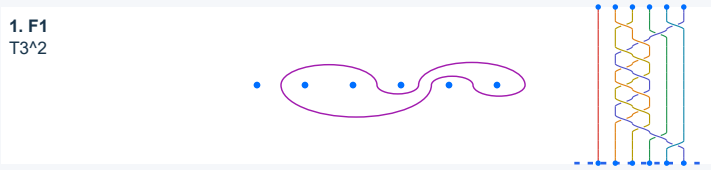
1 - step 002b

Split F1 into ordinary roots

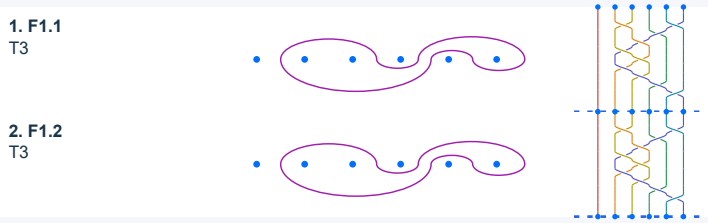
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 12 below. Only the changed interval is shown. PARALLEL STEP 2: moves a, b are independent and each starts from the SAME input. Both orders are certified.

Before



After



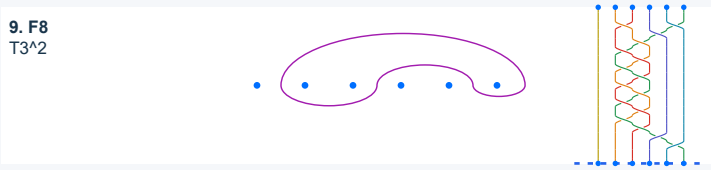
1 - step 003a

Split F8 into ordinary roots

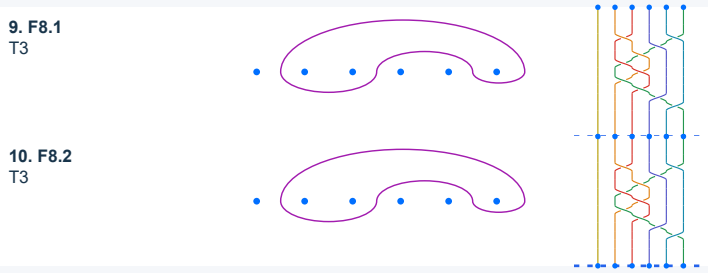
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 5 below. Only the changed interval is shown. PARALLEL STEP 3: moves a, b, c are independent and each starts from the SAME input. Both orders are certified.

Before



After



1 - step 003b

Split F10 into ordinary roots

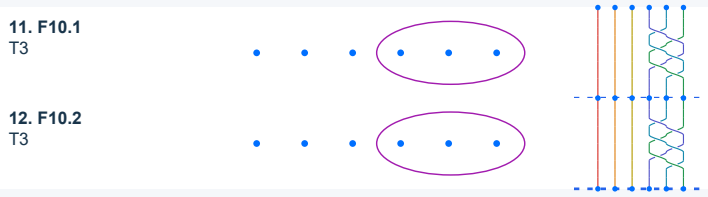
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

10 unchanged displayed factors above; 3 below. Only the changed interval is shown. PARALLEL STEP 3: moves a, b, c are independent and each starts from the SAME input. Both orders are certified.

Before



After



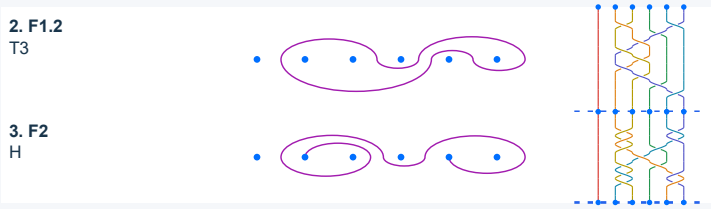
1 - step 003c

F1.2 down past F2

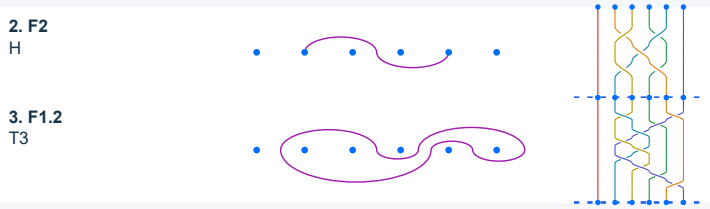
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 11 below. Only the changed interval is shown. PARALLEL STEP 3: moves a, b, c are independent and each starts from the SAME input. Both orders are certified.

Before



After



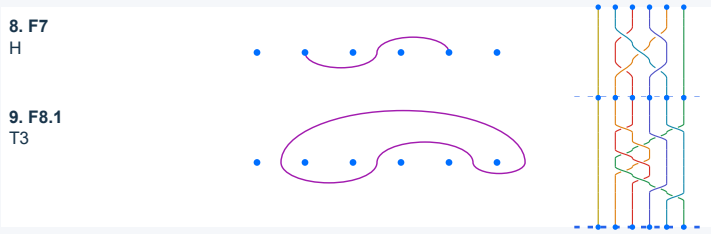
1 - step 004a

F8.1 up past F7

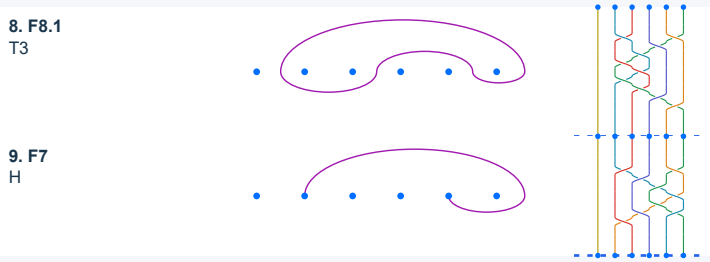
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 7 below. Only the changed interval is shown. PARALLEL STEP 4: moves a, b are independent and each starts from the SAME input. Both orders are certified.

Before



After



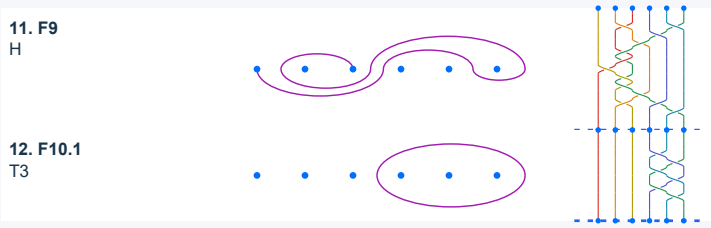
1 - step 004b

F10.1 up past F9

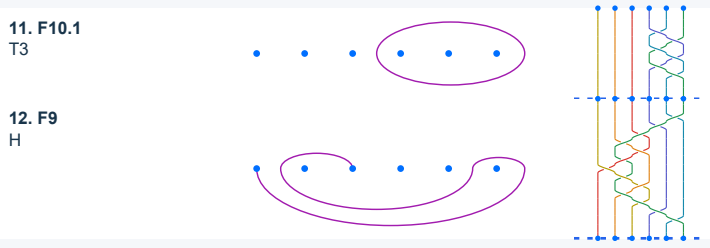
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

10 unchanged displayed factors above; 4 below. Only the changed interval is shown. PARALLEL STEP 4: moves a, b are independent and each starts from the SAME input. Both orders are certified.

Before



After



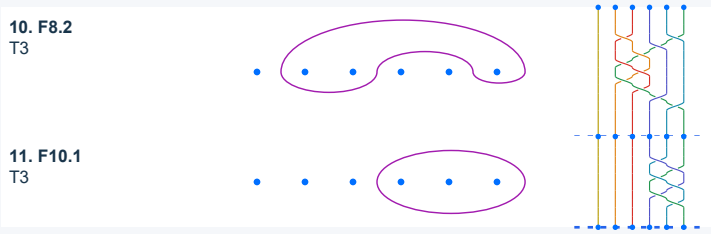
1 - step 005

F10.1 up past F8.2

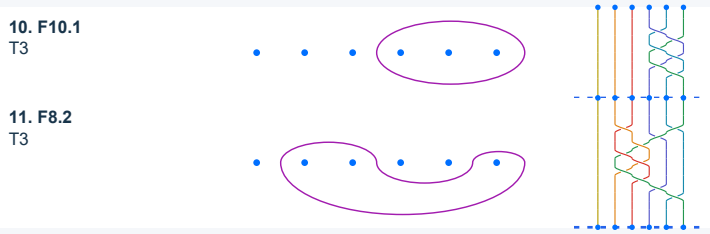
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

9 unchanged displayed factors above; 5 below. Only the changed interval is shown.

Before



After



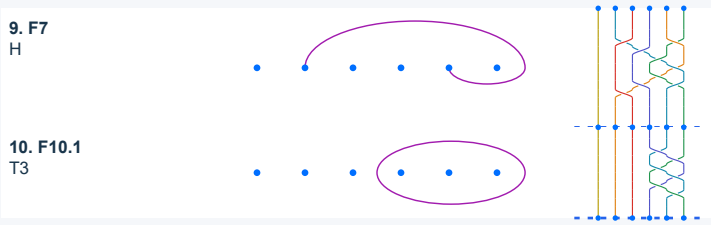
1 - step 006

F10.1 up past F7

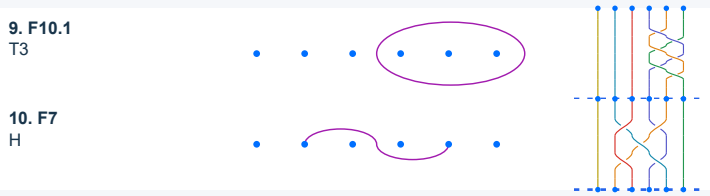
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 6 below. Only the changed interval is shown.

Before



After



1 - step 007

Use the exact disk equality of the matched arcs

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

9 unchanged displayed factors above; 6 below. Only the changed interval is shown.

Before



After



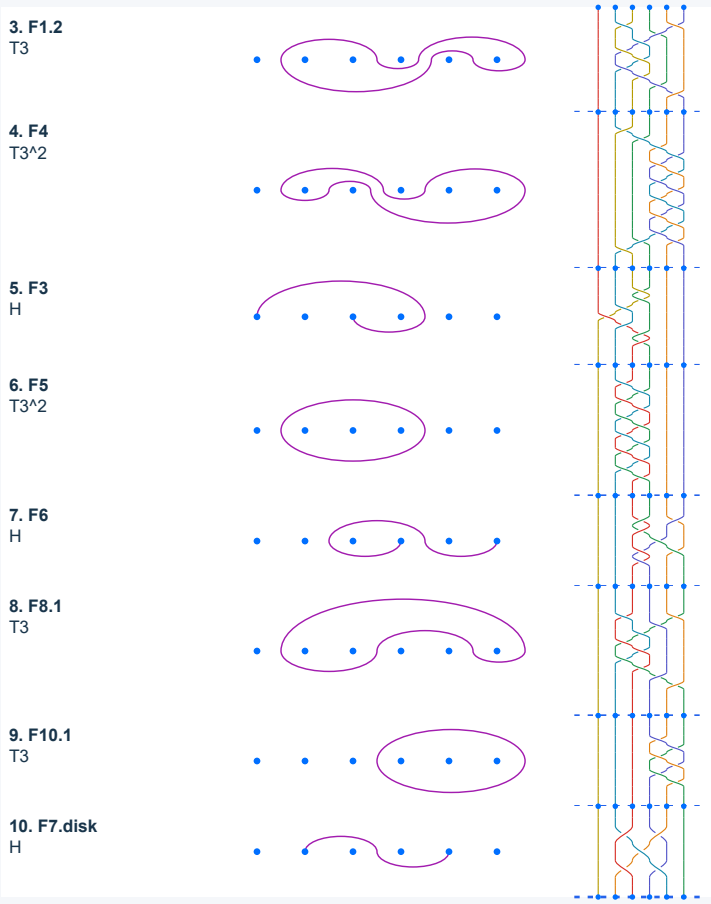
1 - step 008

Drag the matched arc up across SEVEN factors (one Hurwitz drag)

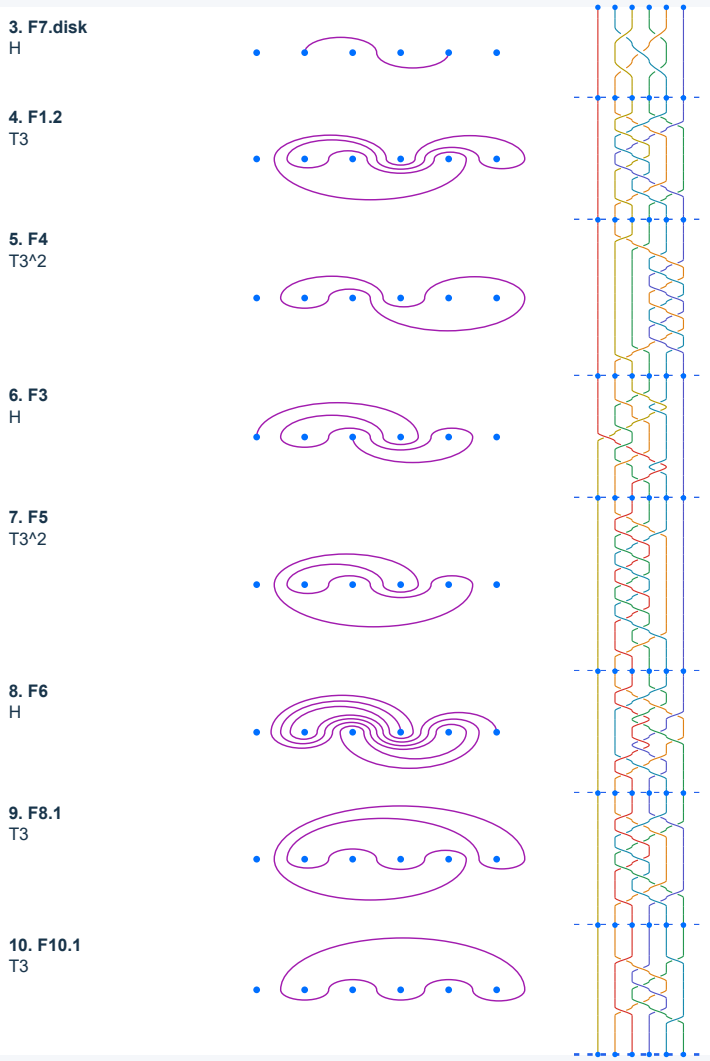
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 6 below. Only the changed interval is shown.

Before



After



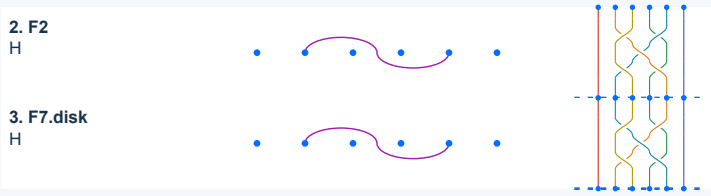
1 - step 009

Combine the two matched half twists

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 13 below. Only the changed interval is shown.

Before



After



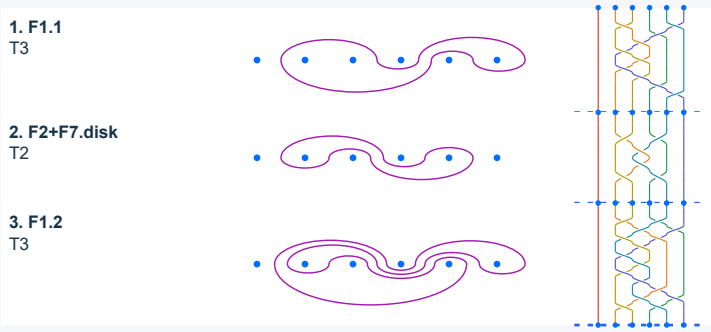
1 - step 010

Disk lantern: retain the actual four-point outer curve

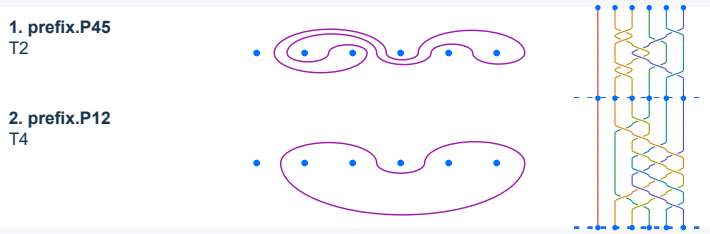
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 12 below. Only the changed interval is shown.

Before



After



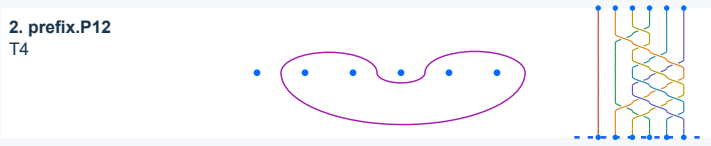
1 - step 011

Redraw ONE outer curve from its complementary side

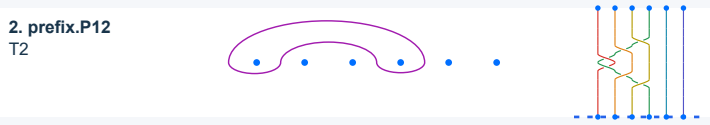
SPHERE REDRAW ONLY: the same curve or arc, with the same factor and power. Framing ledger retained separately.

1 unchanged displayed factors above; 12 below. Only the changed interval is shown. REDRAW ONLY: factor identity, position, power and sphere curve are retained.

Before



After



II. Sphere redraws and the separate framing ledger

Every substitution and sphere redraw is separate. Independent moves share a step number; their individual diagrams remain visible.

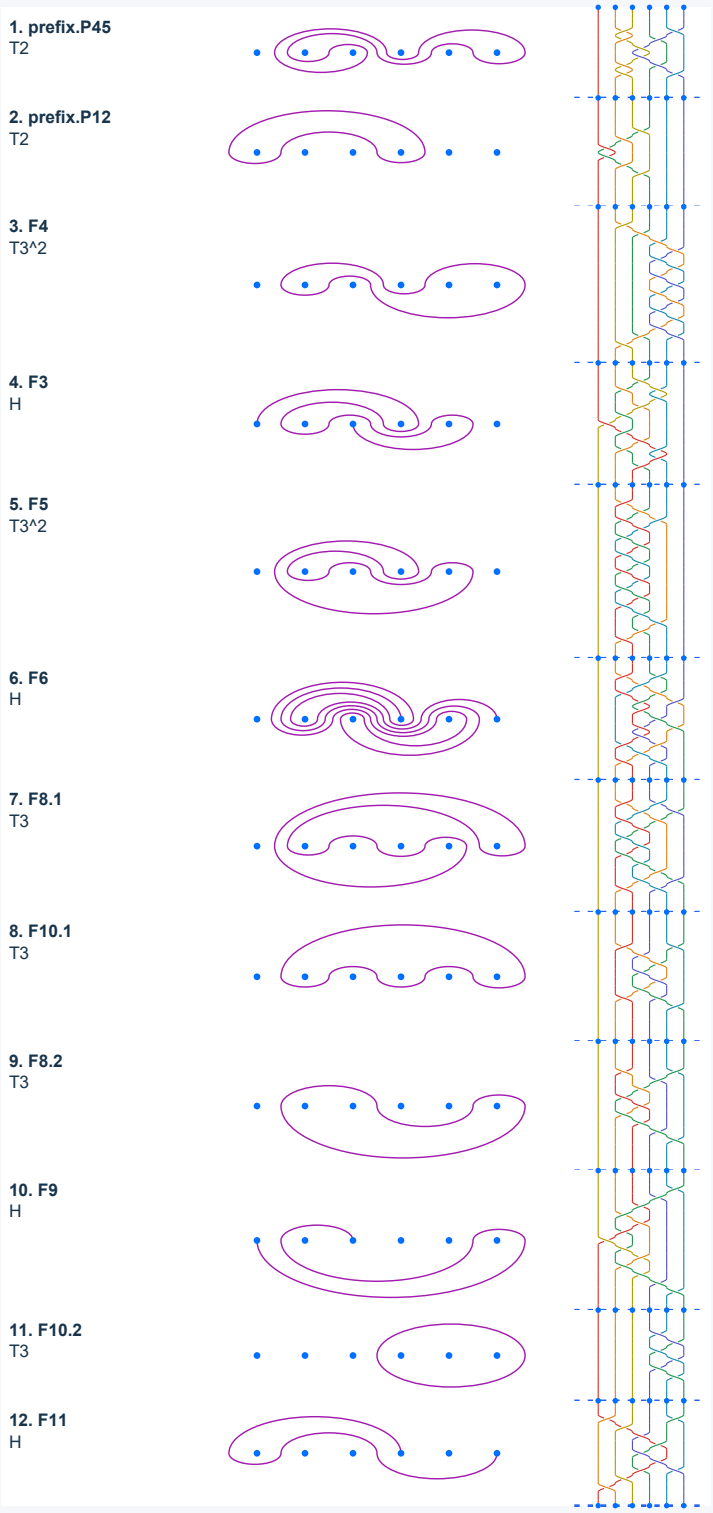
2 - step 012

Global frame: fixed working point 2 becomes displayed point 6

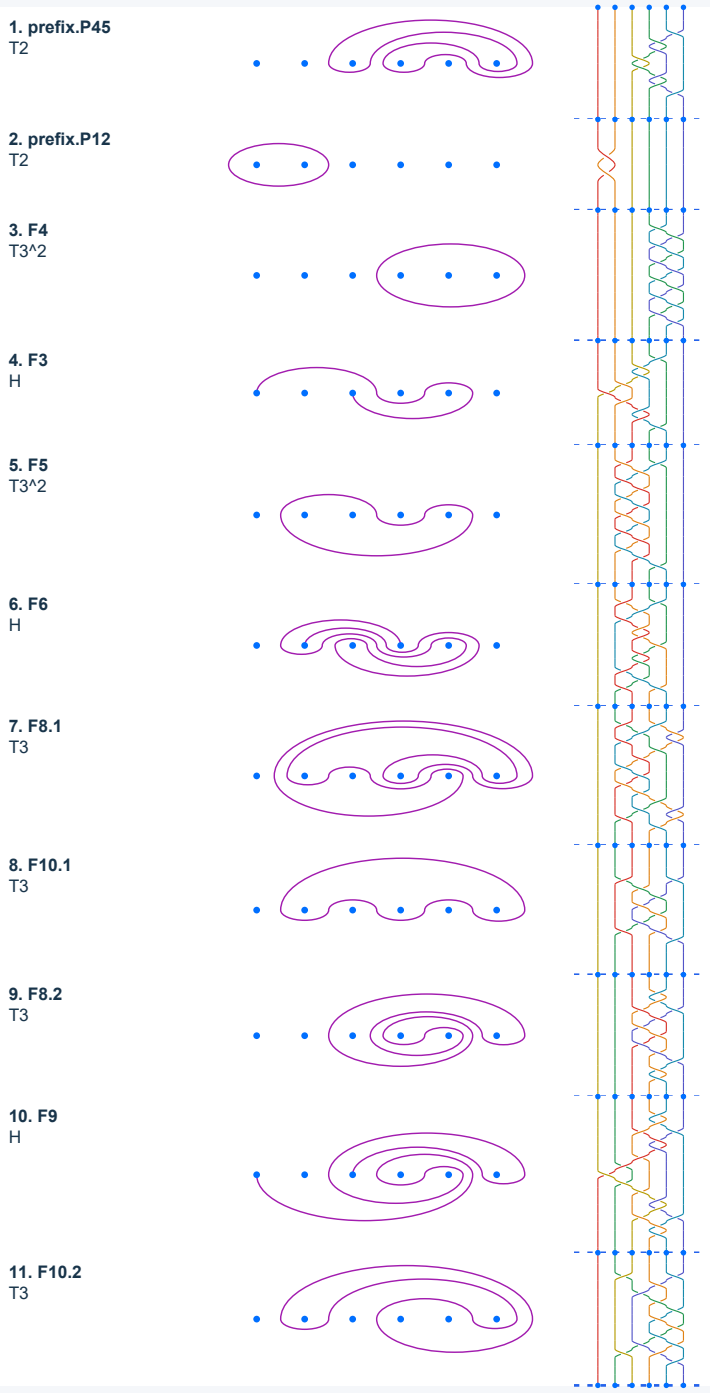
EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



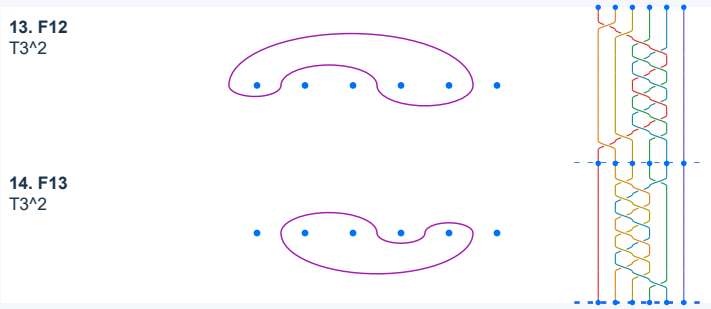
2 - step 012 - continued

Global frame: fixed working point 2 becomes displayed point 6

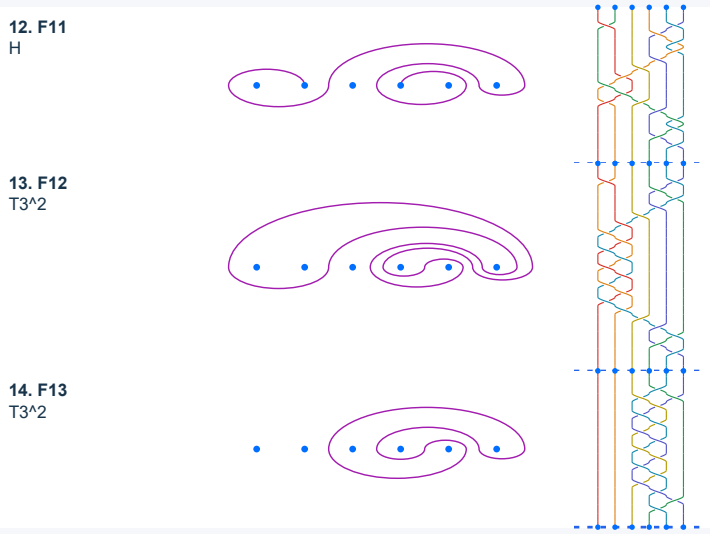
EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



2 - step 013a

Redraw prefix.P45 in the five-point chart

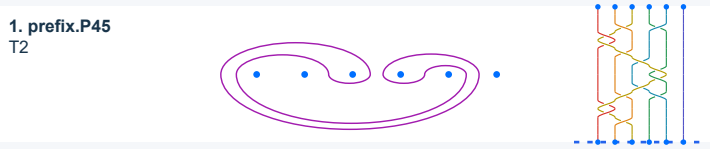
SPHERE REDRAW ONLY: the same curve or arc, with the same factor and power. Framing ledger retained separately.

0 unchanged displayed factors above; 13 below. Only the changed interval is shown. PARALLEL STEP 13: moves a, b, c, d, e, f, g, h, i, j, k, l, m, n are independent and each starts from the SAME input. Both orders are certified. REDRAW ONLY: factor identity, position, power and sphere curve are retained.

Before



After



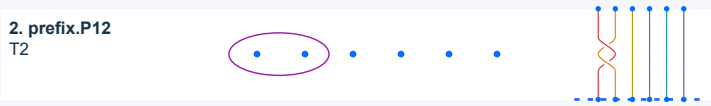
2 - step 013b

Redraw prefix.P12 in the five-point chart

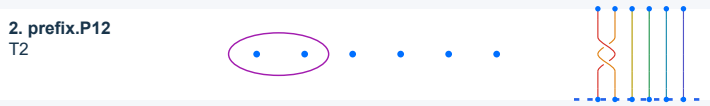
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 12 below. Only the changed interval is shown. PARALLEL STEP 13: moves a, b, c, d, e, f, g, h, i, j, k, l, m, n are independent and each starts from the SAME input. Both orders are certified. REDRAW ONLY: factor identity, position, power and sphere curve are retained.

Before



After



2 - step 013c

Redraw F4 in the five-point chart

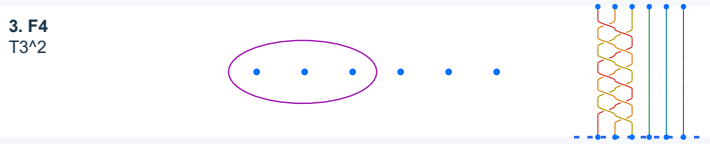
SPHERE REDRAW ONLY: the same curve or arc, with the same factor and power. Framing ledger retained separately.

2 unchanged displayed factors above; 11 below. Only the changed interval is shown. PARALLEL STEP 13: moves a, b, c, d, e, f, g, h, i, j, k, l, m, n are independent and each starts from the SAME input. Both orders are certified. REDRAW ONLY: factor identity, position, power and sphere curve are retained. Separate framing ledger: add 2 boundary twist(s); no factor is inserted into the main diagram. Ledger before: empty; ledger after: T5^2.

Before



After



2 - step 013d

Redraw F3 in the five-point chart

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

3 unchanged displayed factors above; 10 below. Only the changed interval is shown. PARALLEL STEP 13: moves a, b, c, d, e, f, g, h, i, j, k, l, m, n are independent and each starts from the SAME input. Both orders are certified. REDRAW ONLY: factor identity, position, power and sphere curve are retained.

Before



After



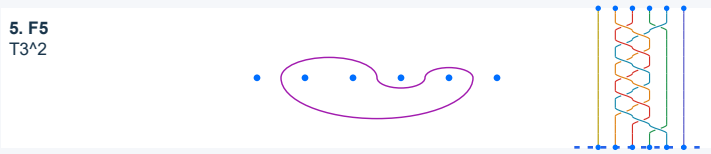
2 - step 013e

Redraw F5 in the five-point chart

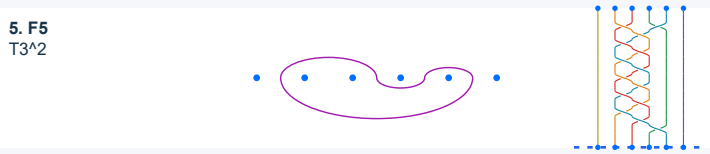
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

4 unchanged displayed factors above; 9 below. Only the changed interval is shown. PARALLEL STEP 13: moves a, b, c, d, e, f, g, h, i, j, k, l, m, n are independent and each starts from the SAME input. Both orders are certified. REDRAW ONLY: factor identity, position, power and sphere curve are retained.

Before



After



2 - step 013f

Redraw F6 in the five-point chart

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

5 unchanged displayed factors above; 8 below. Only the changed interval is shown. PARALLEL STEP 13: moves a, b, c, d, e, f, g, h, i, j, k, l, m, n are independent and each starts from the SAME input. Both orders are certified. REDRAW ONLY: factor identity, position, power and sphere curve are retained.

Before



After



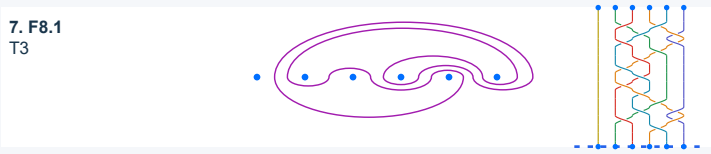
2 - step 013g

Redraw F8.1 in the five-point chart

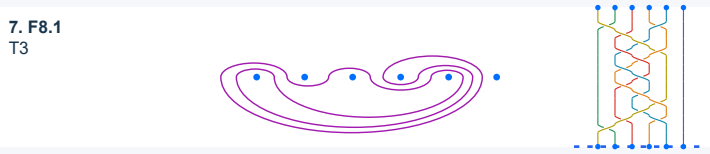
SPHERE REDRAW ONLY: the same curve or arc, with the same factor and power. Framing ledger retained separately.

6 unchanged displayed factors above; 7 below. Only the changed interval is shown. PARALLEL STEP 13: moves a, b, c, d, e, f, g, h, i, j, k, l, m, n are independent and each starts from the SAME input. Both orders are certified. REDRAW ONLY: factor identity, position, power and sphere curve are retained.

Before



After



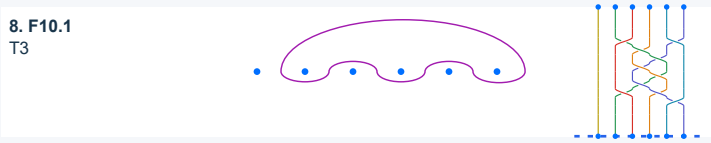
2 - step 013h

Redraw F10.1 in the five-point chart

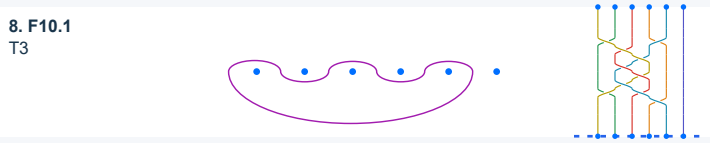
SPHERE REDRAW ONLY: the same curve or arc, with the same factor and power. Framing ledger retained separately.

7 unchanged displayed factors above; 6 below. Only the changed interval is shown. PARALLEL STEP 13: moves a, b, c, d, e, f, g, h, i, j, k, l, m, n are independent and each starts from the SAME input. Both orders are certified. REDRAW ONLY: factor identity, position, power and sphere curve are retained. Separate framing ledger: add 1 boundary twist(s); no factor is inserted into the main diagram. Ledger before: empty; ledger after: T5.

Before



After



2 - step 013i

Redraw F8.2 in the five-point chart

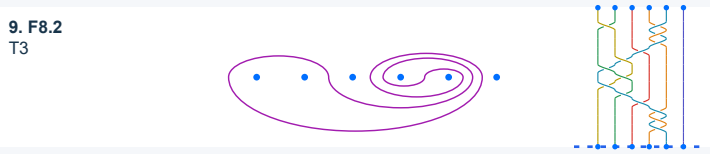
SPHERE REDRAW ONLY: the same curve or arc, with the same factor and power. Framing ledger retained separately.

8 unchanged displayed factors above; 5 below. Only the changed interval is shown. PARALLEL STEP 13: moves a, b, c, d, e, f, g, h, i, j, k, l, m, n are independent and each starts from the SAME input. Both orders are certified. REDRAW ONLY: factor identity, position, power and sphere curve are retained. Separate framing ledger: add 1 boundary twist(s); no factor is inserted into the main diagram. Ledger before: empty; ledger after: T5.

Before



After



2 - step 013j

Redraw F9 in the five-point chart

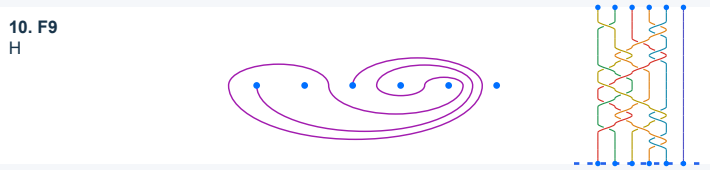
SPHERE REDRAW ONLY: the same curve or arc, with the same factor and power. Framing ledger retained separately.

9 unchanged displayed factors above; 4 below. Only the changed interval is shown. PARALLEL STEP 13: moves a, b, c, d, e, f, g, h, i, j, k, l, m, n are independent and each starts from the SAME input. Both orders are certified. REDRAW ONLY: factor identity, position, power and sphere curve are retained.

Before



After



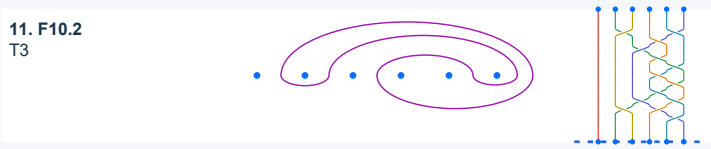
2 - step 013k

Redraw F10.2 in the five-point chart

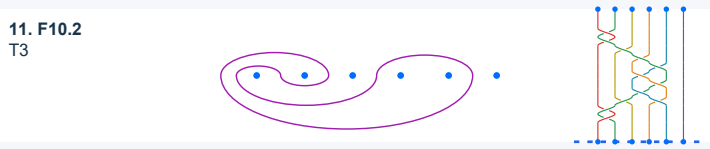
SPHERE REDRAW ONLY: the same curve or arc, with the same factor and power. Framing ledger retained separately.

10 unchanged displayed factors above; 3 below. Only the changed interval is shown. PARALLEL STEP 13: moves a, b, c, d, e, f, g, h, i, j, k, l, m, n are independent and each starts from the SAME input. Both orders are certified. REDRAW ONLY: factor identity, position, power and sphere curve are retained.

Before



After



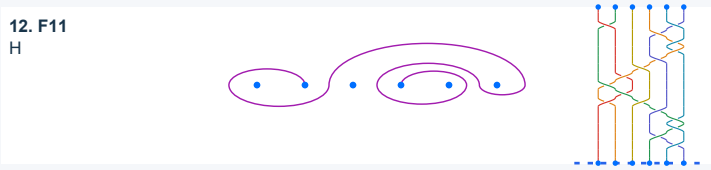
2 - step 013l

Redraw F11 in the five-point chart

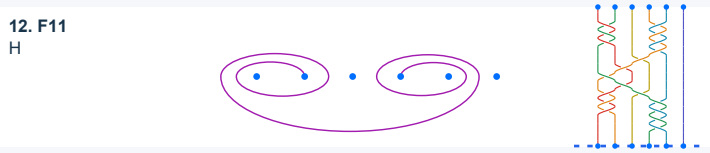
SPHERE REDRAW ONLY: the same curve or arc, with the same factor and power. Framing ledger retained separately.

11 unchanged displayed factors above; 2 below. Only the changed interval is shown. PARALLEL STEP 13: moves a, b, c, d, e, f, g, h, i, j, k, l, m, n are independent and each starts from the SAME input. Both orders are certified. REDRAW ONLY: factor identity, position, power and sphere curve are retained.

Before



After



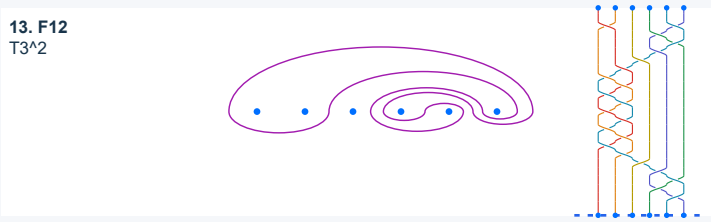
2 - step 013m

Redraw F12 in the five-point chart

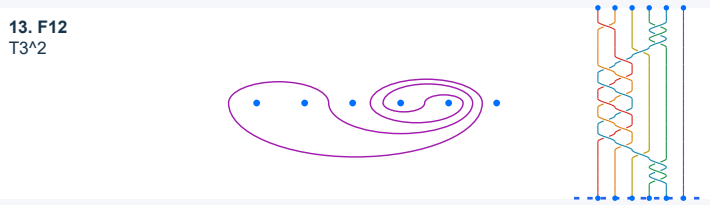
SPHERE REDRAW ONLY: the same curve or arc, with the same factor and power. Framing ledger retained separately.

12 unchanged displayed factors above; 1 below. Only the changed interval is shown. PARALLEL STEP 13: moves a, b, c, d, e, f, g, h, i, j, k, l, m, n are independent and each starts from the SAME input. Both orders are certified. REDRAW ONLY: factor identity, position, power and sphere curve are retained.

Before



After



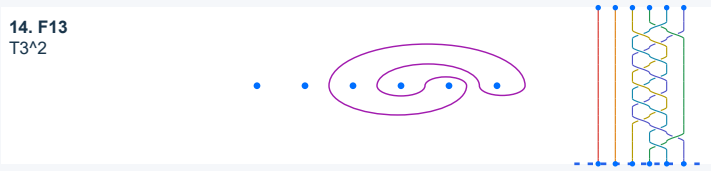
2 - step 013n

Redraw F13 in the five-point chart

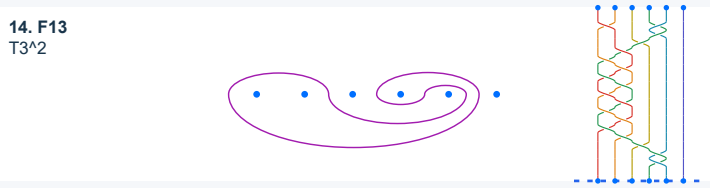
SPHERE REDRAW ONLY: the same curve or arc, with the same factor and power. Framing ledger retained separately.

13 unchanged displayed factors above; 0 below. Only the changed interval is shown. PARALLEL STEP 13: moves a, b, c, d, e, f, g, h, i, j, k, l, m, n are independent and each starts from the SAME input. Both orders are certified. REDRAW ONLY: factor identity, position, power and sphere curve are retained. Separate framing ledger: add 2 boundary twist(s); no factor is inserted into the main diagram. Ledger before: empty; ledger after: T5^2.

Before



After



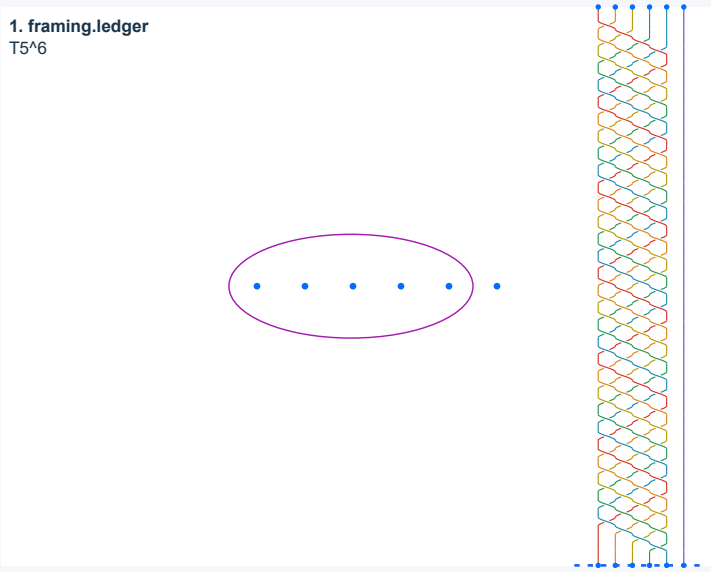
2 - step 014

Redraw the SIX-correction ledger curve around point6

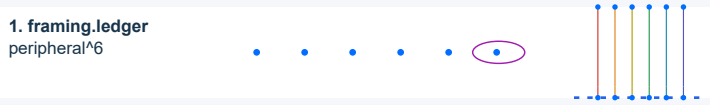
SPHERE REDRAW ONLY: the same curve or arc, with the same factor and power. Framing ledger retained separately.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. REDRAW ONLY: factor identity, position, power and sphere curve are retained. THIS ROW IS THE SEPARATE FRAMING LEDGER; the main factor list is unchanged.

Before



After



2 - step 015

Clear the already-peripheral SIX-correction ledger

PERIPHERAL CLEANUP: the curve has already been redrawn around one point; its even powered twist is trivial in the chosen lift.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. THIS ROW IS THE SEPARATE FRAMING LEDGER; the main factor list is unchanged.

Before



After

Identity / no factors in this interval.

III. Positive disk transports and named relations

Every substitution and sphere redraw is separate. Independent moves share a step number; their individual diagrams remain visible.

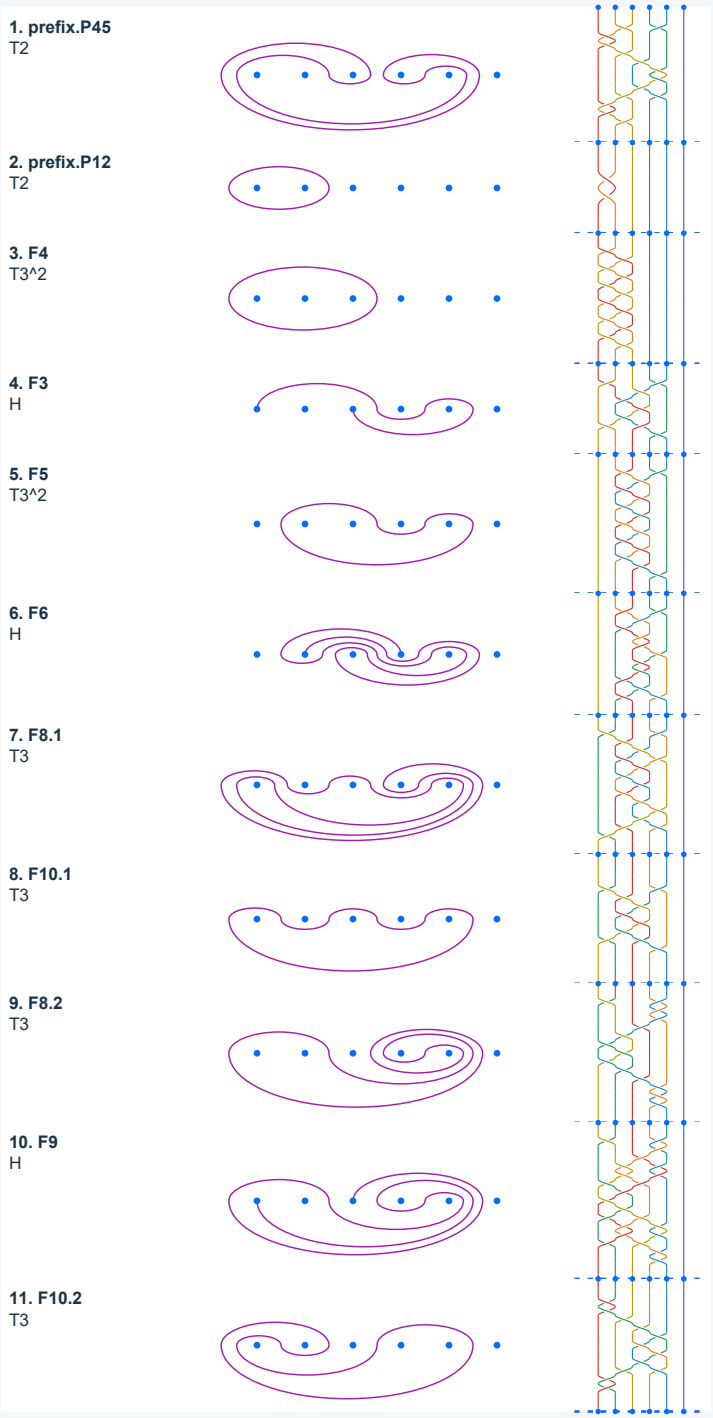
3 - step 016

Cyclic start at the WHOLE original124 square

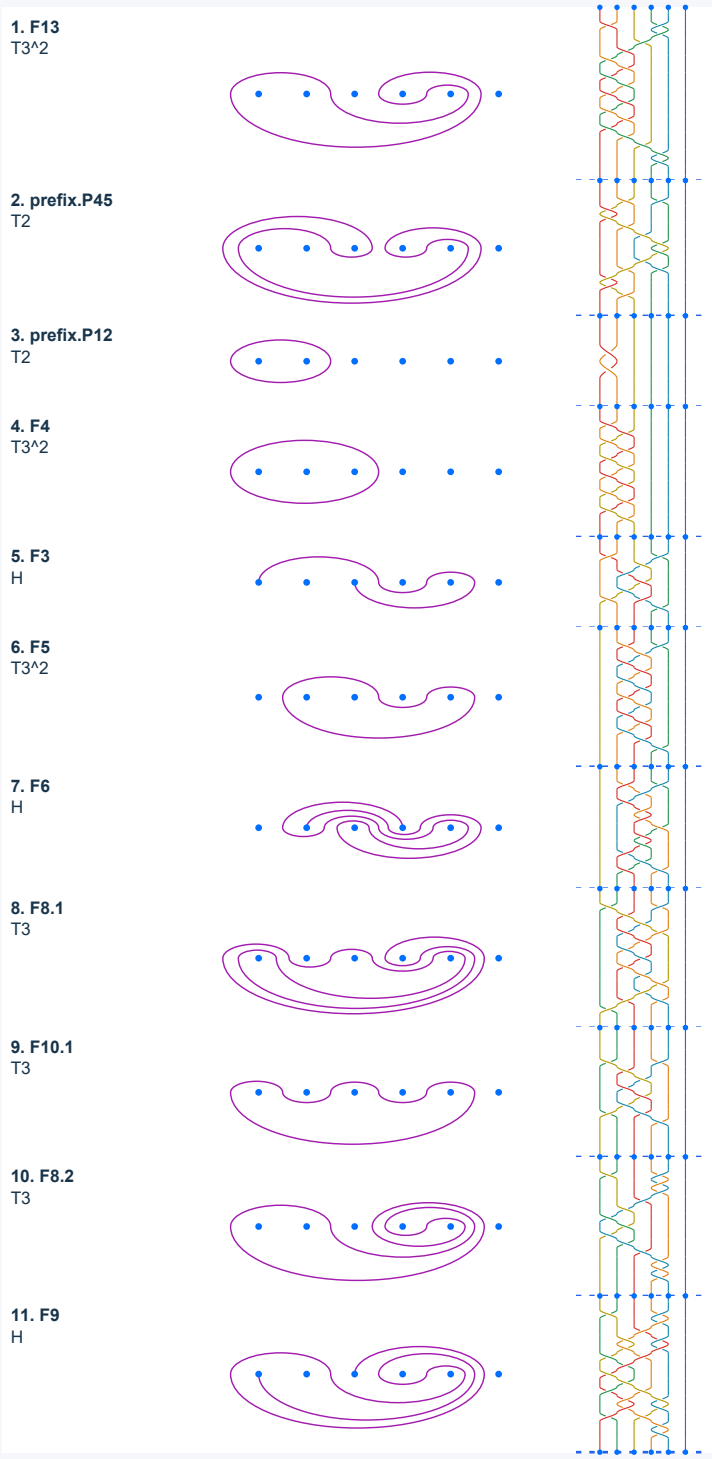
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



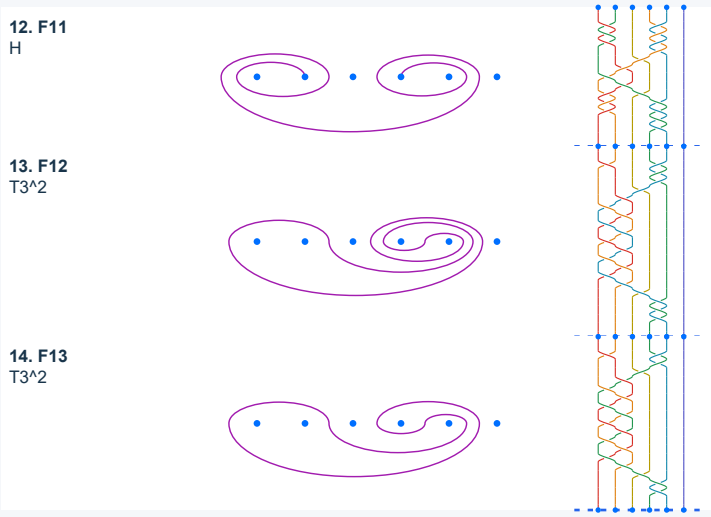
3 - step 016 - continued

Cyclic start at the WHOLE original124 square

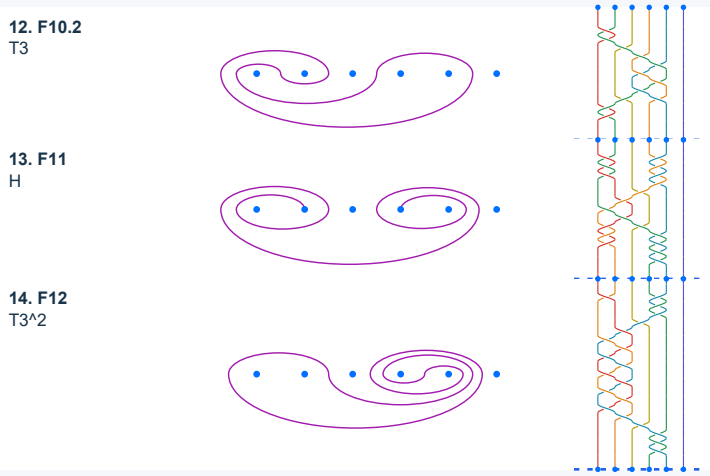
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



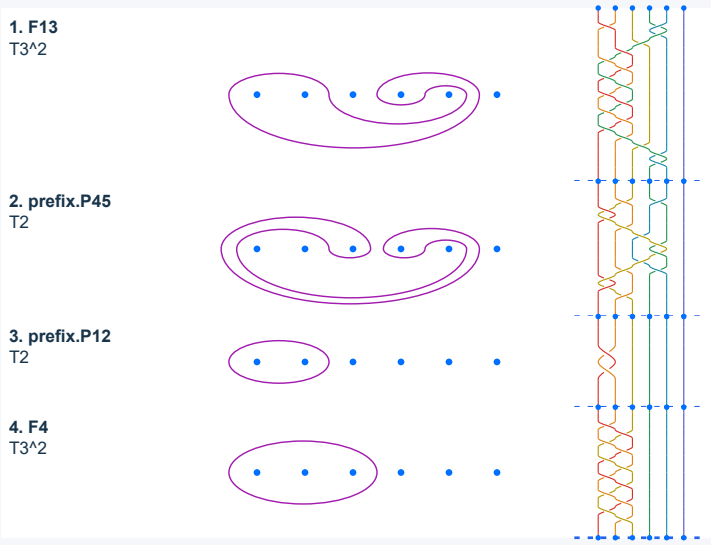
3 - step 017

Powered cabled PC lantern keeps both root squares intact and absorbs12

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 10 below. Only the changed interval is shown.

Before



After



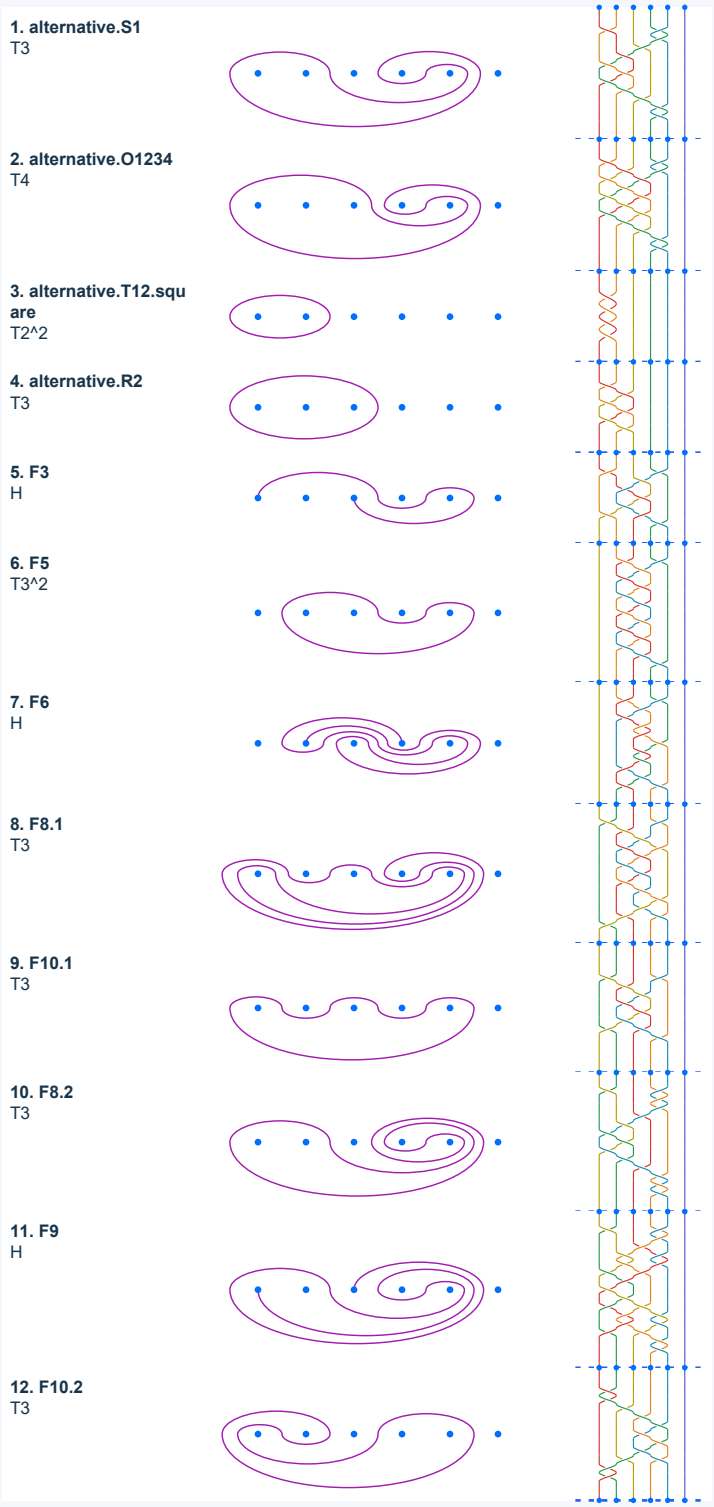
3 - step 018

Cyclic start after the retained actual124 unit

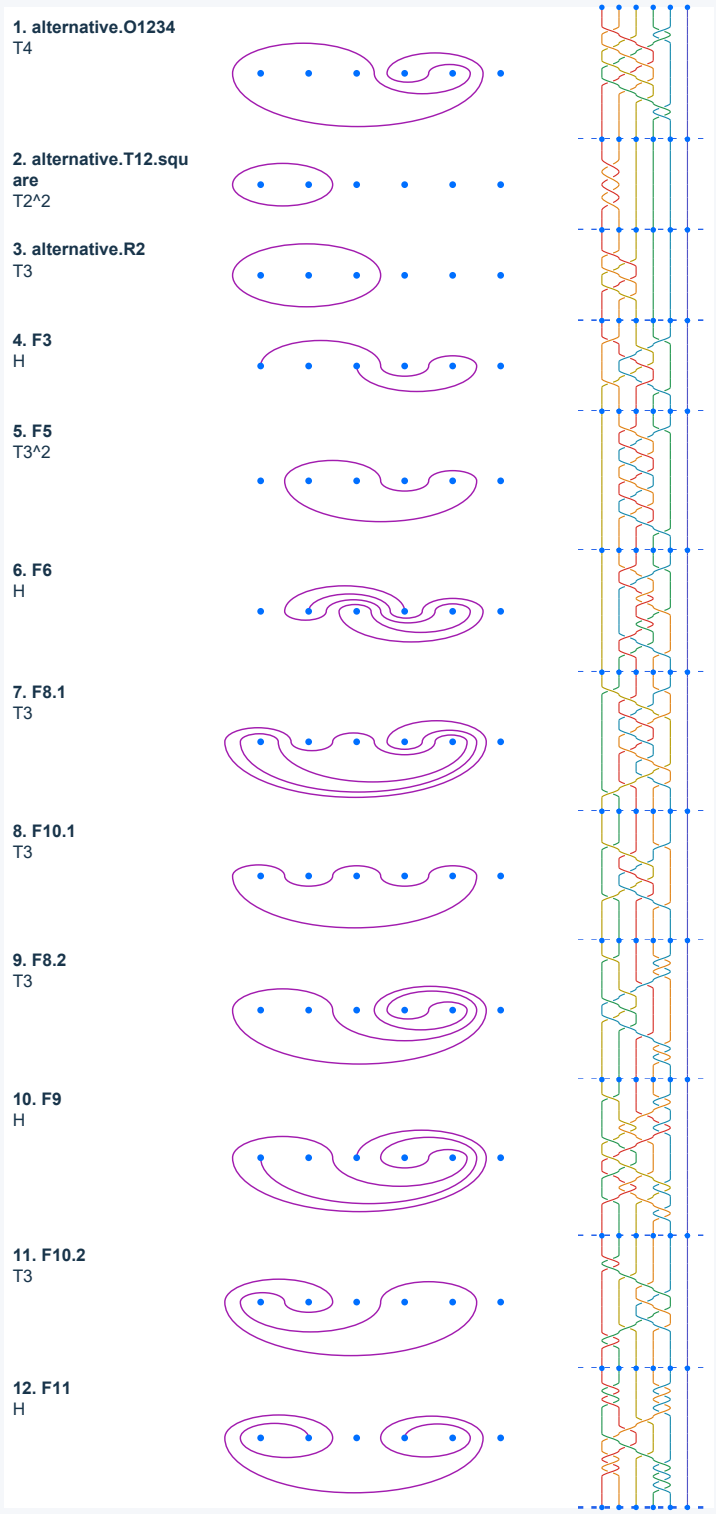
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



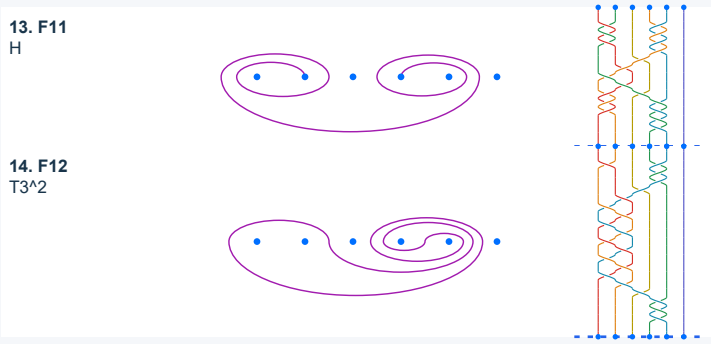
3 - step 018 - continued

Cyclic start after the retained actual124 unit

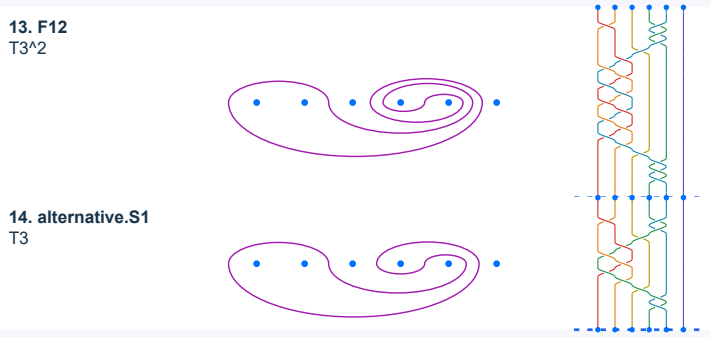
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



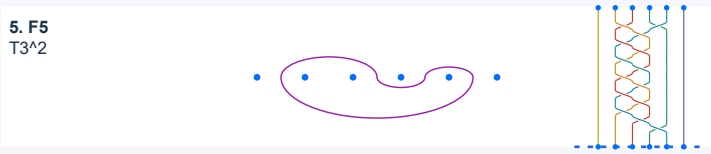
3 - step 019

Split F5 into its two positive units

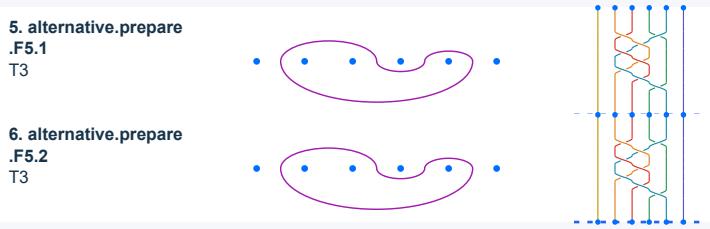
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

4 unchanged displayed factors above; 9 below. Only the changed interval is shown.

Before



After



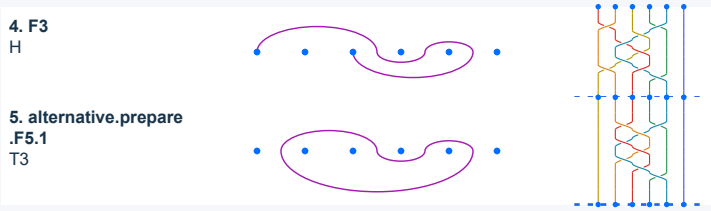
3 - step 020a

One 235 root up over F3

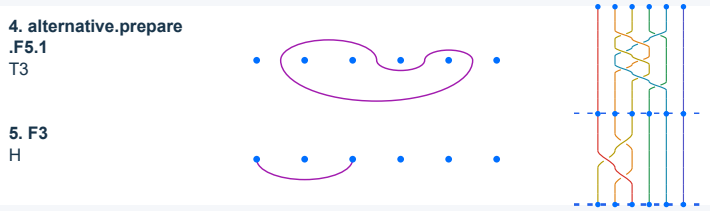
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

3 unchanged displayed factors above; 10 below. Only the changed interval is shown. PARALLEL STEP 20: moves a, b, c are independent and each starts from the SAME input. Both orders are certified.

Before



After



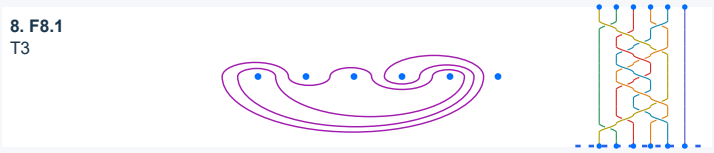
3 - step 020b

Original 345 daisy supplies actual P35

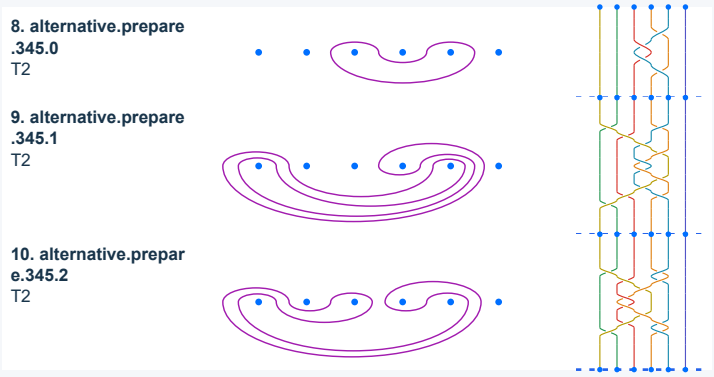
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 7 below. Only the changed interval is shown. PARALLEL STEP 20: moves a, b, c are independent and each starts from the SAME input. Both orders are certified.

Before



After



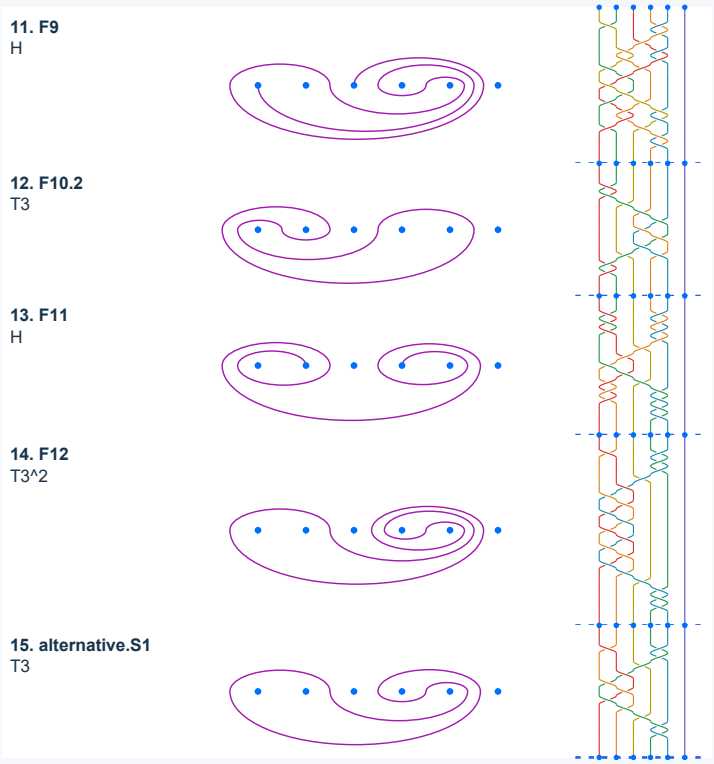
3 - step 020c

Remaining 124 root up over F9

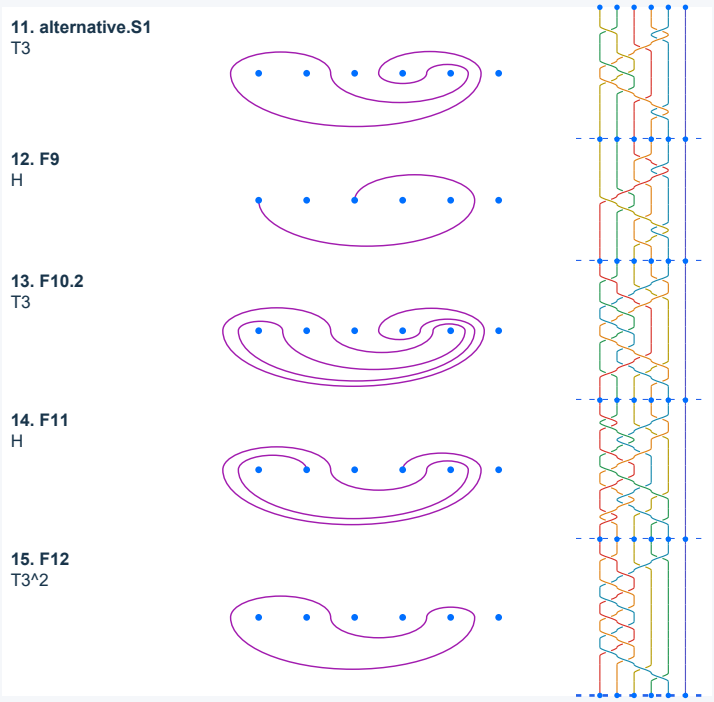
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

10 unchanged displayed factors above; 0 below. Only the changed interval is shown. PARALLEL STEP 20: moves a, b, c are independent and each starts from the SAME input. Both orders are certified.

Before



After



3 - step 021a

Carry P35 up over F3; a Q14 transport is still needed

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

4 unchanged displayed factors above; 9 below. Only the changed interval is shown. PARALLEL STEP 21: moves a, b are independent and each starts from the SAME input. Both orders are certified.

Before



After



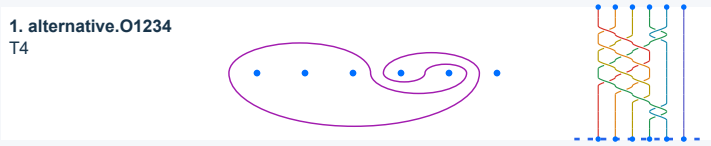
3 - step 021b

PC daisy lends its actual Q14 petal

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 16 below. Only the changed interval is shown. PARALLEL STEP 21: moves a, b are independent and each starts from the SAME input. Both orders are certified.

Before



After



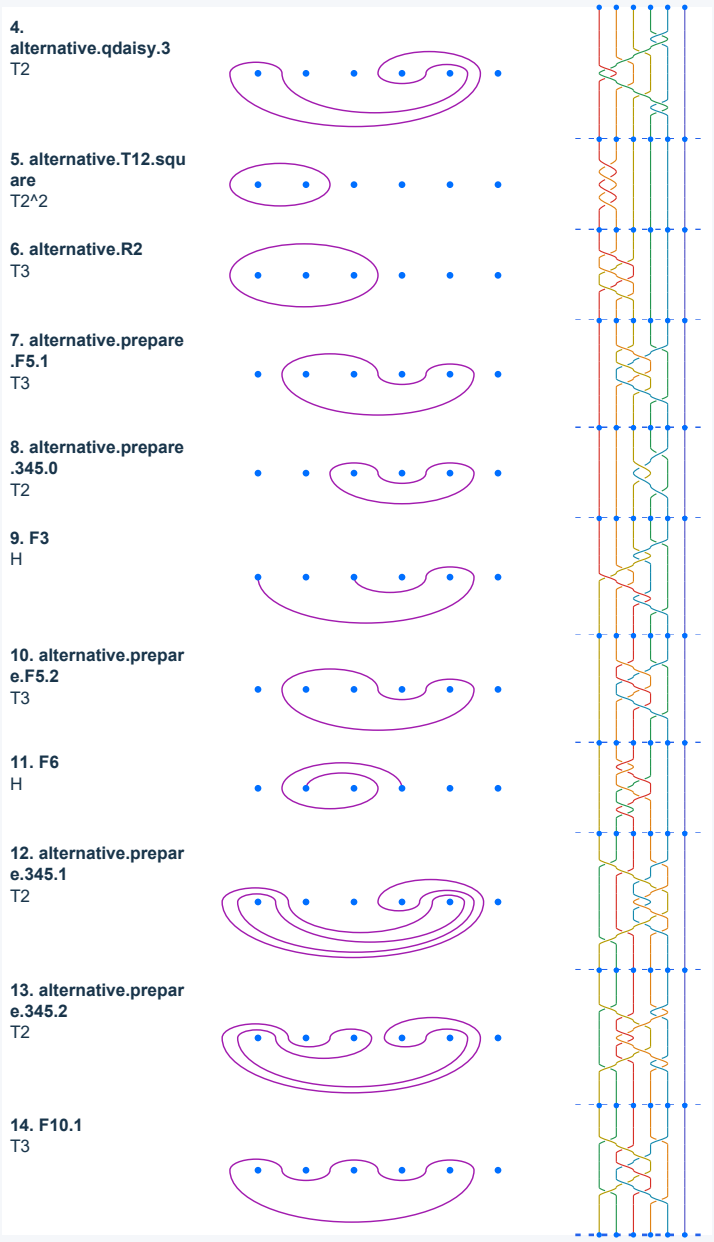
3 - step 022

Carry Q14 down across F3 and the remaining 345 petals

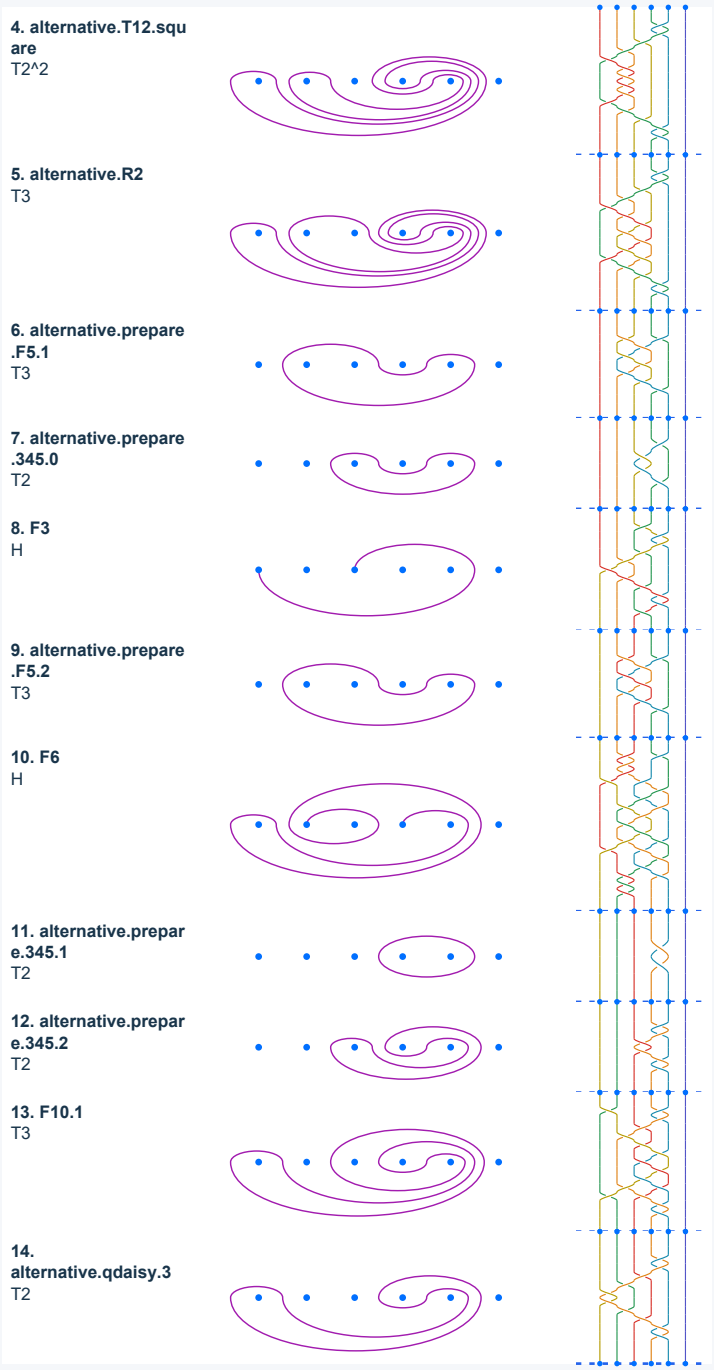
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

3 unchanged displayed factors above; 6 below. Only the changed interval is shown.

Before



After



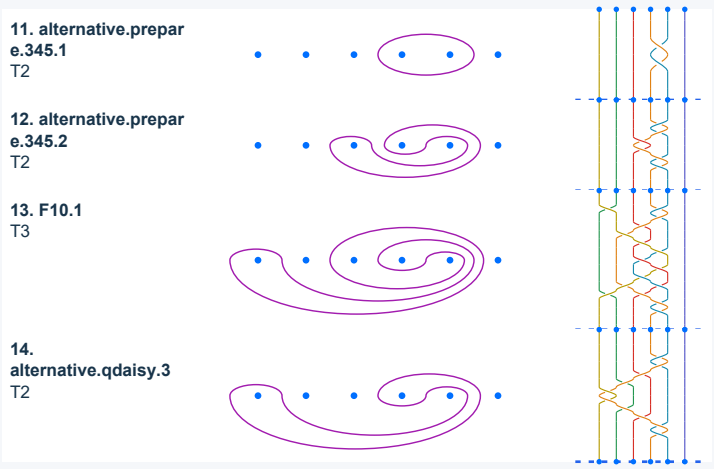
3 - step 023

PC daisy contracts the ACTUAL transported petals to O1345

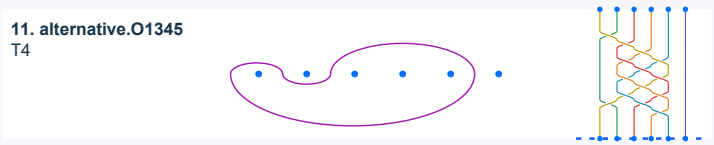
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

10 unchanged displayed factors above; 6 below. Only the changed interval is shown.

Before



After



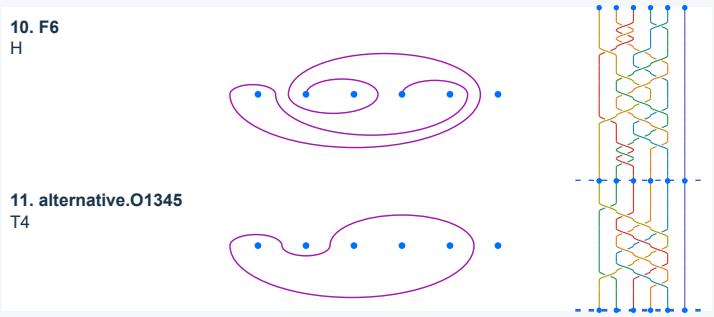
3 - step 024

Carry O1345 up across F6 before the matching-half gather

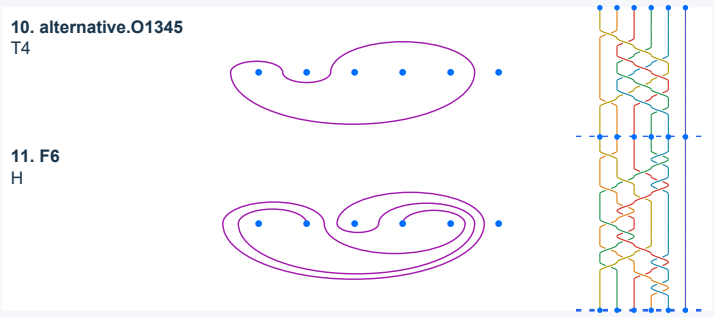
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

9 unchanged displayed factors above; 6 below. Only the changed interval is shown.

Before



After



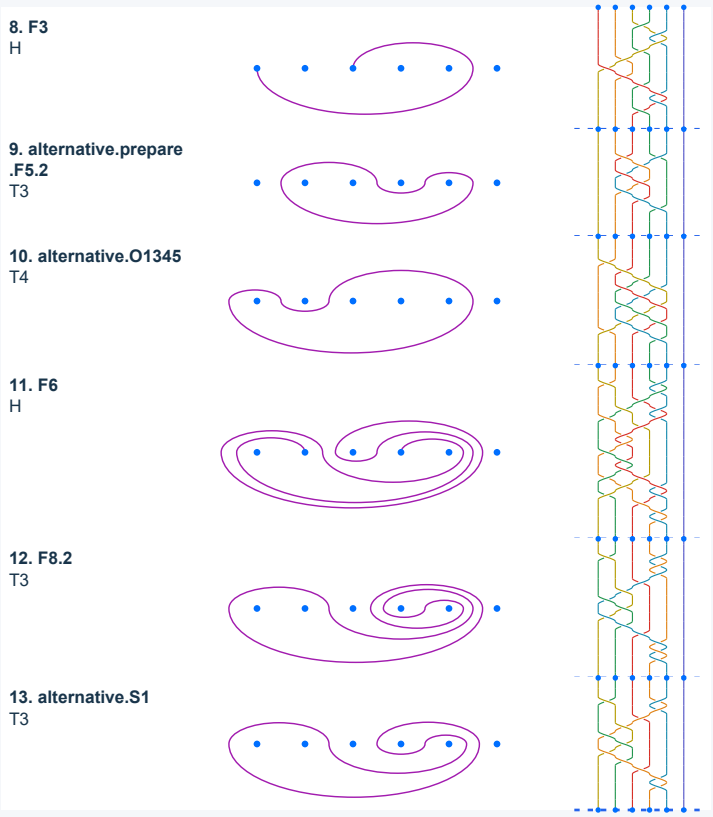
3 - step 025

Carry the matched F3 down to F9; F6 unwinds

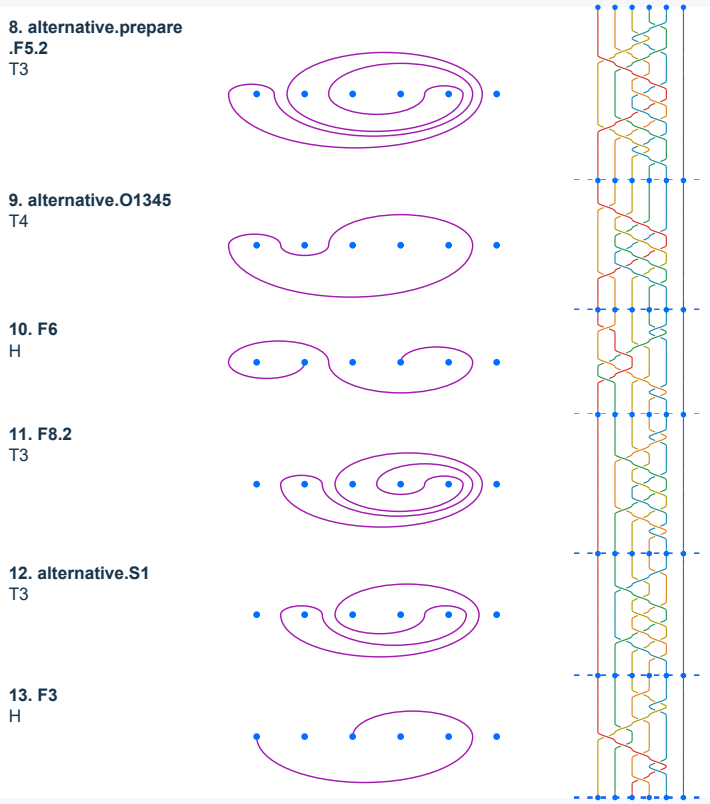
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 4 below. Only the changed interval is shown.

Before



After



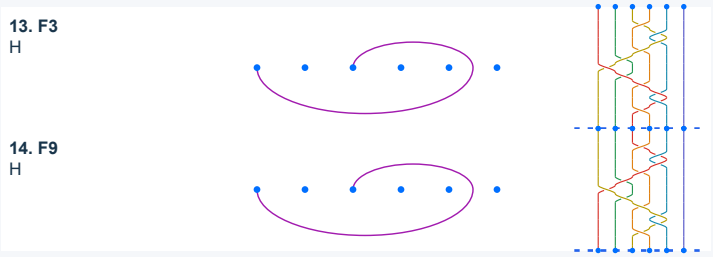
3 - step 026a

Combine the matching F3 and F9

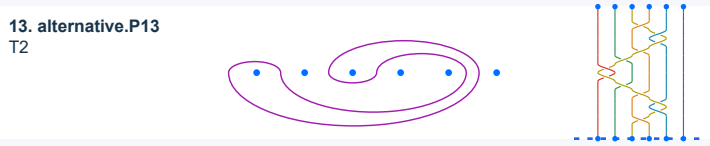
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

12 unchanged displayed factors above; 3 below. Only the changed interval is shown. PARALLEL STEP 26: moves a, b are independent and each starts from the SAME input. Both orders are certified.

Before



After



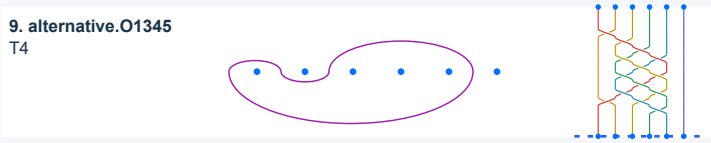
3 - step 026b

The O1345 daisy returns EXACTLY the borrowed Q14

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 8 below. Only the changed interval is shown. PARALLEL STEP 26: moves a, b are independent and each starts from the SAME input. Both orders are certified.

Before



After



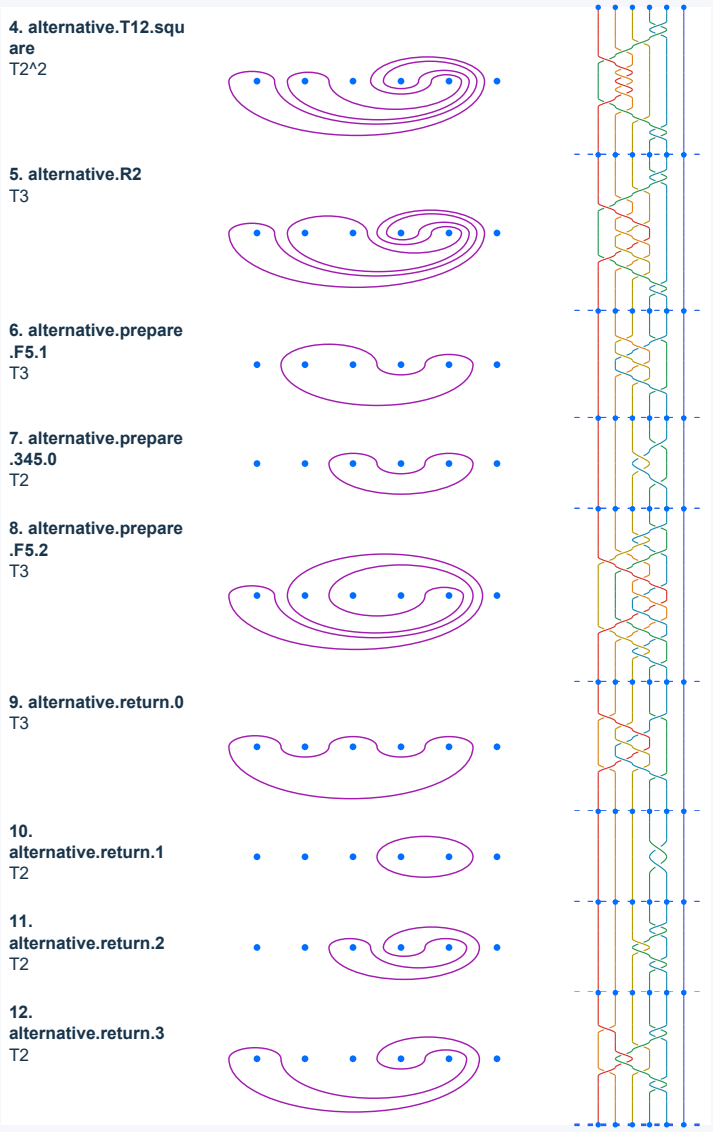
3 - step 027

Return Q14 up to restore the round T12 square and T123

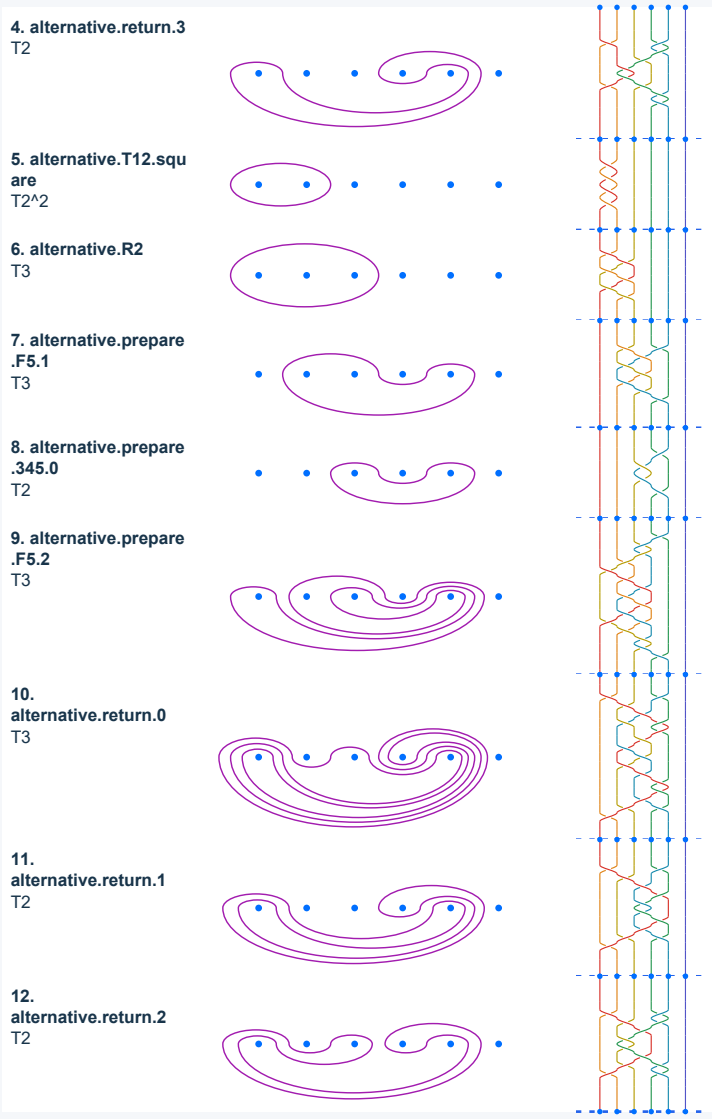
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

3 unchanged displayed factors above; 7 below. Only the changed interval is shown.

Before



After



3 - step 028a

Close the original daisy after its actual Q14 returns

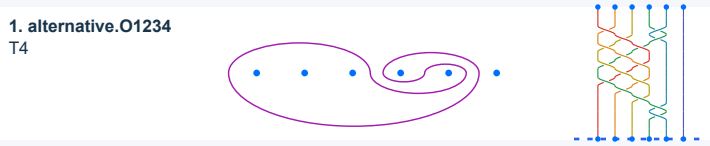
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 15 below. Only the changed interval is shown. PARALLEL STEP 28: moves a, b are independent and each starts from the SAME input. Both orders are certified.

Before



After



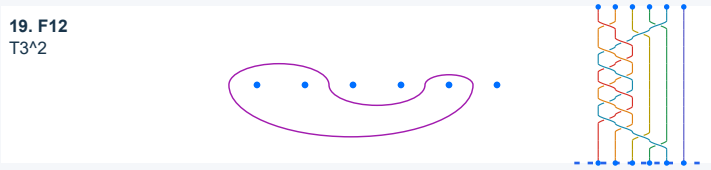
3 - step 028b

Split F12 into its two positive units

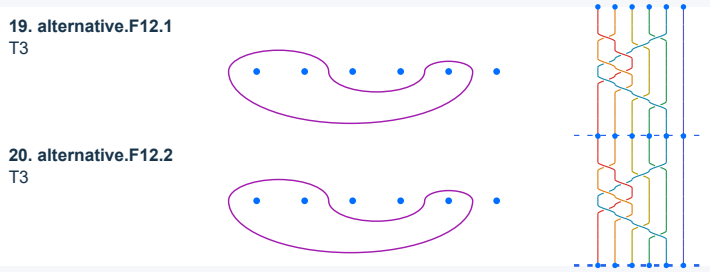
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

18 unchanged displayed factors above; 0 below. Only the changed interval is shown. PARALLEL STEP 28: moves a, b are independent and each starts from the SAME input. Both orders are certified.

Before



After



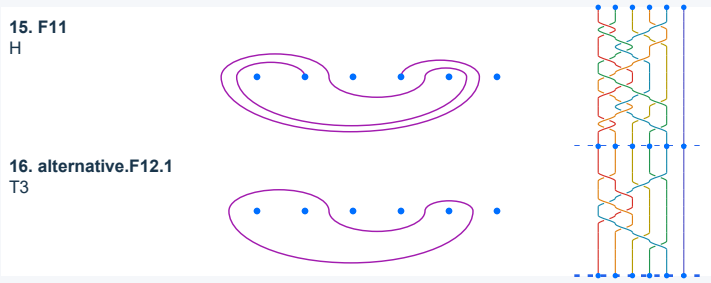
3 - step 029

One C12 root shortens F11

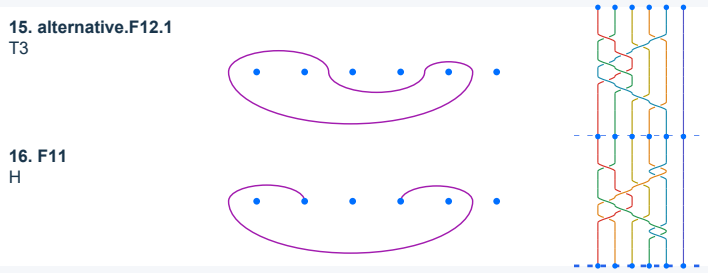
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

14 unchanged displayed factors above; 1 below. Only the changed interval is shown.

Before



After



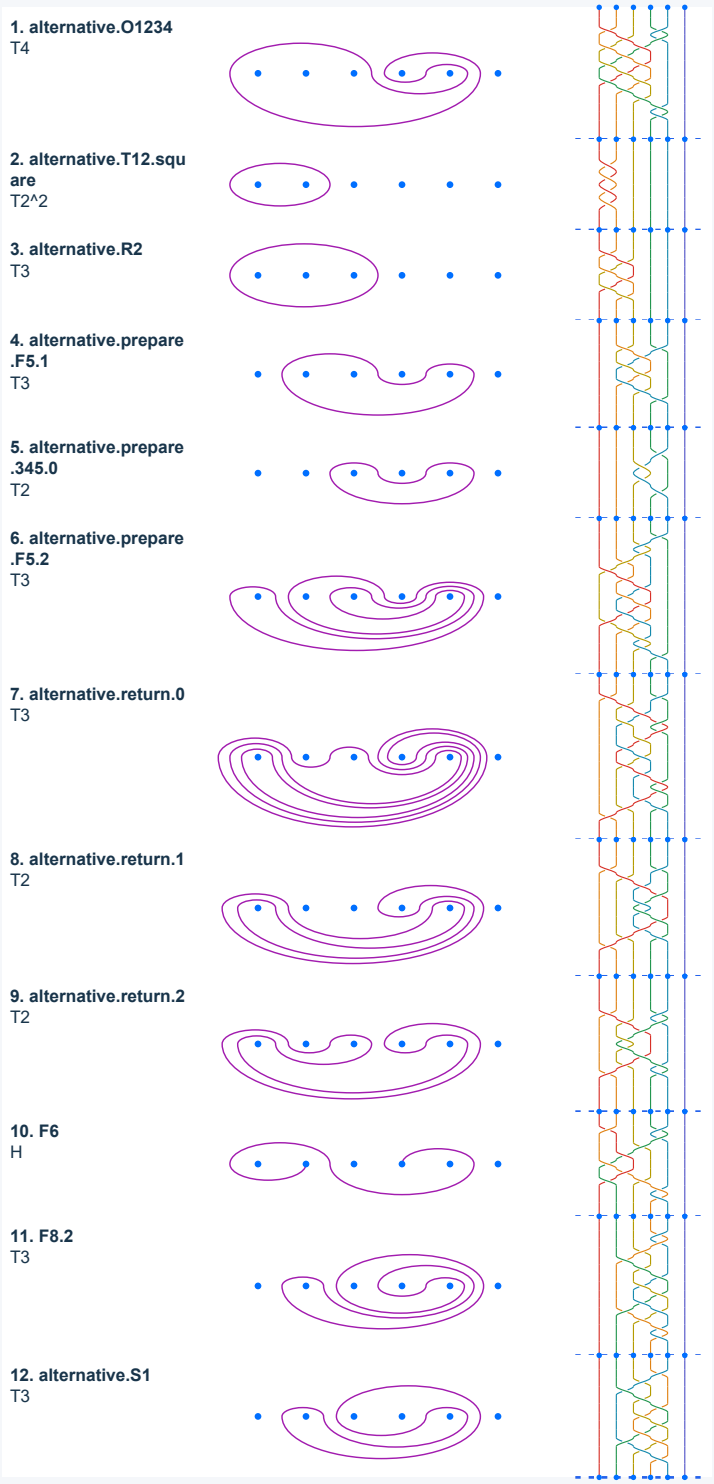
3 - step 030

Cyclic start after BOTH the square and its commuting outer

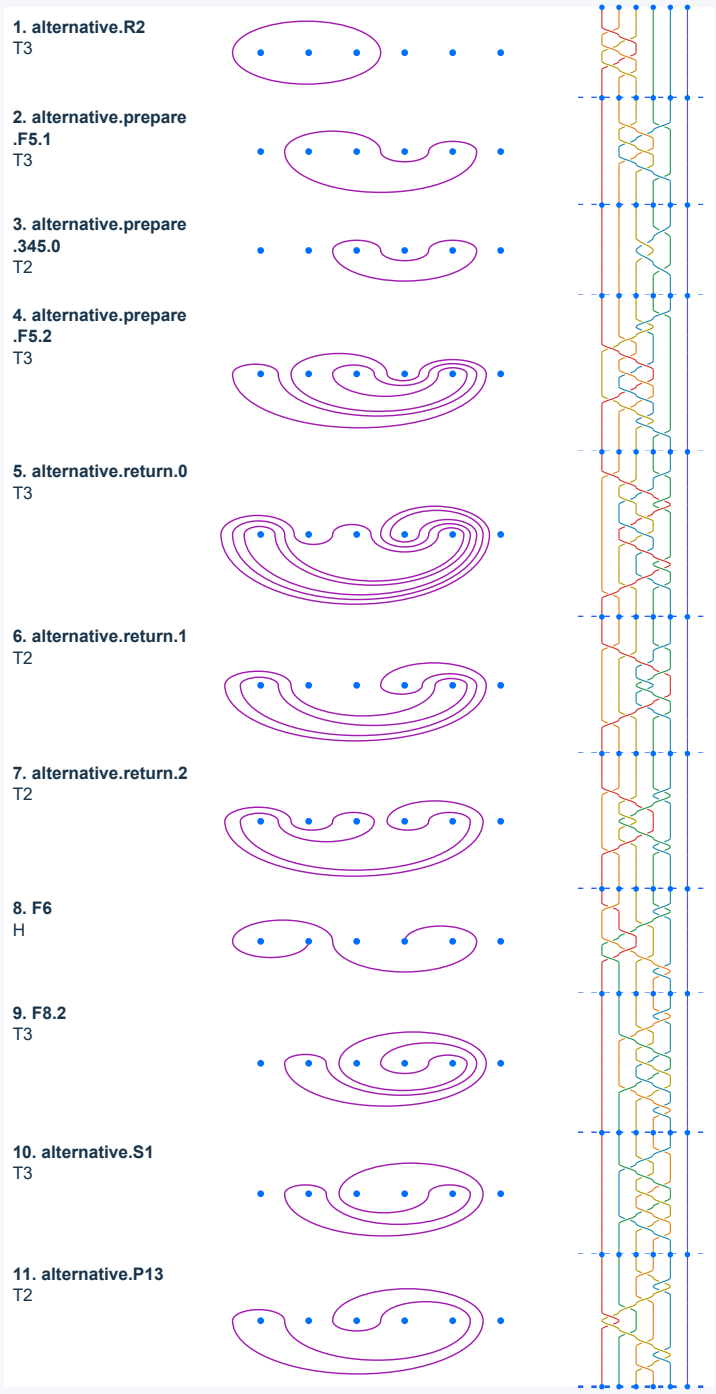
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



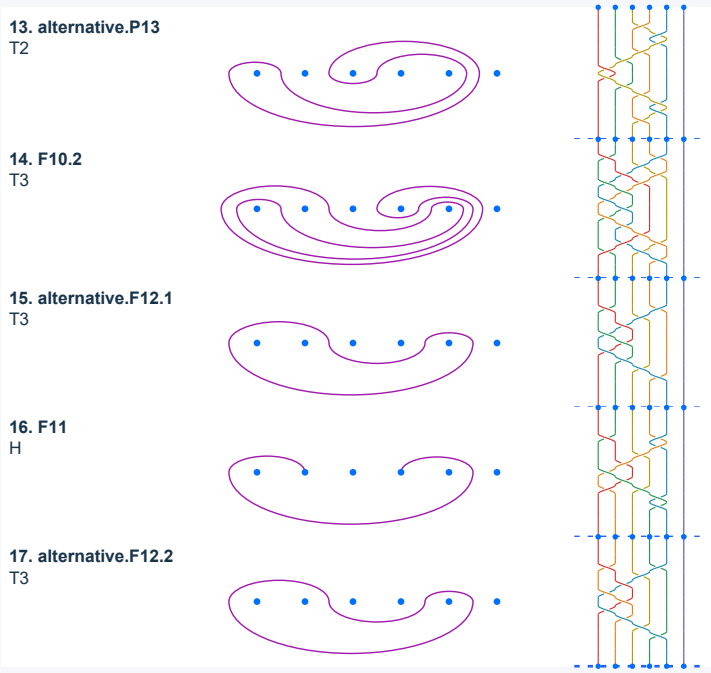
3 - step 030 - continued

Cyclic start after BOTH the square and its commuting outer

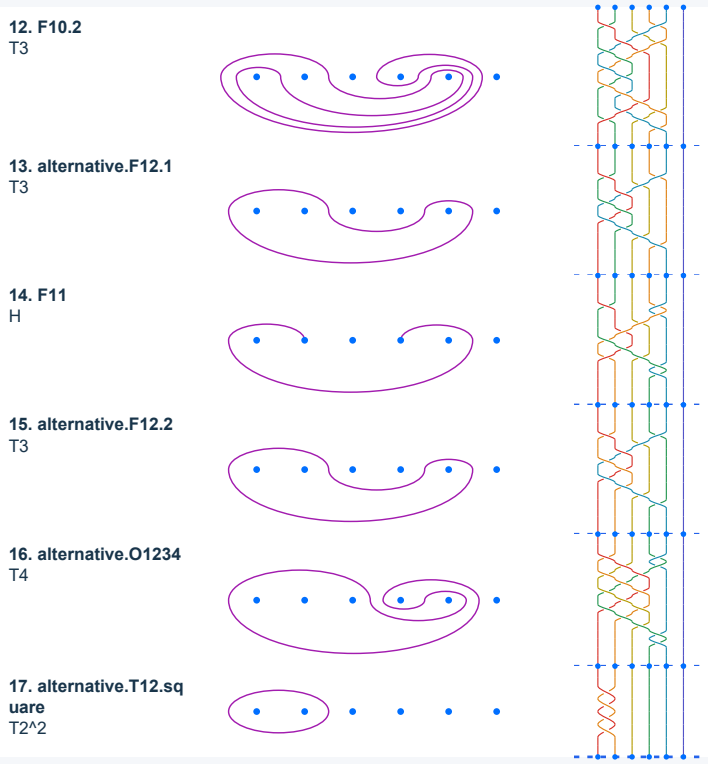
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



3 - step 031

One short-side square drag includes its commuting outer

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

13 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



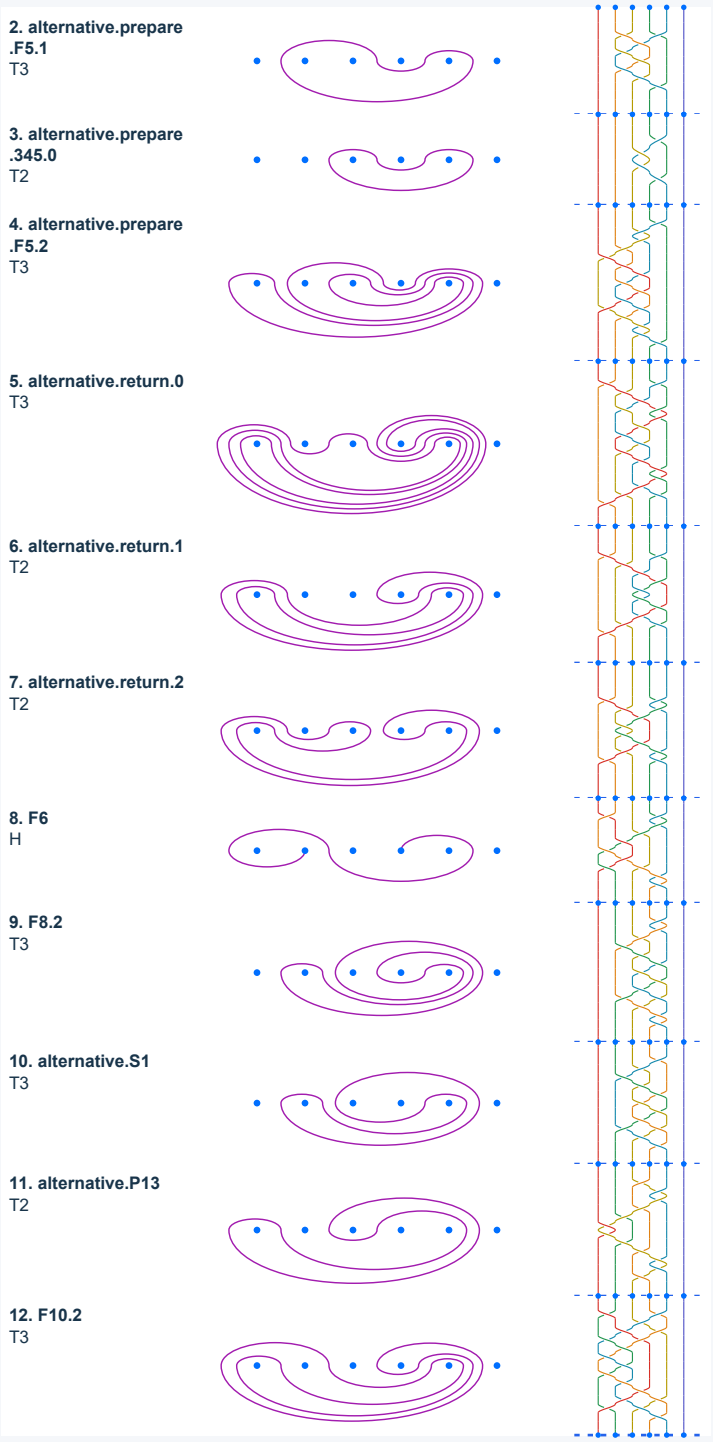
3 - step 032

The same composed sigma4 common frame

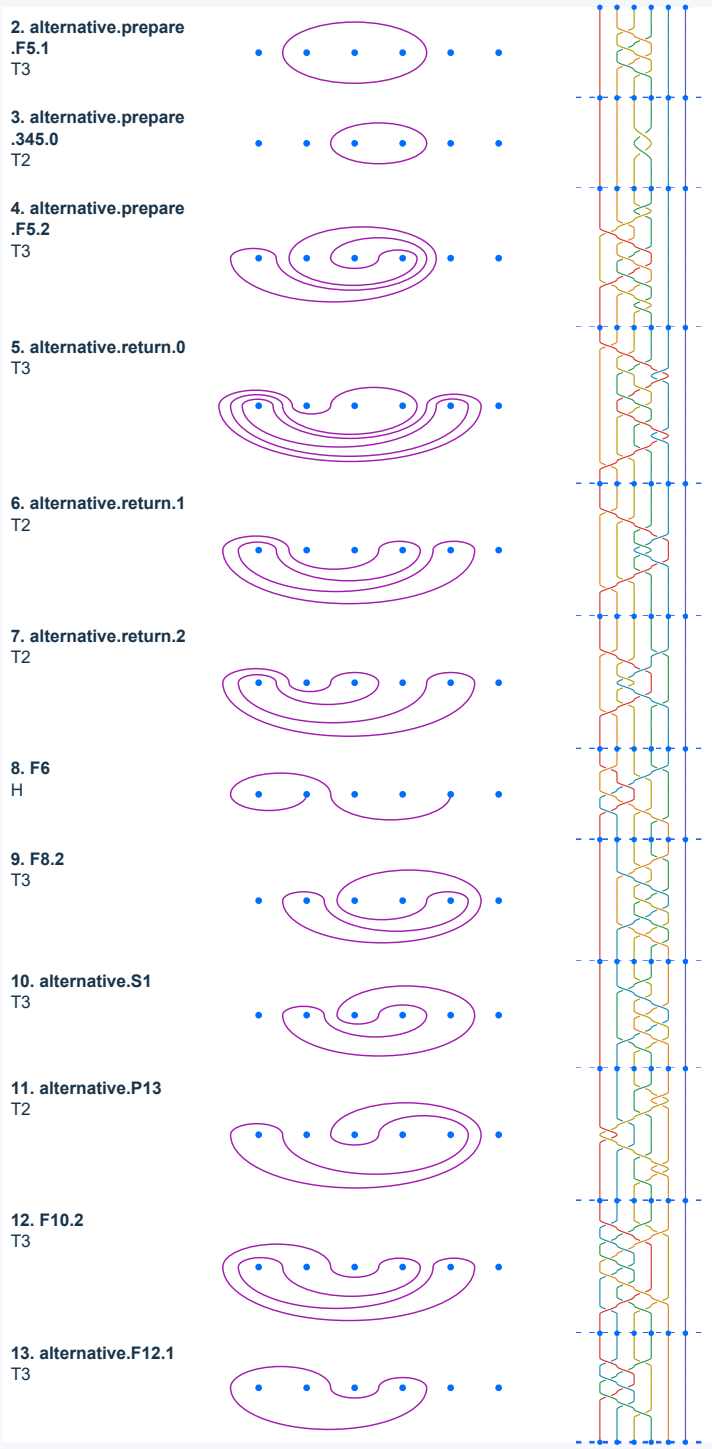
EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

1 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



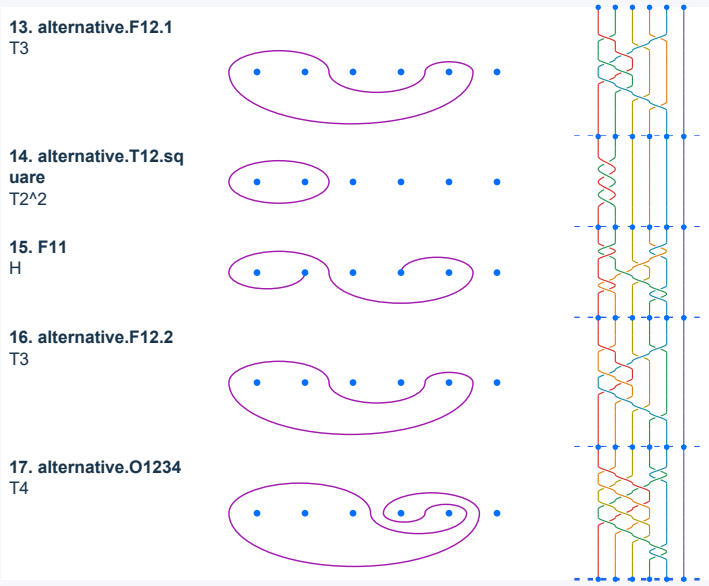
3 - step 032 - continued

The same composed sigma4 common frame

EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

1 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



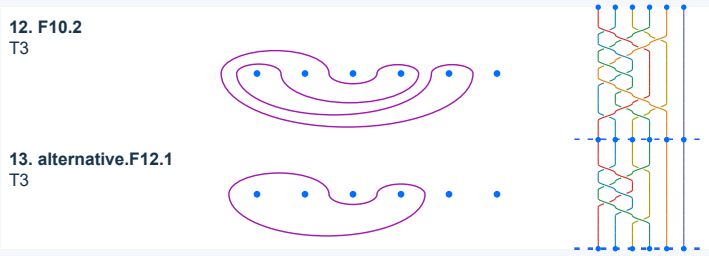
3 - step 033

Reorder C12 before the matching-half gather

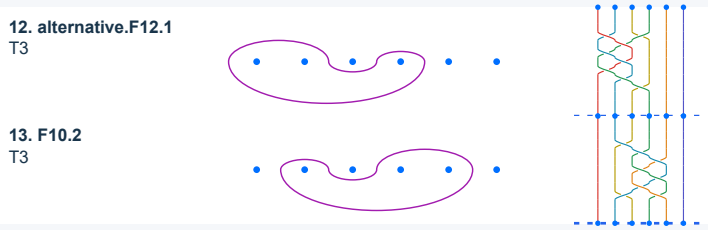
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

11 unchanged displayed factors above; 4 below. Only the changed interval is shown.

Before



After



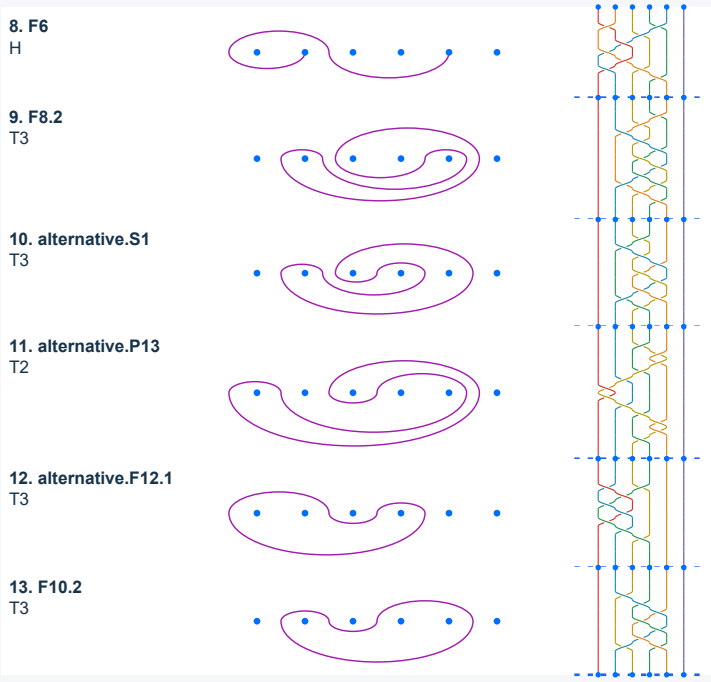
3 - step 034a

Carry F6 DOWN over the first five neighbors; keep F6 fixed

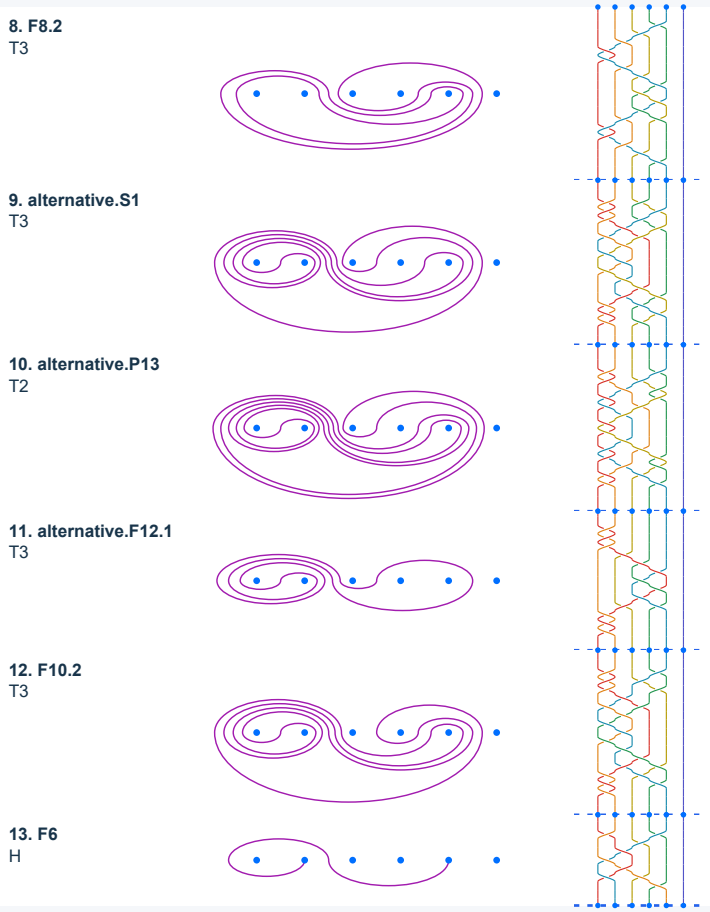
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 4 below. Only the changed interval is shown. PARALLEL STEP 34: moves a, b are independent and each starts from the SAME input. Both orders are certified.

Before



After



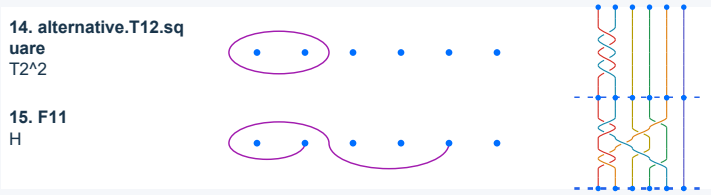
3 - step 034b

Carry F11 UP over the squared pair; keep F11 fixed

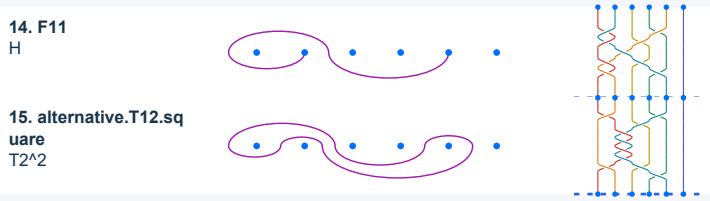
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

13 unchanged displayed factors above; 2 below. Only the changed interval is shown. PARALLEL STEP 34: moves a, b are independent and each starts from the SAME input. Both orders are certified.

Before



After



3 - step 035

Combine the unchanged half twists where they meet

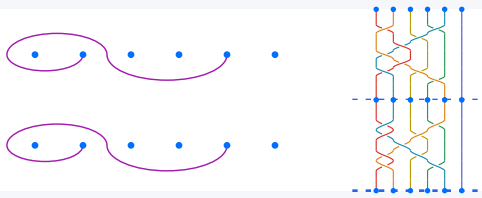
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

12 unchanged displayed factors above; 3 below. Only the changed interval is shown.

Before

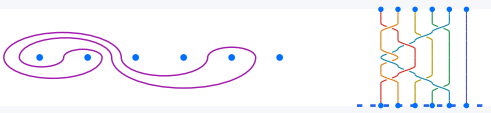
13. F6
H

14. F11
H



After

13. alternative.P24
T2



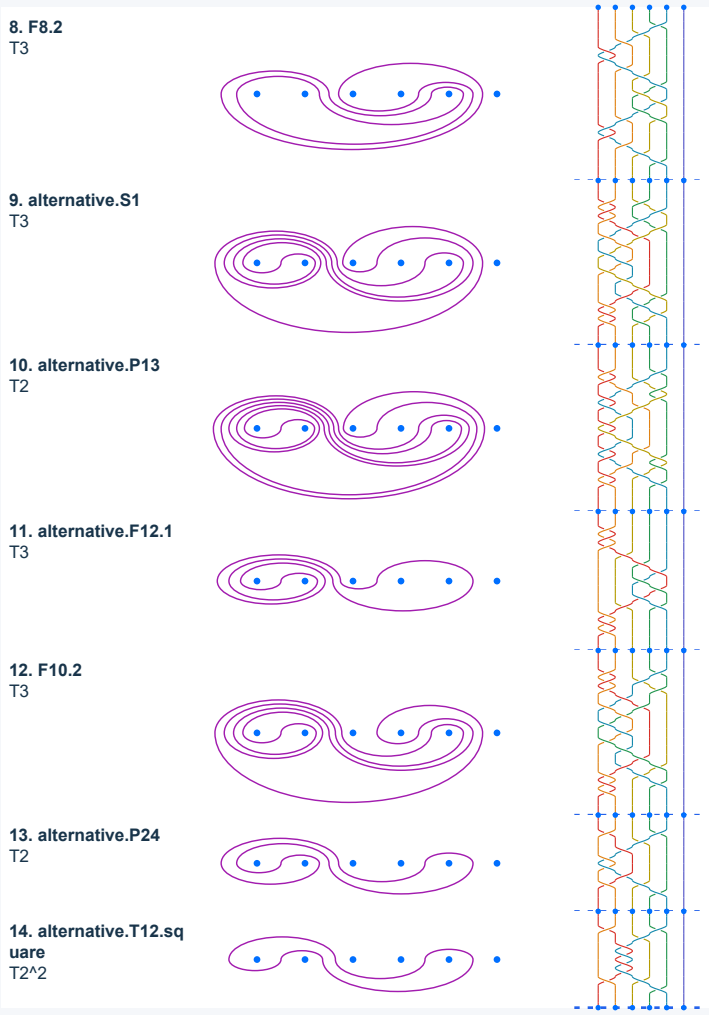
3 - step 036

Carry the combined full twist UP over five neighbors; the lower square stays fixed

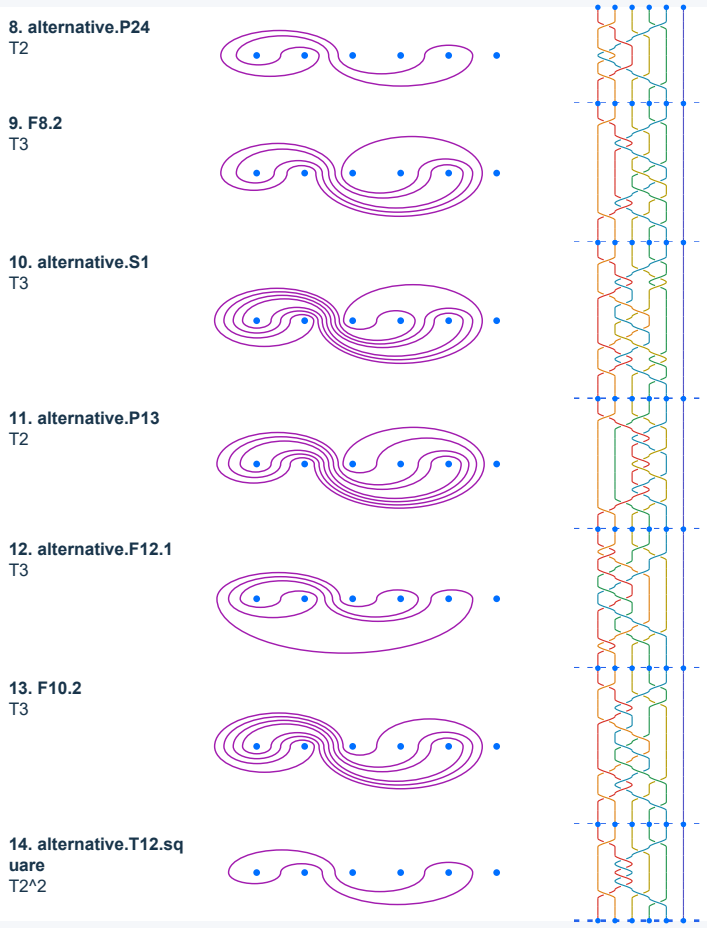
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 2 below. Only the changed interval is shown.

Before



After



3 - step 037

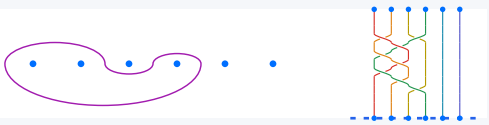
Expand the retained three-point root after the half twists combine

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

14 unchanged displayed factors above; 1 below. Only the changed interval is shown.

Before

15. alternative.F12.2
T3

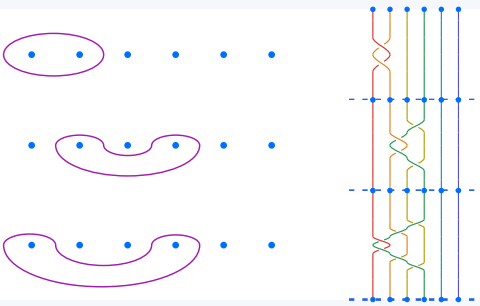


After

15. alternative.next.0
T2

16. alternative.next.1
T2

17. alternative.next.2
T2



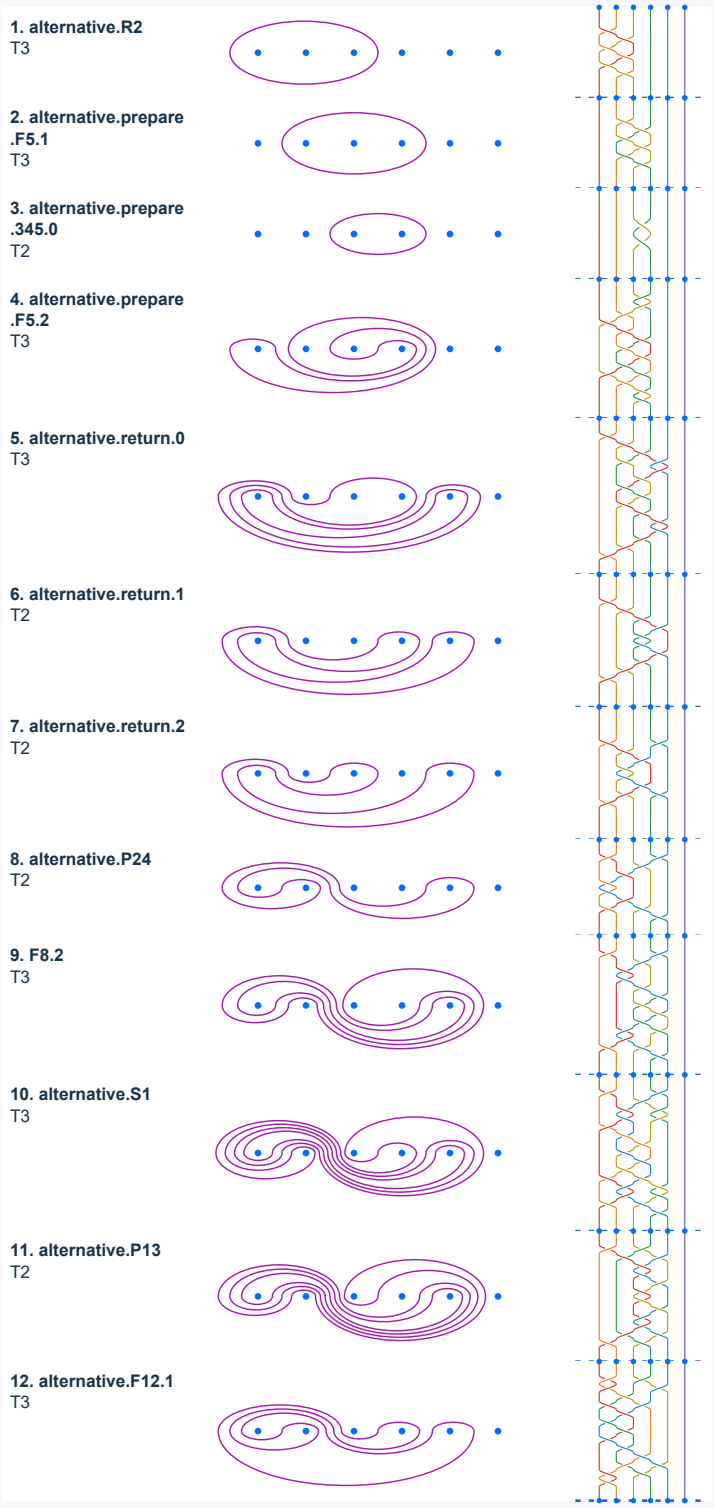
3 - step 038

Cyclic start at Q15 REJOINS the unchanged route

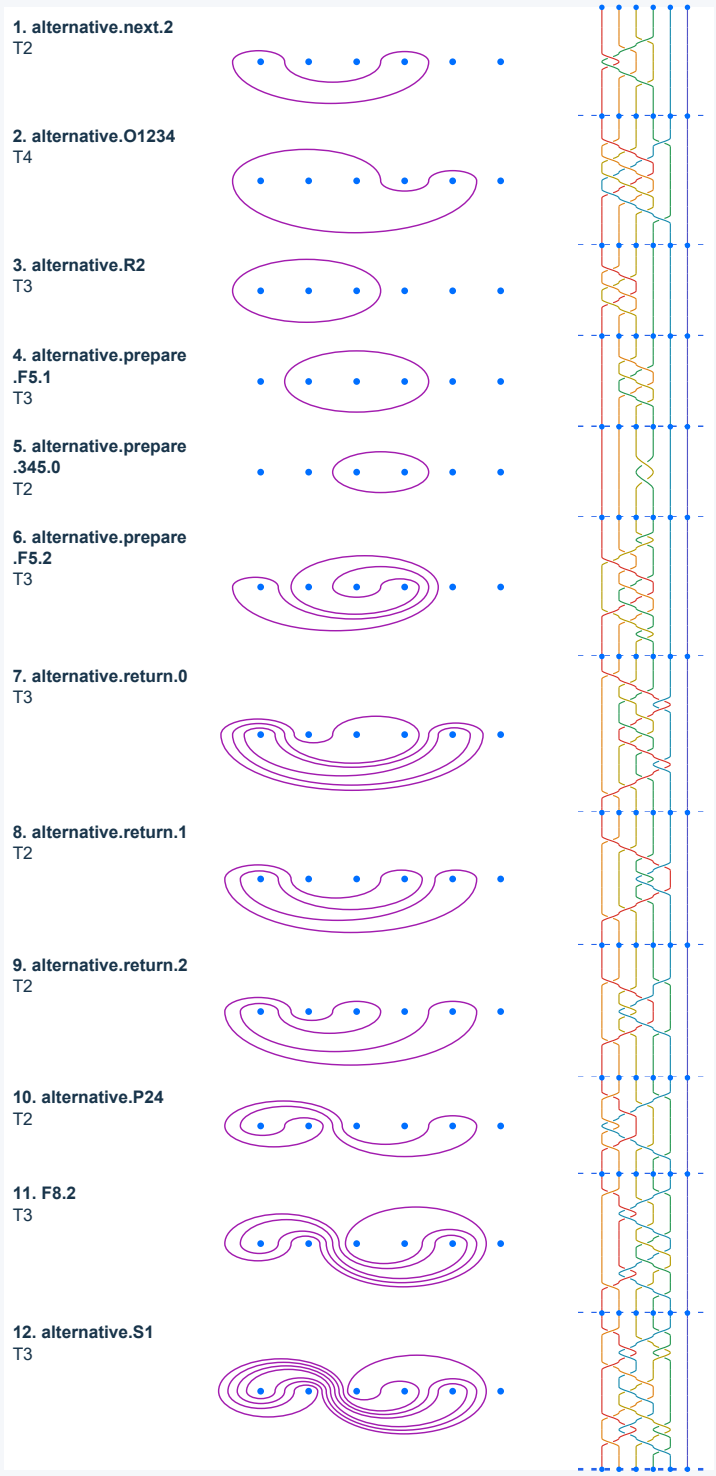
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



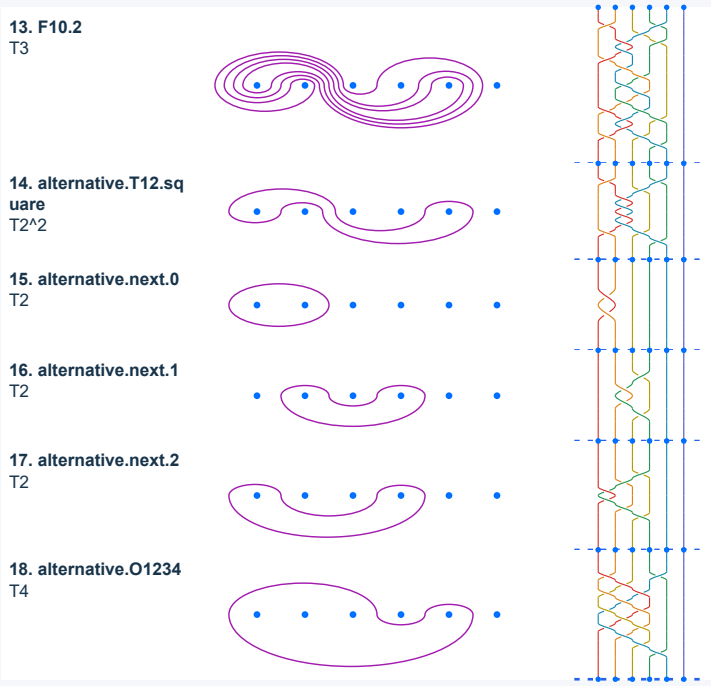
3 - step 038 - continued

Cyclic start at Q15 REJOINS the unchanged route

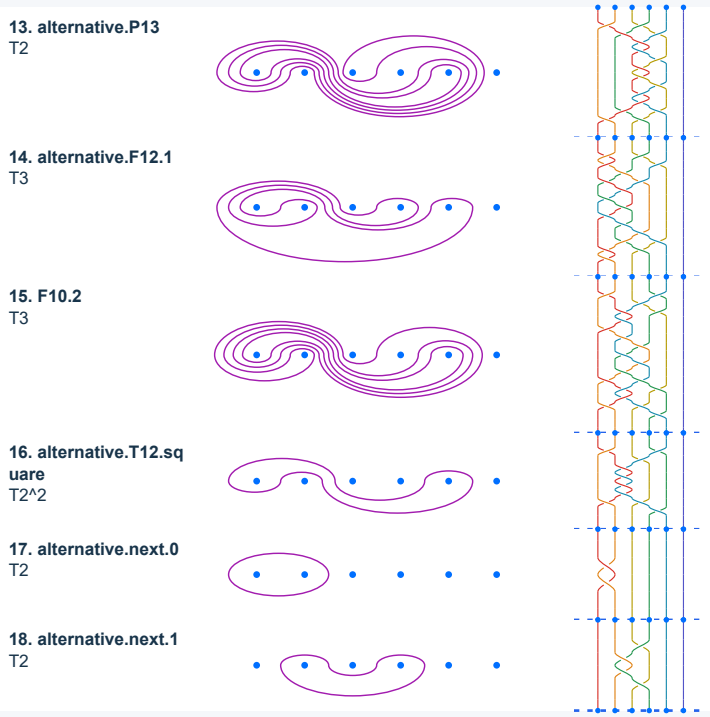
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



3 - step 039

Carry Q15 through the combined four-point prefix and the two roots

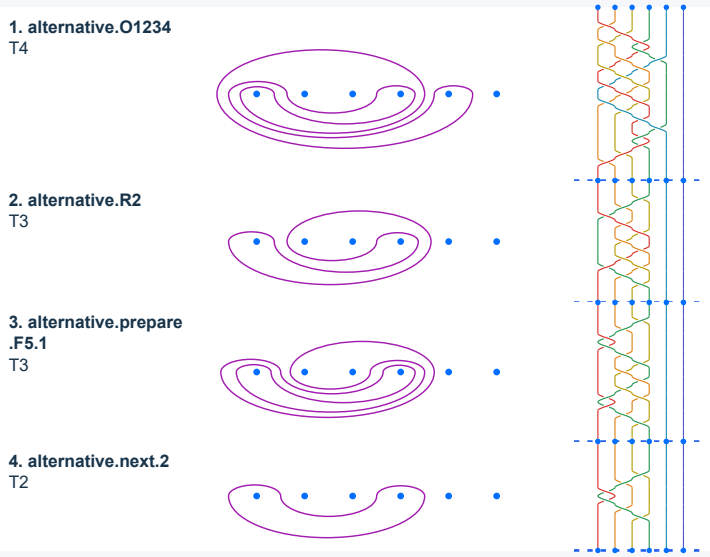
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 14 below. Only the changed interval is shown.

Before



After



3 - step 040

PC cluster lantern gives the second four-point twist

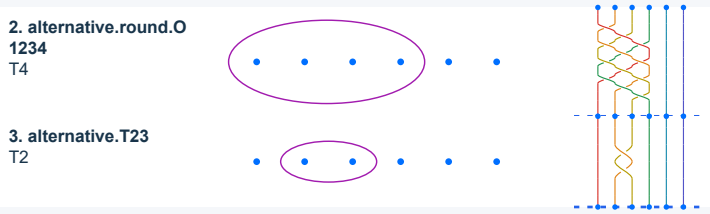
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 14 below. Only the changed interval is shown.

Before



After



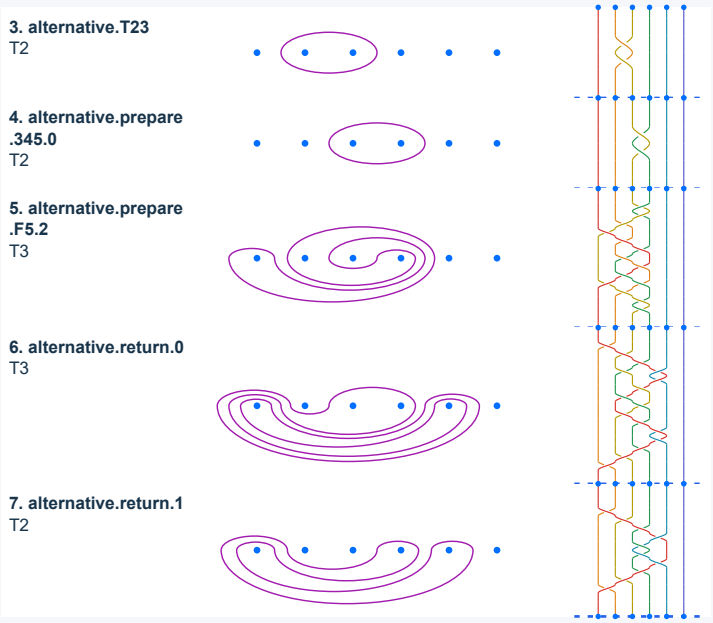
3 - step 041

Gather the ACTUAL missing 45 petal by the short side

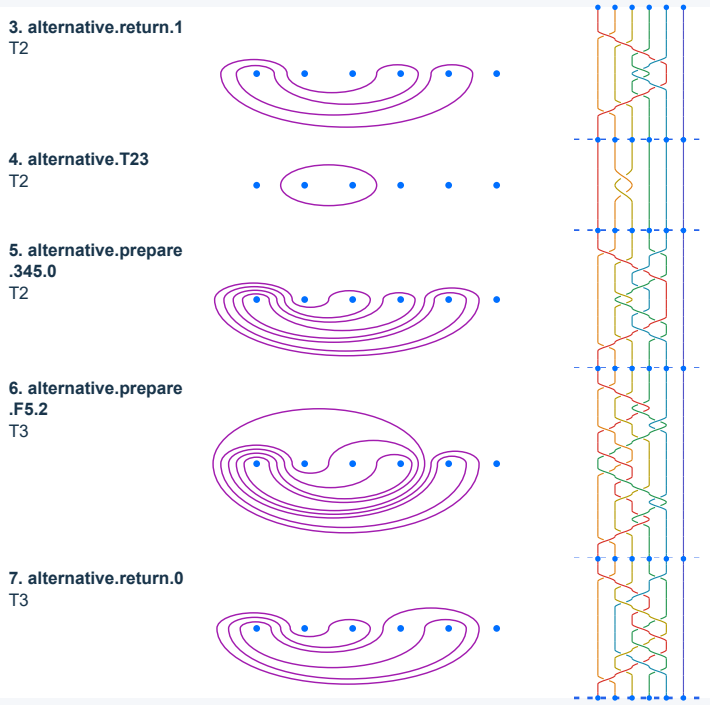
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 10 below. Only the changed interval is shown.

Before



After



3 - step 042a

Cabled PC lantern on the ACTUAL two four-point curves and supplied 45 petal

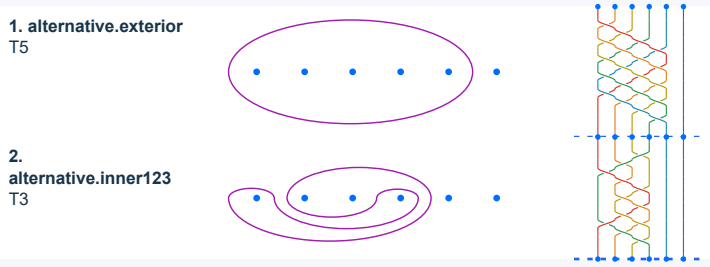
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 14 below. Only the changed interval is shown. PARALLEL STEP 42: moves a, b are independent and each starts from the SAME input. Both orders are certified.

Before



After



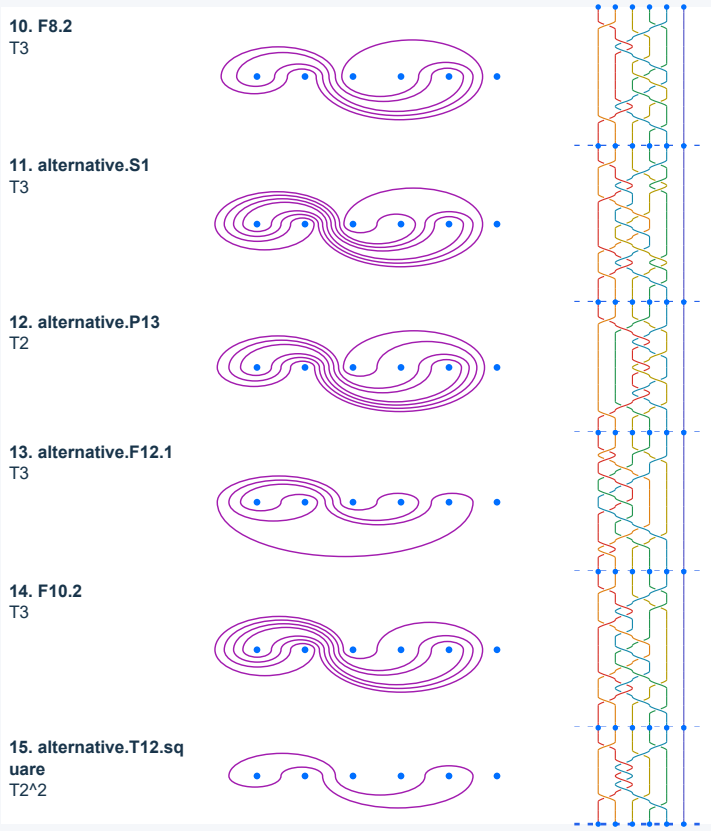
3 - step 042b

Unwind five actual residual curves with the fixed positive square

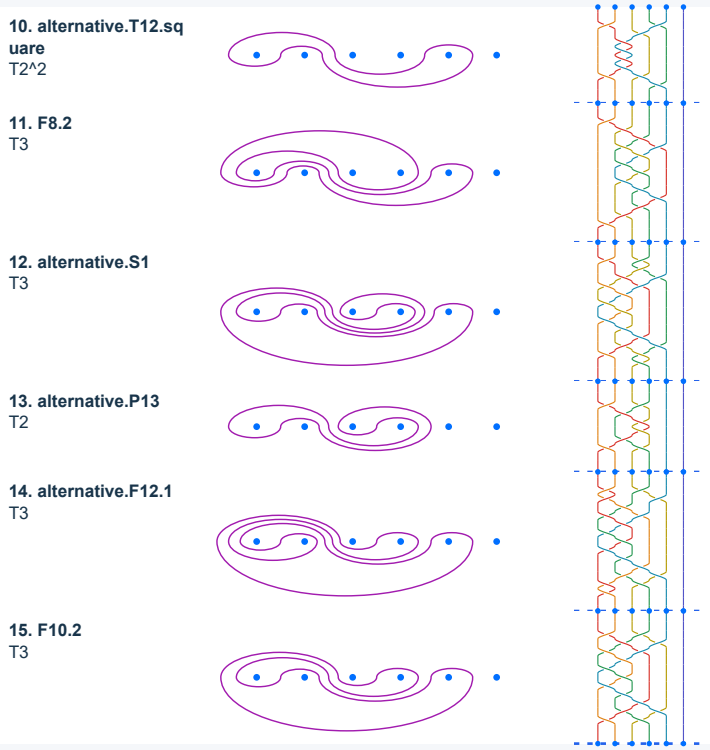
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

9 unchanged displayed factors above; 2 below. Only the changed interval is shown. PARALLEL STEP 42: moves a, b are independent and each starts from the SAME input. Both orders are certified.

Before



After



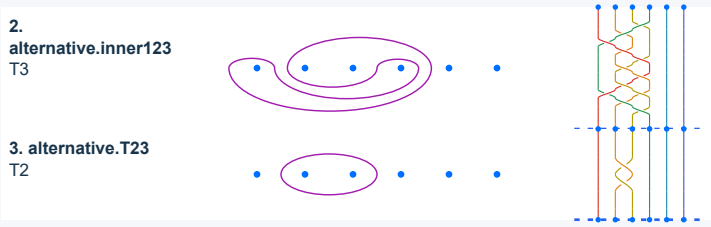
3 - step 043

Commute the 123 root past T23 to its actual 34 and 124 companions

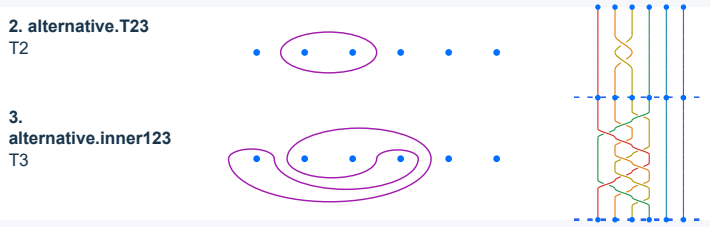
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 13 below. Only the changed interval is shown.

Before



After



3 - step 044a

Cabled PC lantern contracts the ACTUAL 123, 34 and 124 curves

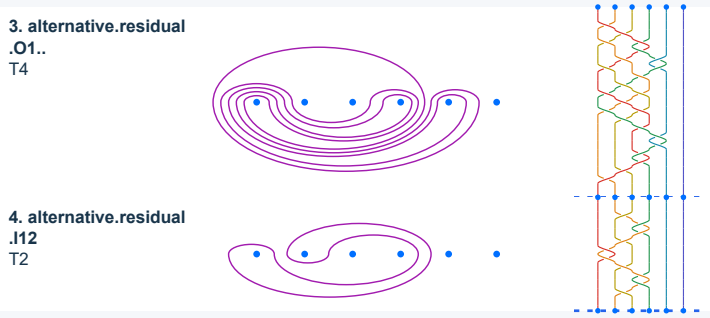
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 11 below. Only the changed interval is shown. PARALLEL STEP 44: moves a, b, c are independent and each starts from the SAME input. Both orders are certified.

Before



After



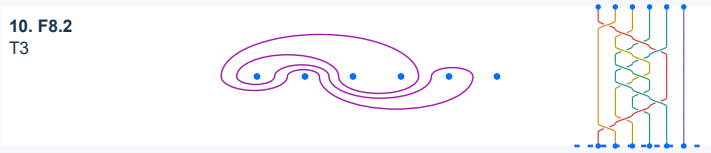
3 - step 044b

The actual 345 daisy lends its round 34 petal

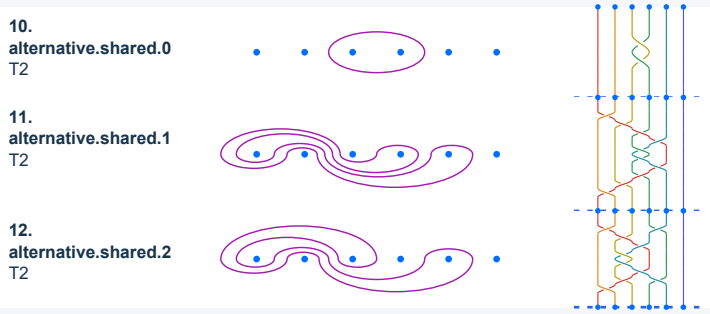
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

9 unchanged displayed factors above; 6 below. Only the changed interval is shown. PARALLEL STEP 44: moves a, b, c are independent and each starts from the SAME input. Both orders are certified.

Before



After



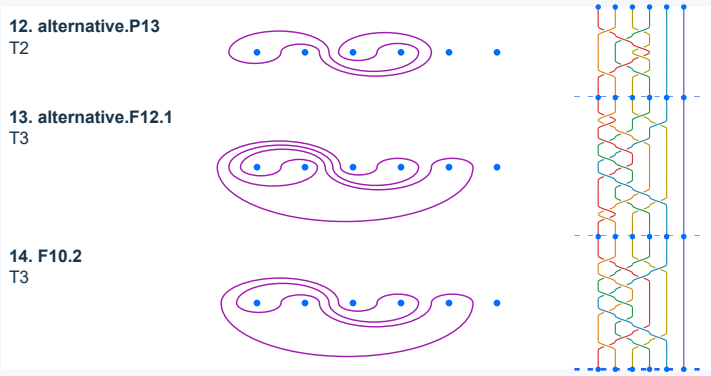
3 - step 044c

Carry the 245 root up across the 145 root and the 13 pair

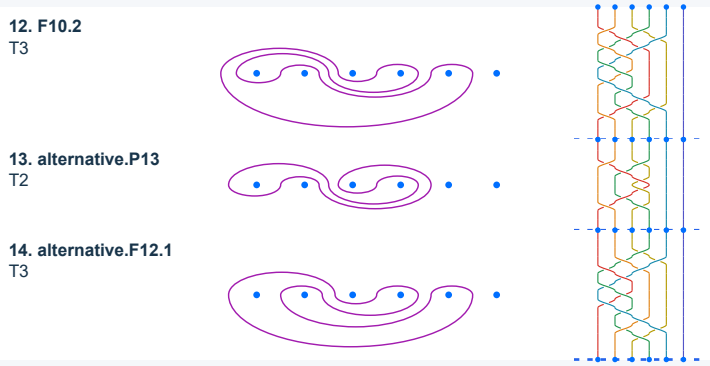
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

11 unchanged displayed factors above; 2 below. Only the changed interval is shown. PARALLEL STEP 44: moves a, b, c are independent and each starts from the SAME input. Both orders are certified.

Before



After



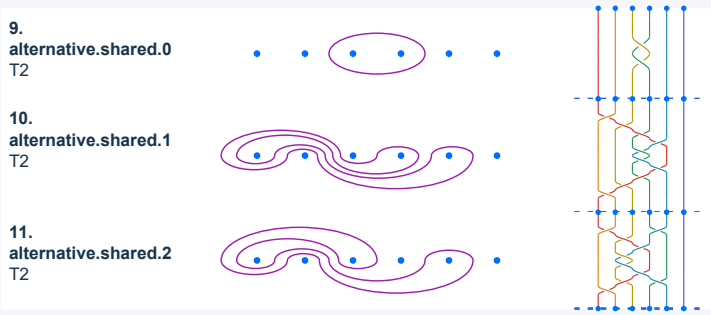
3 - step 045

Carry round 34 down through its own two remaining petals

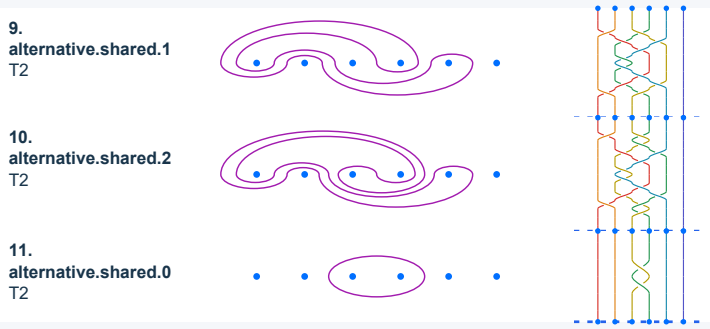
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 6 below. Only the changed interval is shown.

Before



After



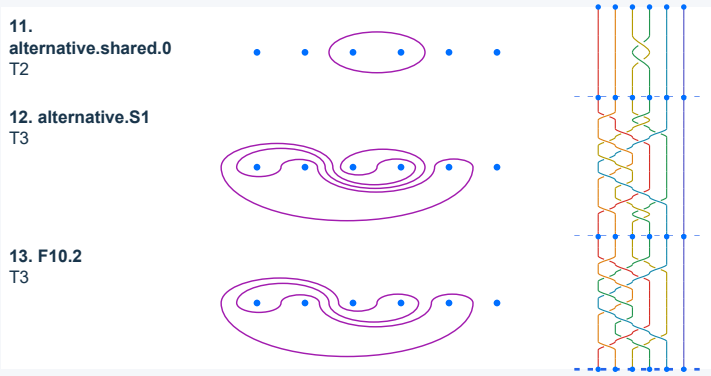
3 - step 046a

Cabled PC lantern closes the actual 34, 235 and 245 cluster

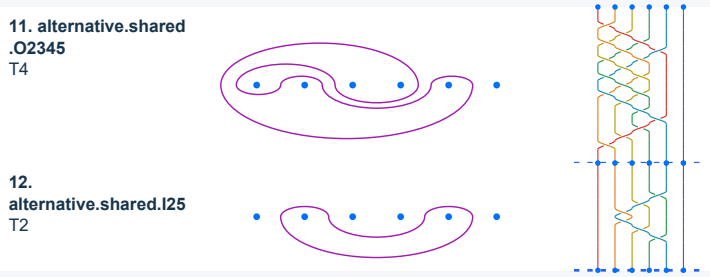
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

10 unchanged displayed factors above; 4 below. Only the changed interval is shown. PARALLEL STEP 46: moves a, b are independent and each starts from the SAME input. Both orders are certified.

Before



After



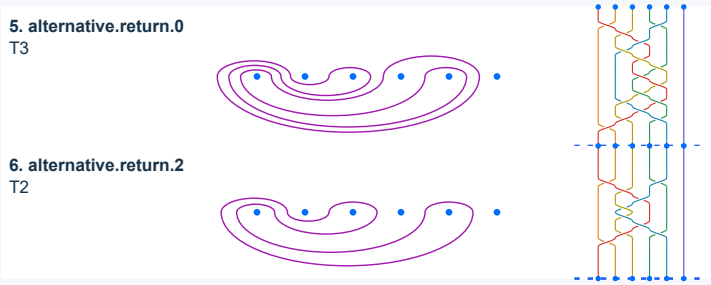
3 - step 046b

The actual 35 pair unwinds the 134 root to a short chart

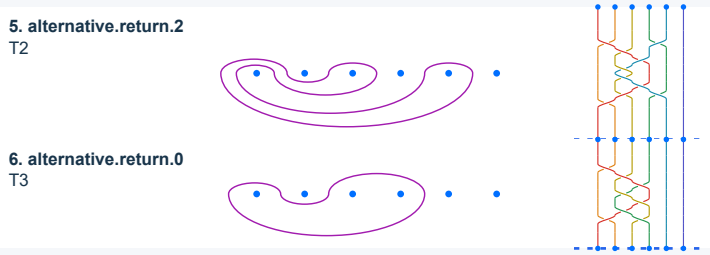
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

4 unchanged displayed factors above; 11 below. Only the changed interval is shown. PARALLEL STEP 46: moves a, b are independent and each starts from the SAME input. Both orders are certified.

Before



After



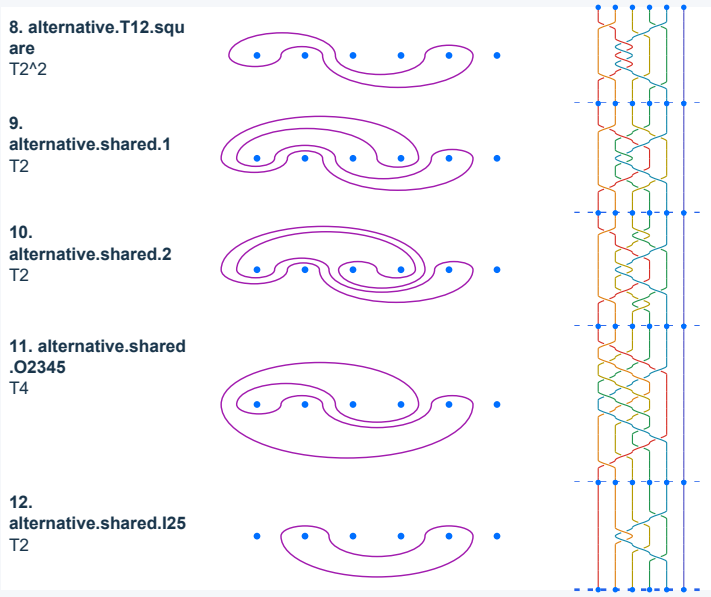
3 - step 047

Carry the shared 25 pair up; unwind the square and its two petals

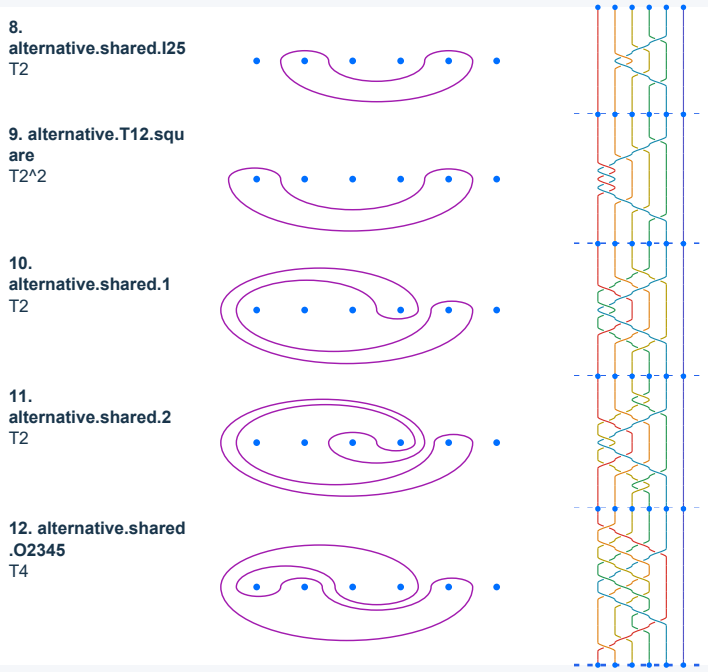
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 4 below. Only the changed interval is shown.

Before



After



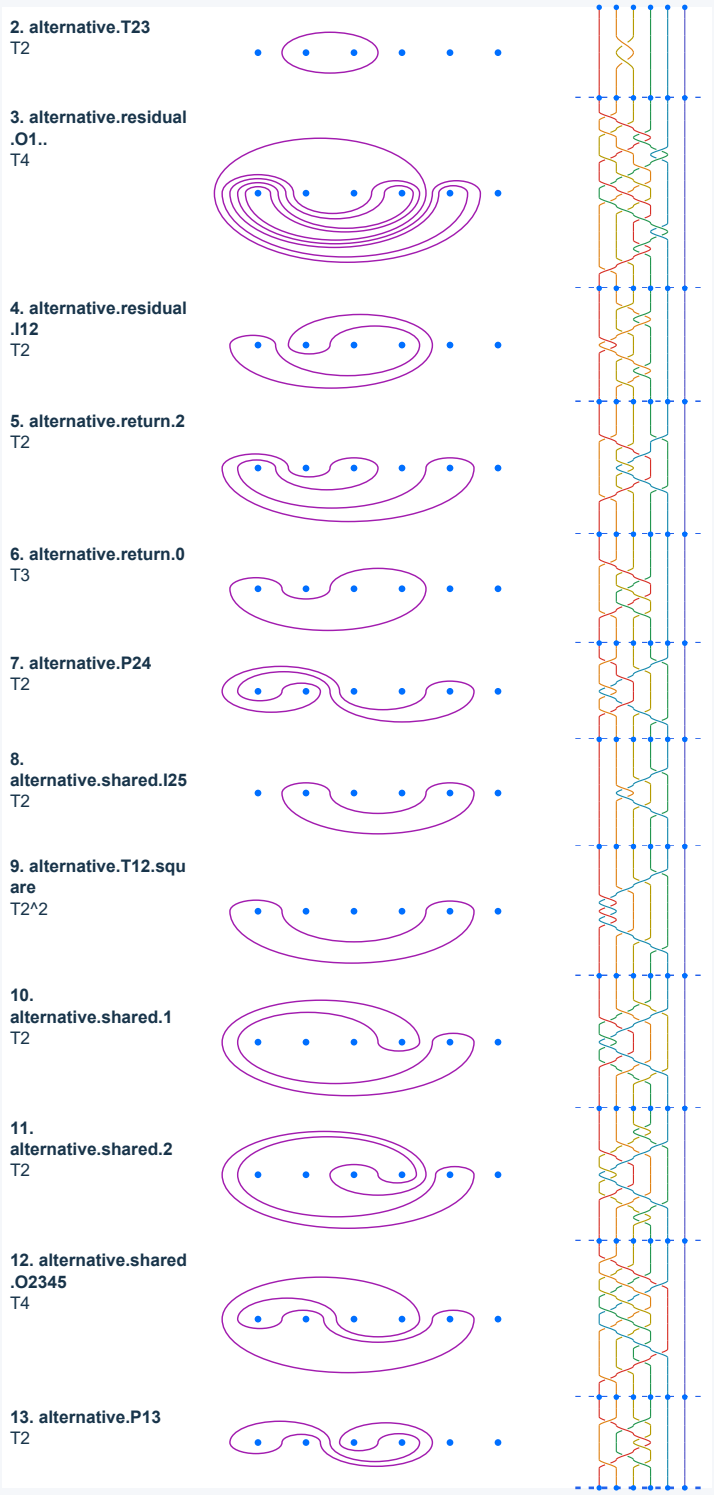
3 - step 048

Common disk frame keeps the inside-four daisy readable

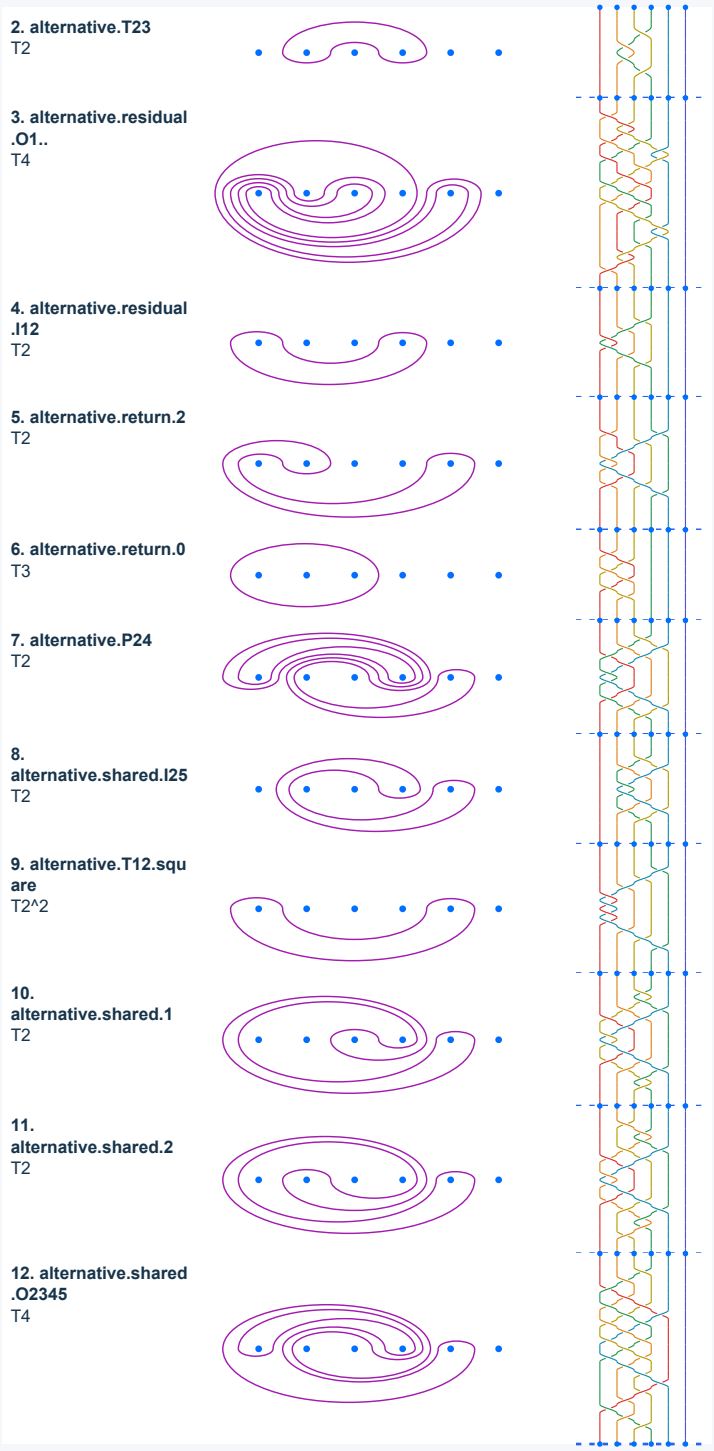
EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

1 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



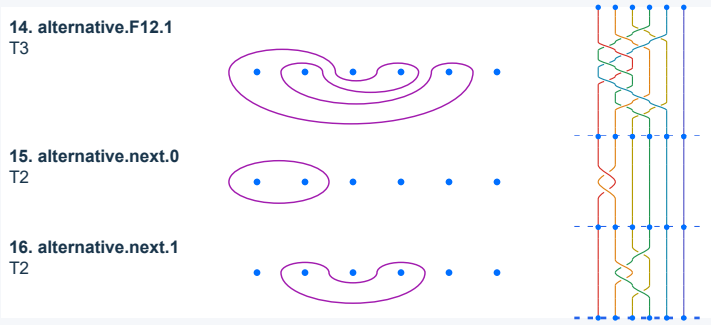
3 - step 048 - continued

Common disk frame keeps the inside-four daisy readable

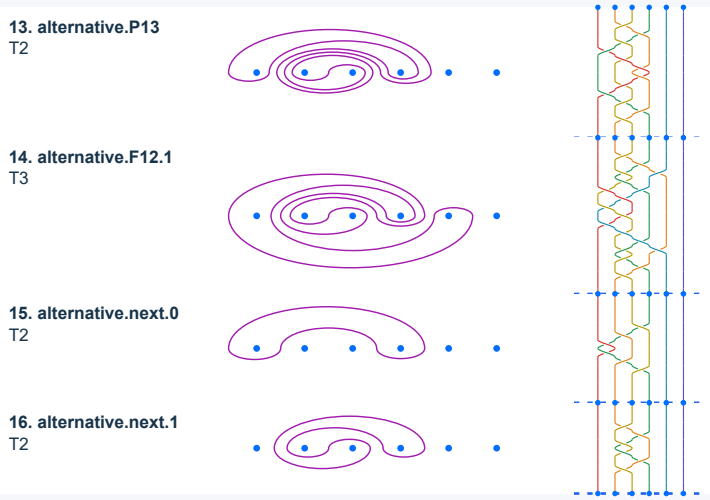
EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

1 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



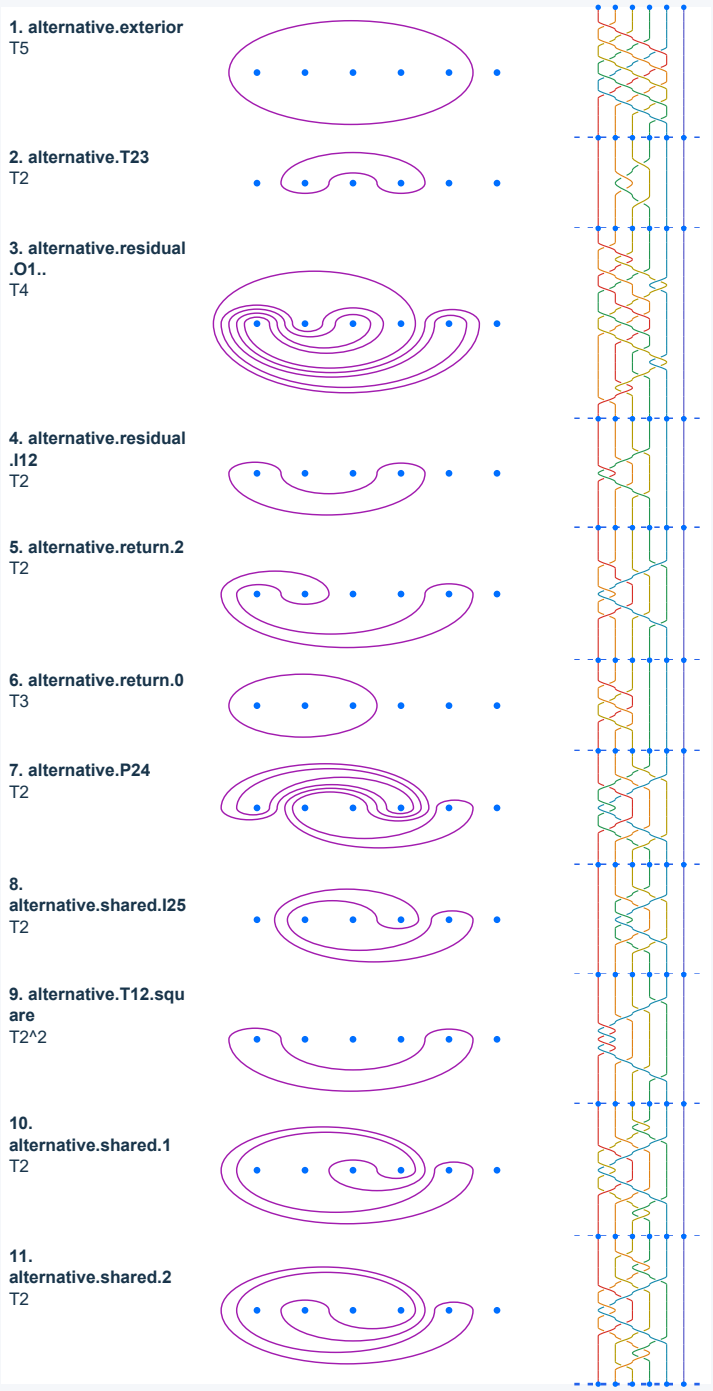
3 - step 049

Choose the cyclic start at the actual three-point root

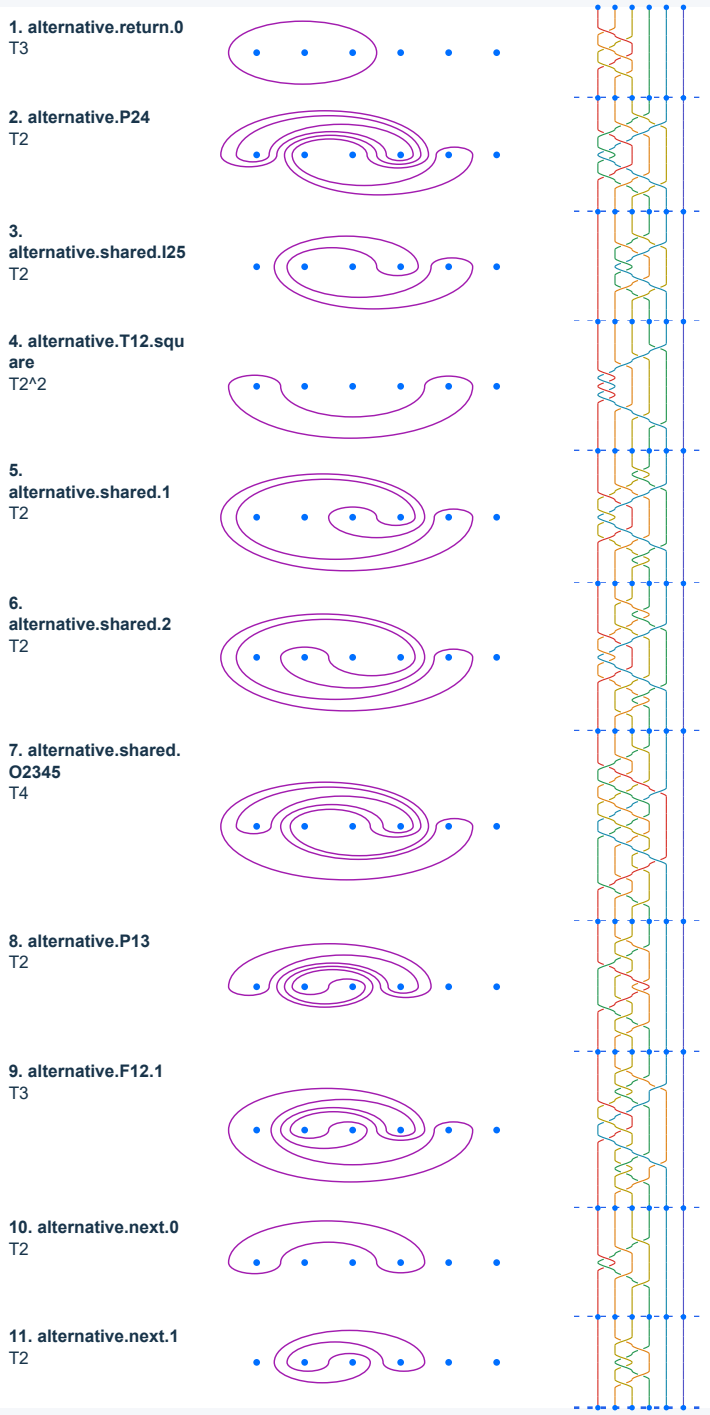
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



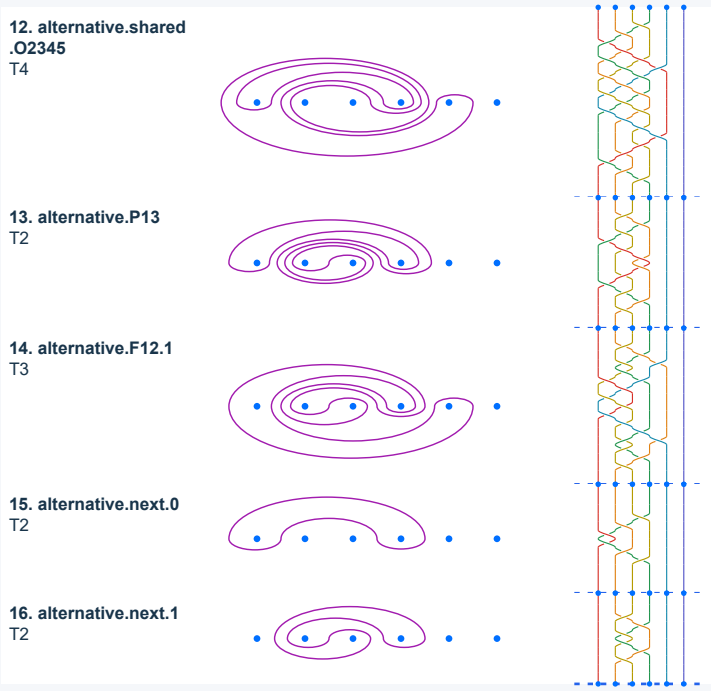
3 - step 049 - continued

Choose the cyclic start at the actual three-point root

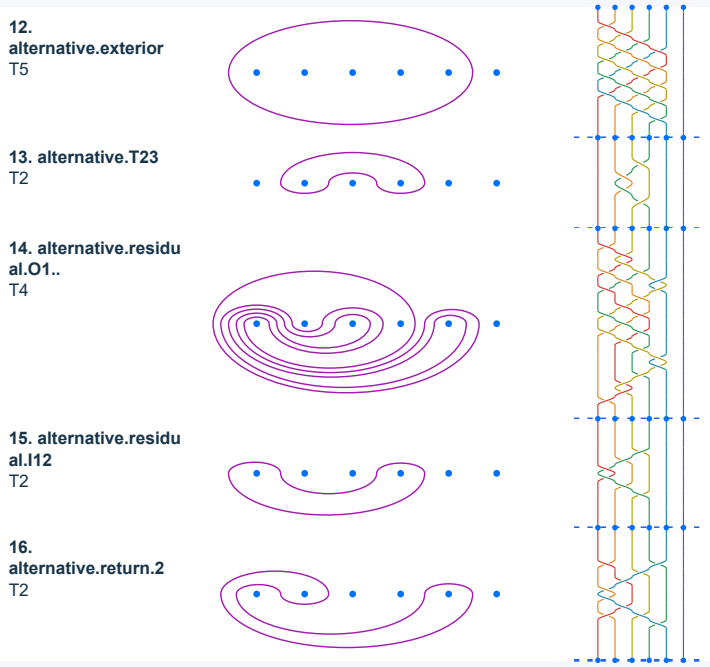
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



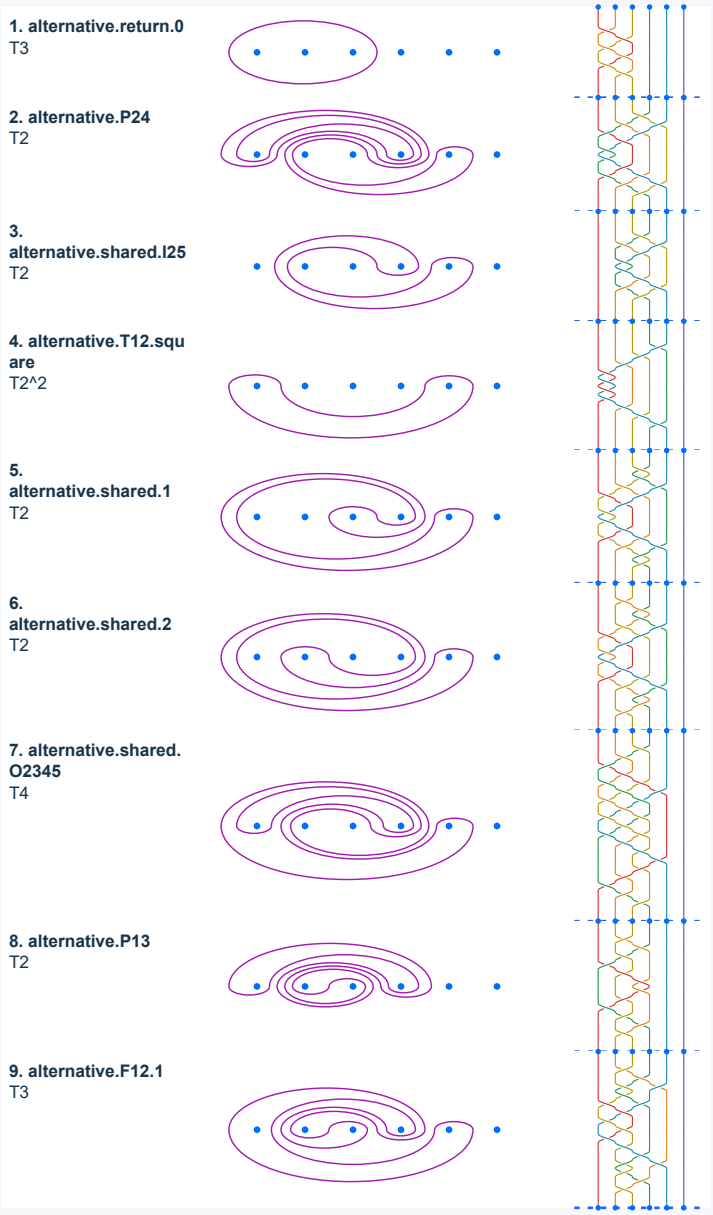
3 - step 050a

Carry the three-point root down through the eight residual factors

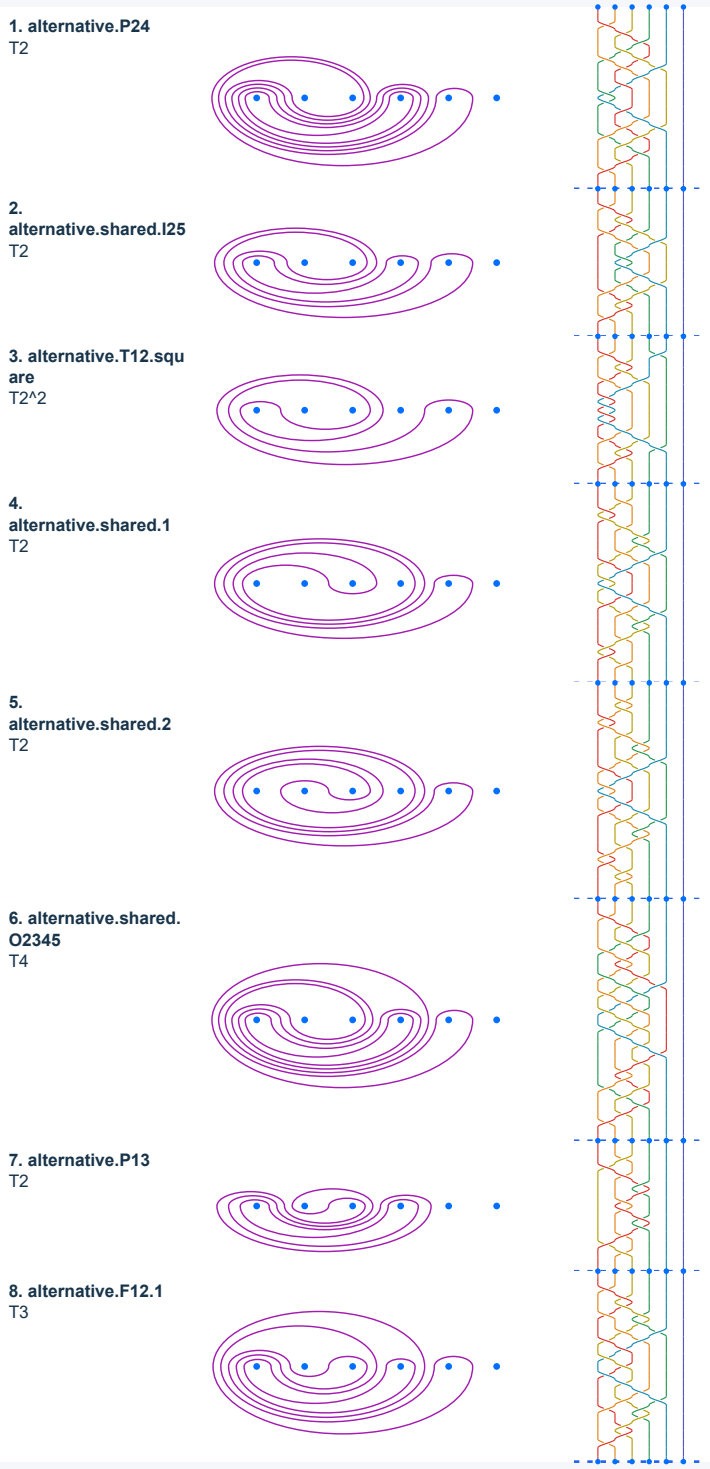
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 7 below. Only the changed interval is shown. PARALLEL STEP 50: moves a, b are independent and each starts from the SAME input. Both orders are certified.

Before



After



3 - step 050a - continued

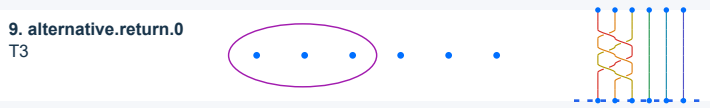
Carry the three-point root down through the eight residual factors

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 7 below. Only the changed interval is shown. PARALLEL STEP 50: moves a, b are independent and each starts from the SAME input. Both orders are certified.

Before

After



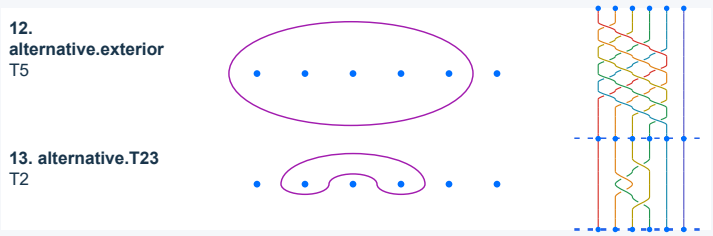
3 - step 050b

Commute the final pair petal past the visible exterior

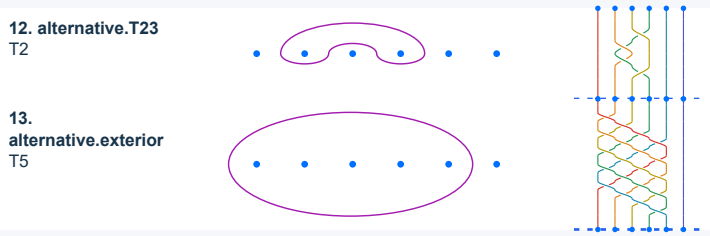
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

11 unchanged displayed factors above; 3 below. Only the changed interval is shown. PARALLEL STEP 50: moves a, b are independent and each starts from the SAME input. Both orders are certified.

Before



After



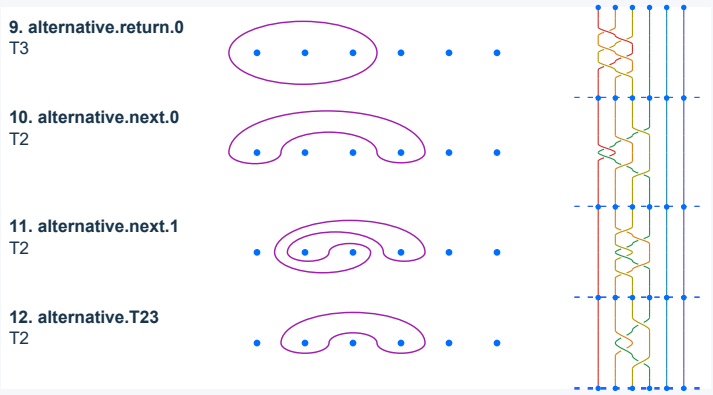
3 - step 051

Transported PC daisy closes the actual root and its three pair petals

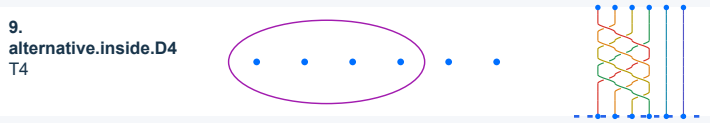
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 4 below. Only the changed interval is shown.

Before



After



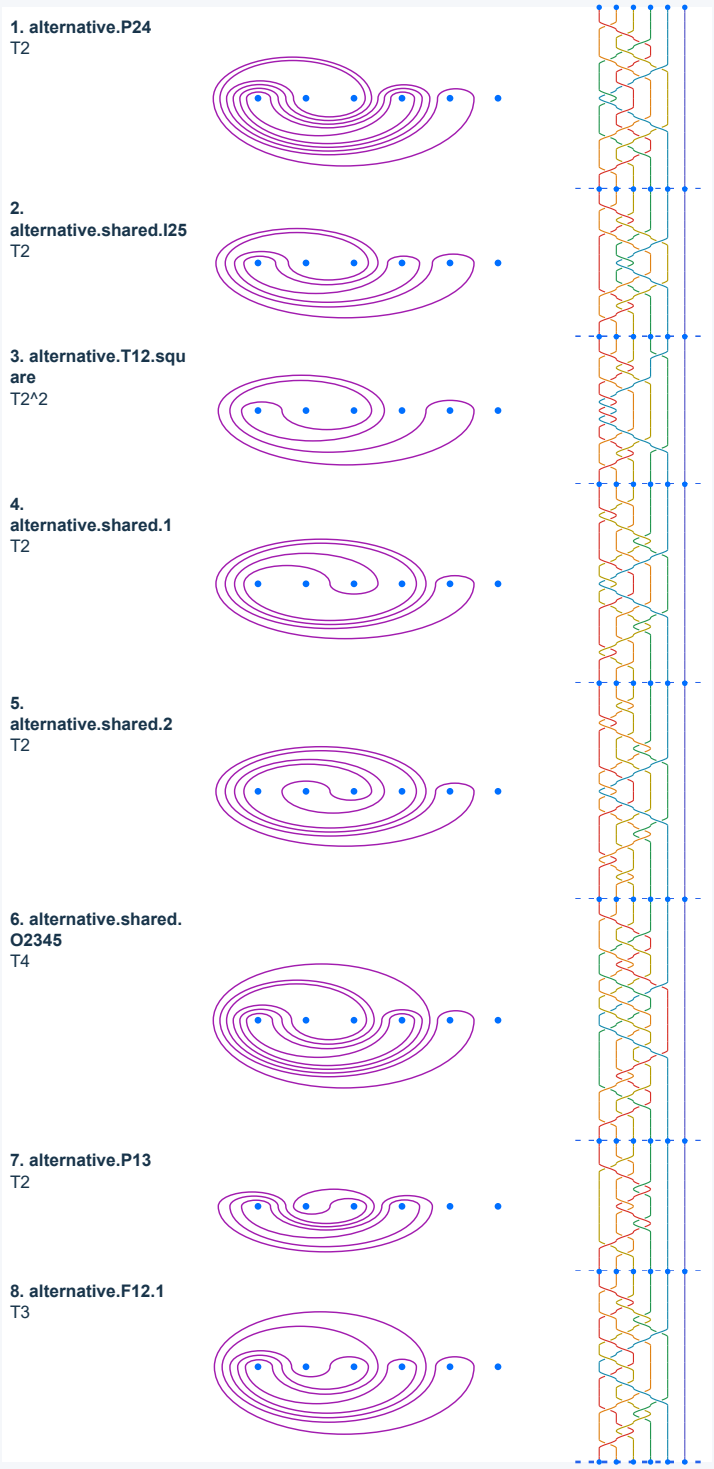
3 - step 052

Carry the new round four-point twist up to unwind the residual block

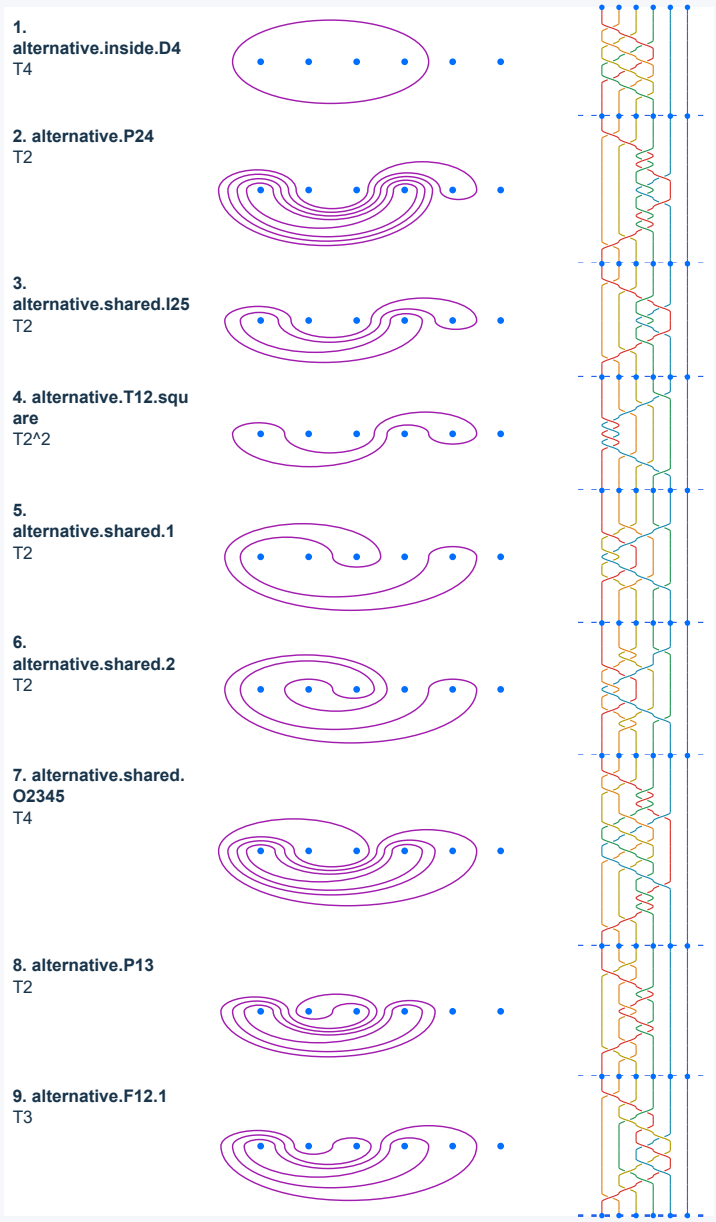
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 4 below. Only the changed interval is shown.

Before



After



3 - step 052 - continued

Carry the new round four-point twist up to unwind the residual block

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 4 below. Only the changed interval is shown.

Before

9.
alternative.inside.D4
T4



After

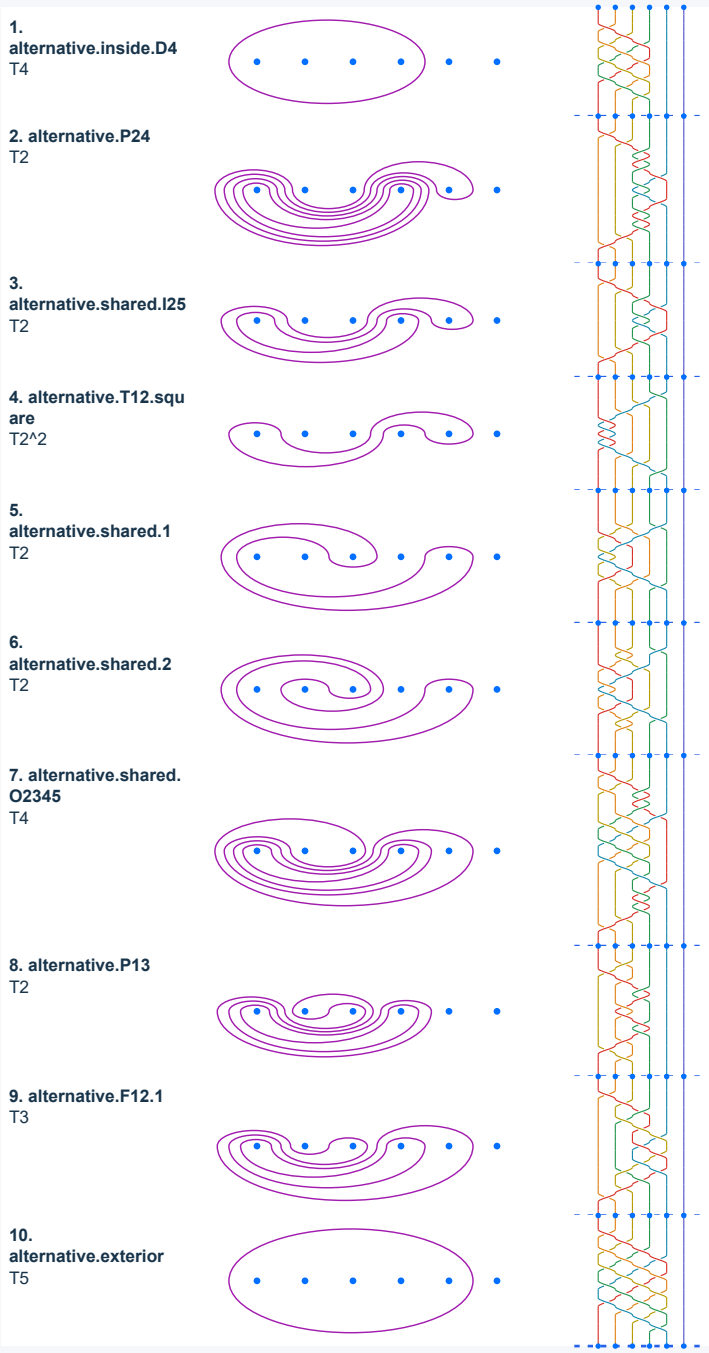
3 - step 053

ONE positive T45 lantern chart replaces the two former frames

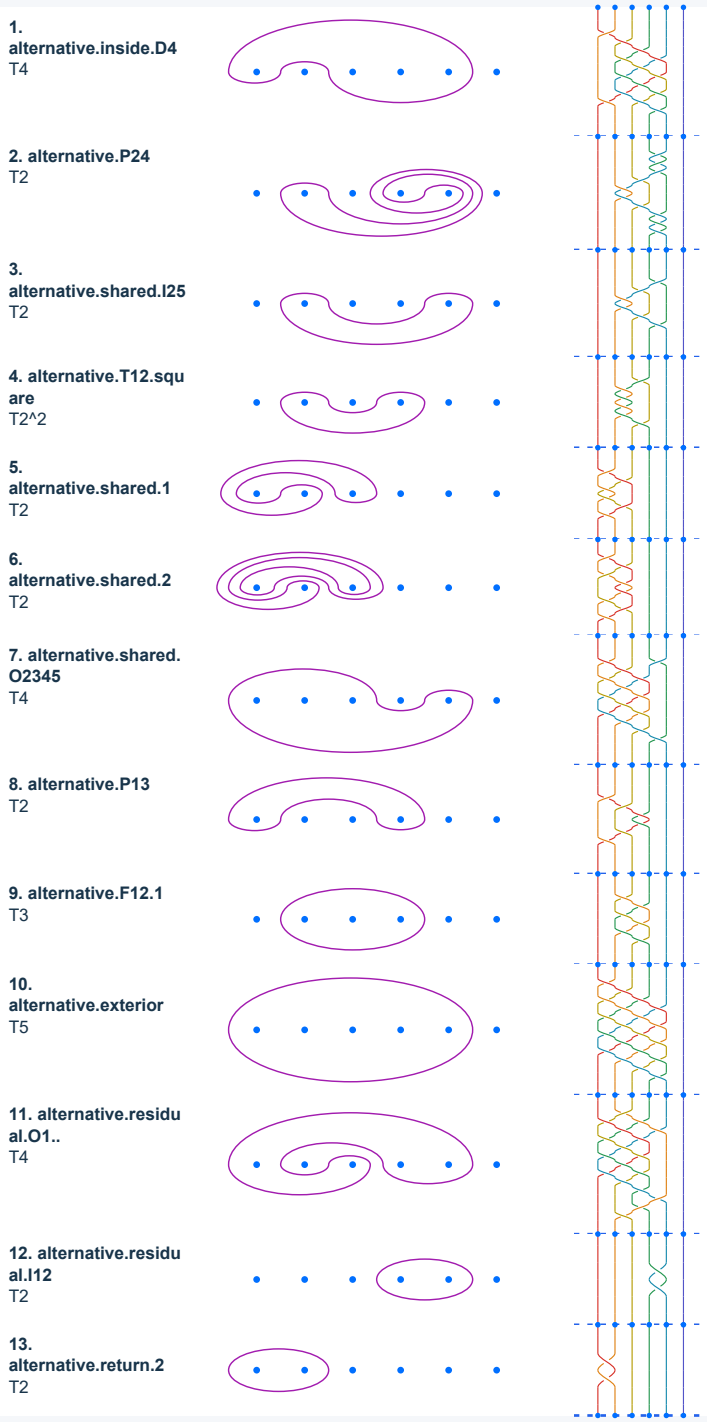
EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



3 - step 053 - continued

ONE positive T45 lantern chart replaces the two former frames

EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After

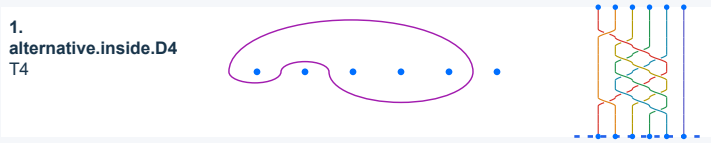
3 - step 054

The four-point PC daisy lends its actual three-point root

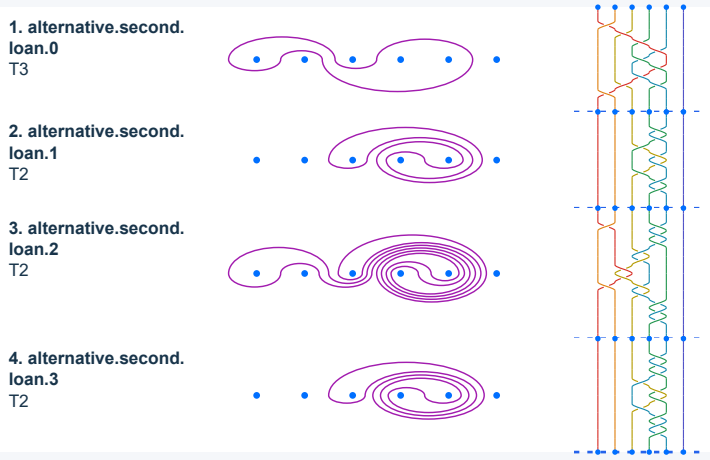
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 12 below. Only the changed interval is shown.

Before



After



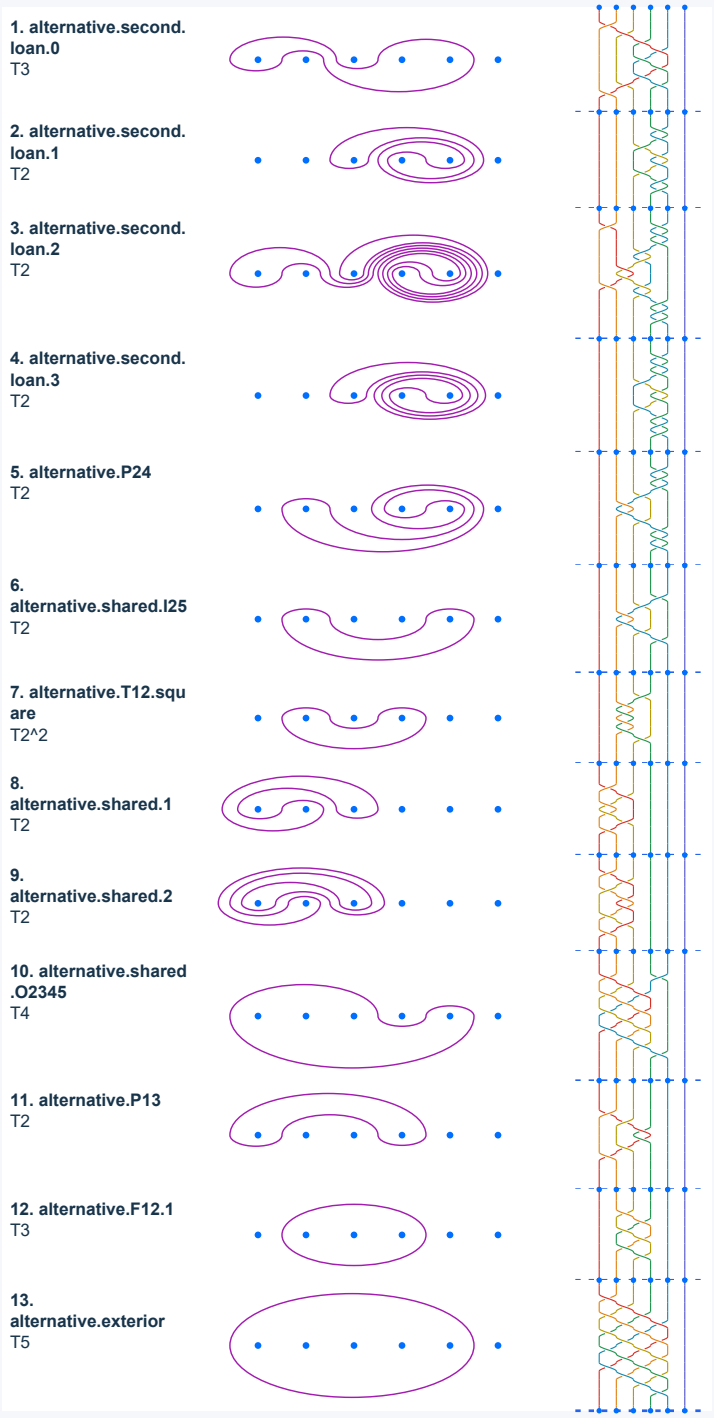
3 - step 055

Cyclic start at the actual pair that transports the borrowed root

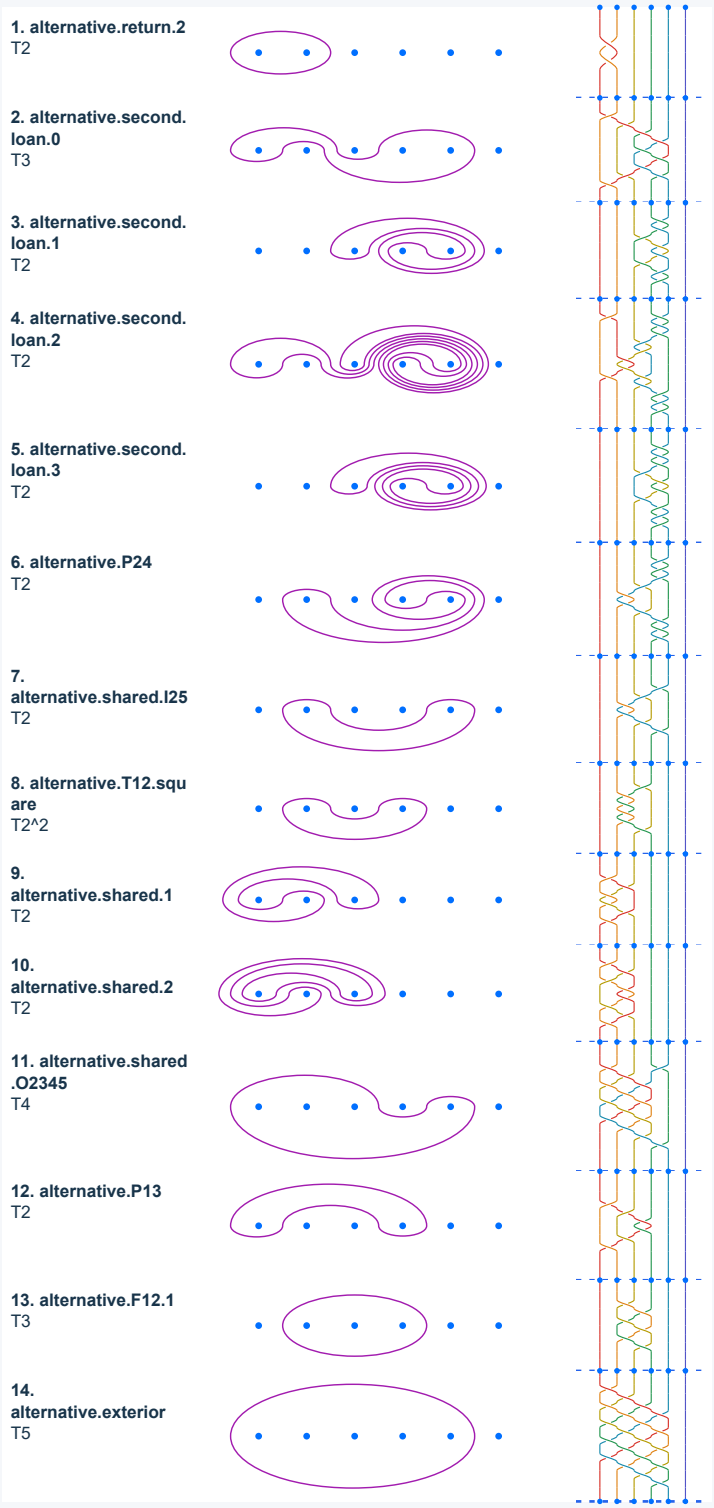
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



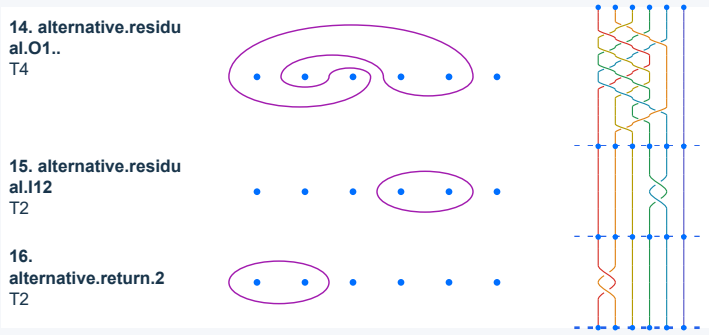
3 - step 055 - continued

Cyclic start at the actual pair that transports the borrowed root

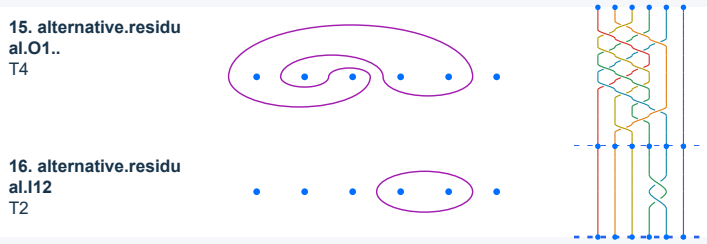
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



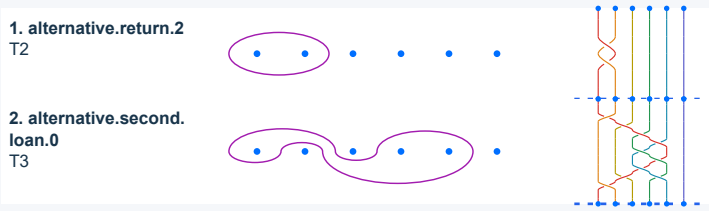
3 - step 056a

The actual pair carries the borrowed root to the exact missing class

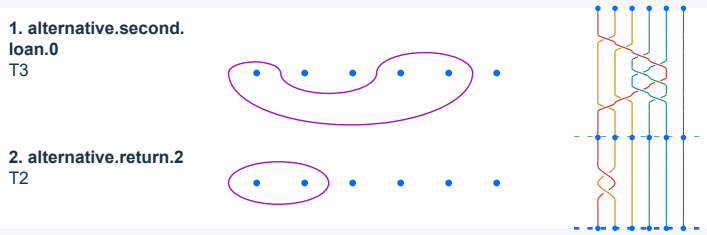
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 14 below. Only the changed interval is shown. PARALLEL STEP 56: moves a, b are independent and each starts from the SAME input. Both orders are certified.

Before



After



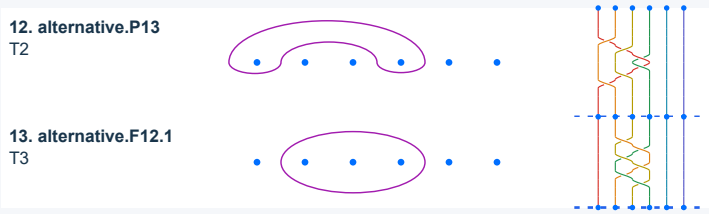
3 - step 056b

Bring the three-point curve up across one neighboring pair

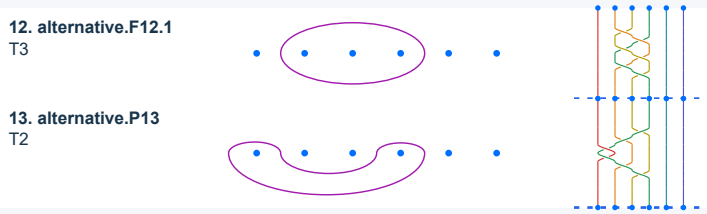
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

11 unchanged displayed factors above; 3 below. Only the changed interval is shown. PARALLEL STEP 56: moves a, b are independent and each starts from the SAME input. Both orders are certified.

Before



After



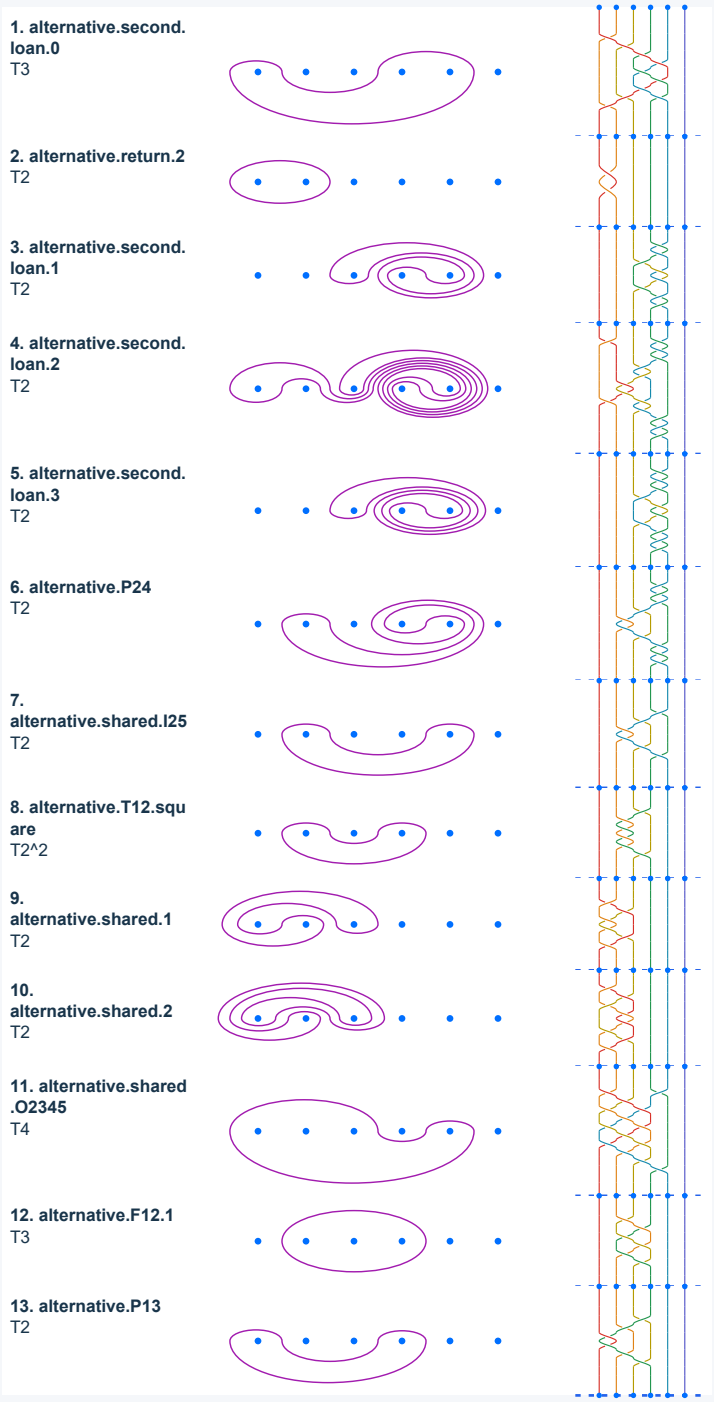
3 - step 057

Cyclic start at the matched four-point curve

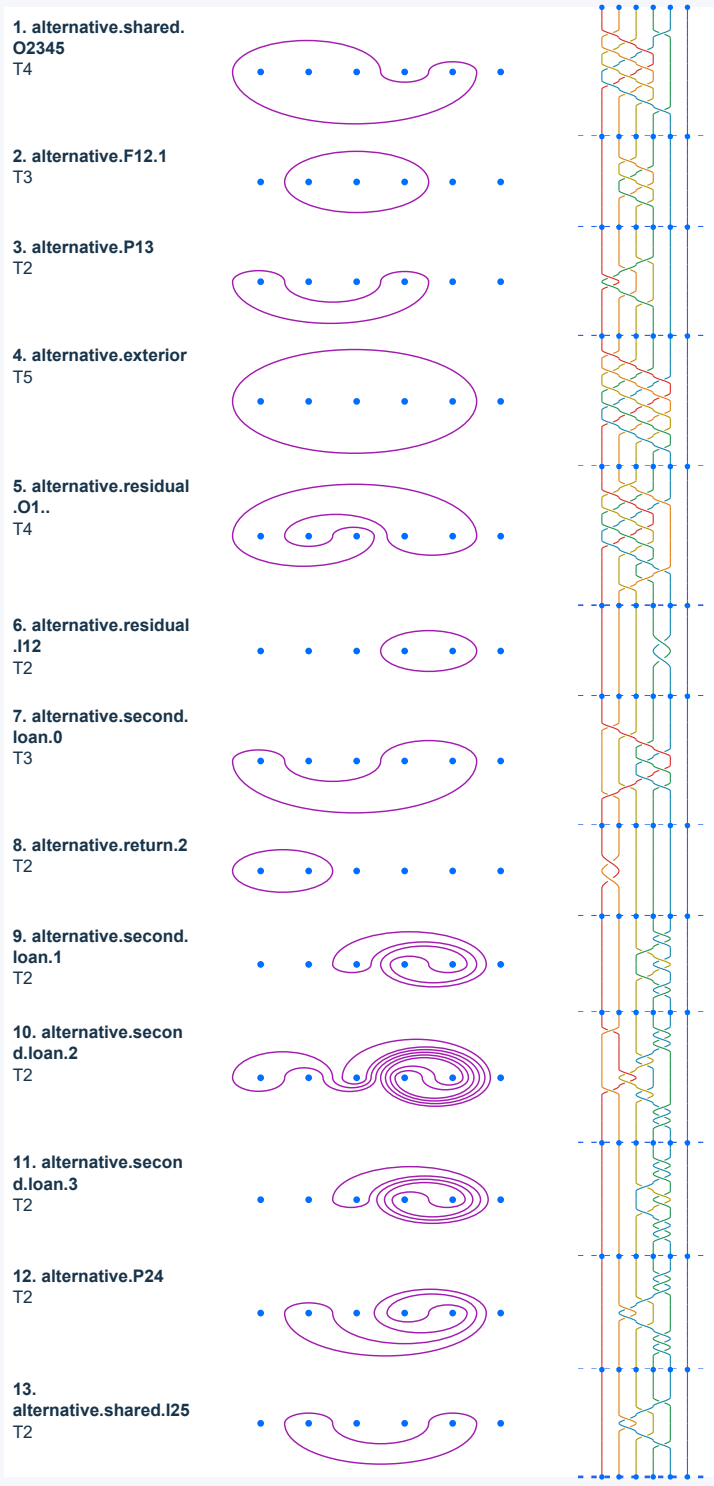
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



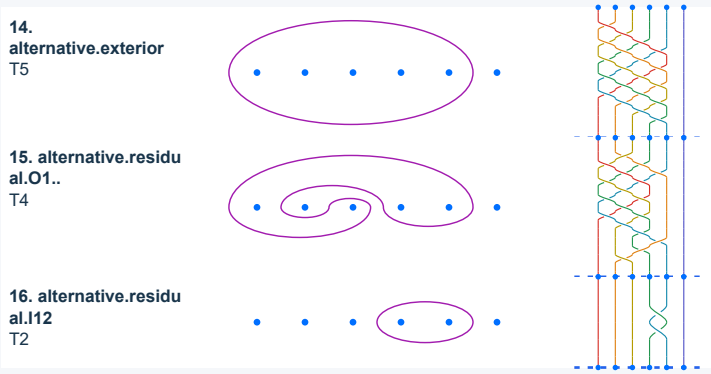
3 - step 057 - continued

Cyclic start at the matched four-point curve

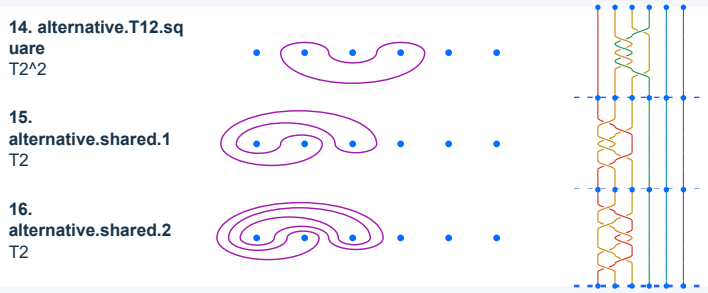
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



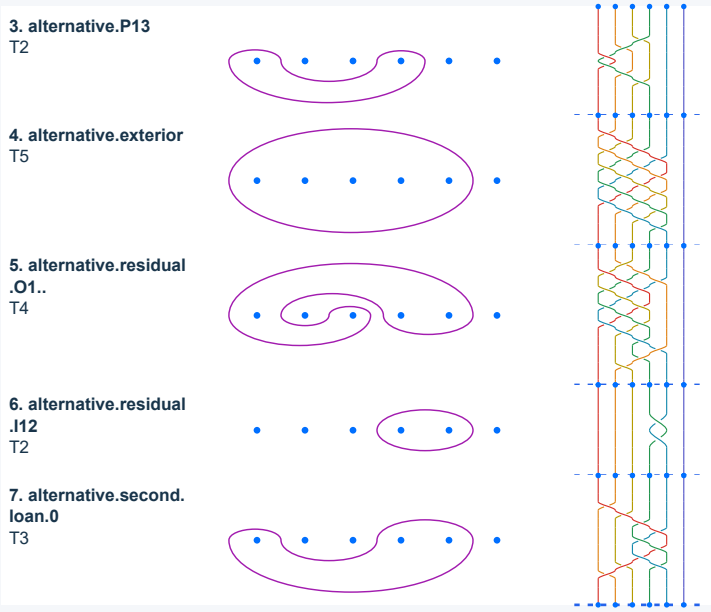
3 - step 058

Gather the borrowed root up the four-neighbor short side

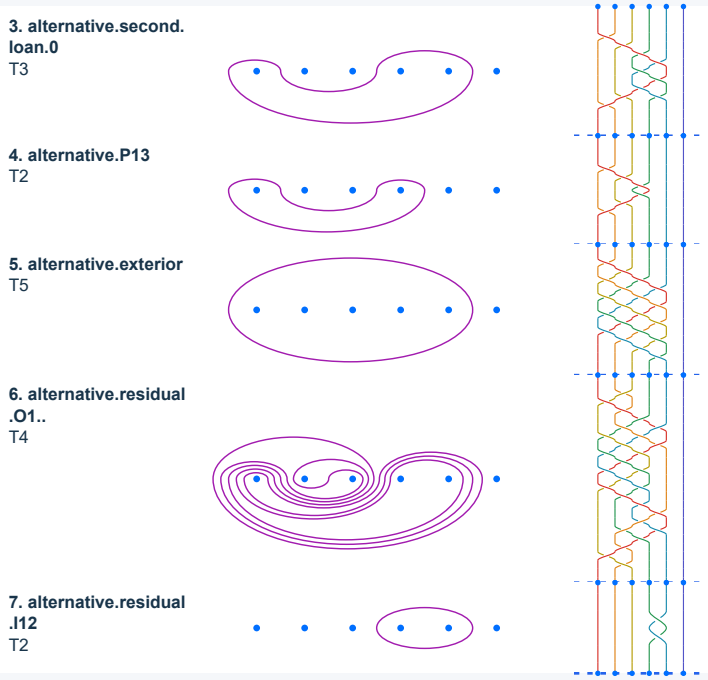
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 9 below. Only the changed interval is shown.

Before



After



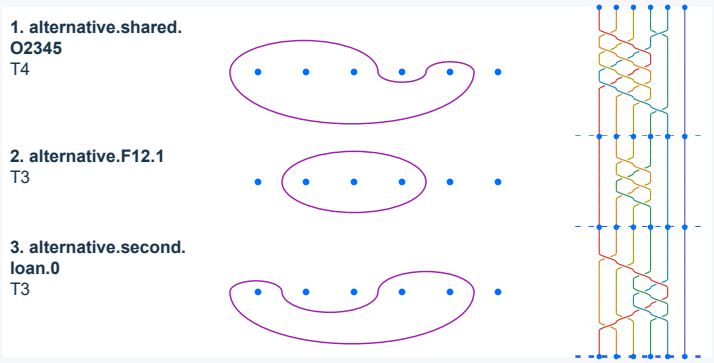
3 - step 059

Cabled PC lantern reaches a SECOND visible exterior twist

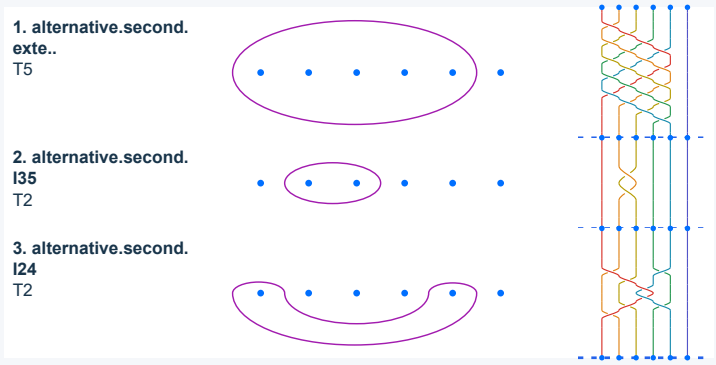
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 13 below. Only the changed interval is shown.

Before



After



3 - step 060

Commute the exterior out of the expanded loan

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 11 below. Only the changed interval is shown.

Before



After



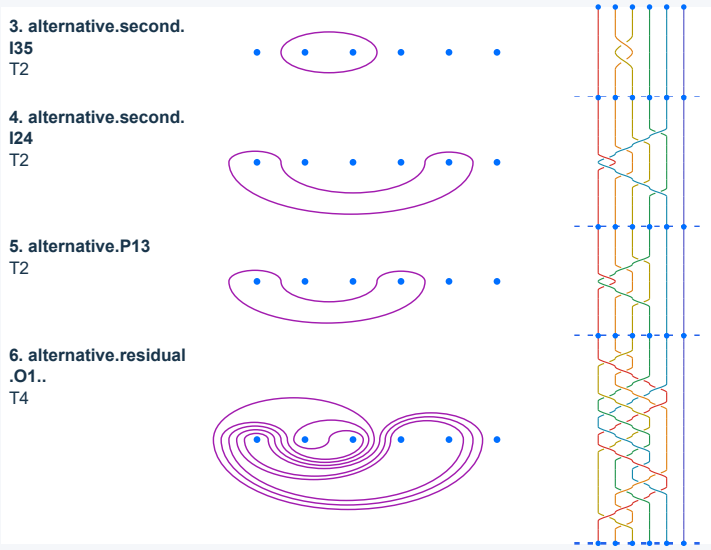
3 - step 061a

Inverse Hurwitz transport of the SAME outer past three petals

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 10 below. Only the changed interval is shown. PARALLEL STEP 61: moves a, b are independent and each starts from the SAME input. Both orders are certified.

Before



After



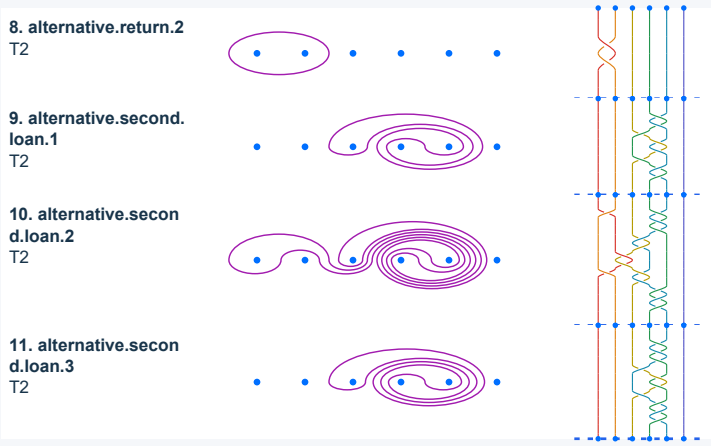
3 - step 061b

Return the12 petal through the three actual loan petals

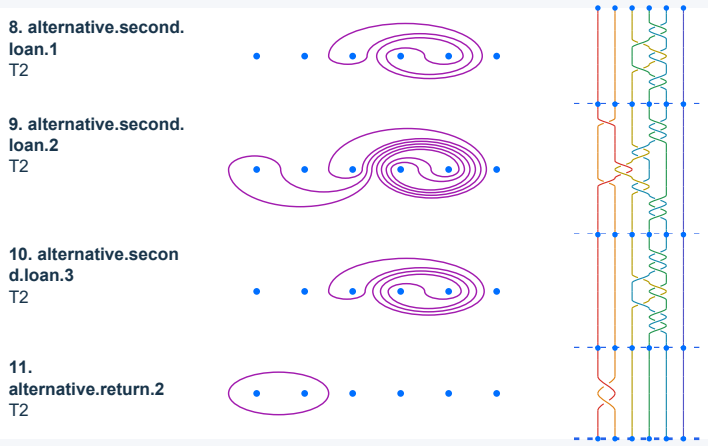
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 5 below. Only the changed interval is shown. PARALLEL STEP 61: moves a, b are independent and each starts from the SAME input. Both orders are certified.

Before



After



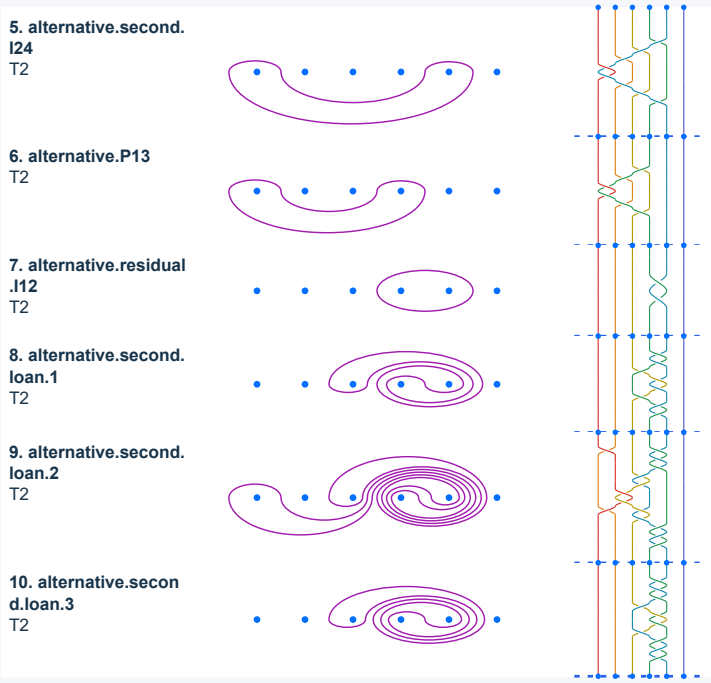
3 - step 062

Expanded PC rose returns the loan WITHOUT a petal reorder

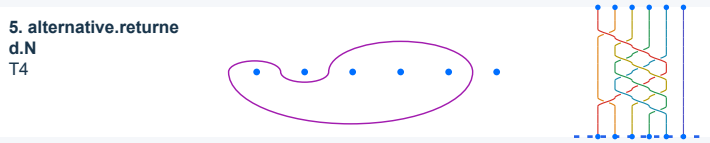
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

4 unchanged displayed factors above; 6 below. Only the changed interval is shown.

Before



After



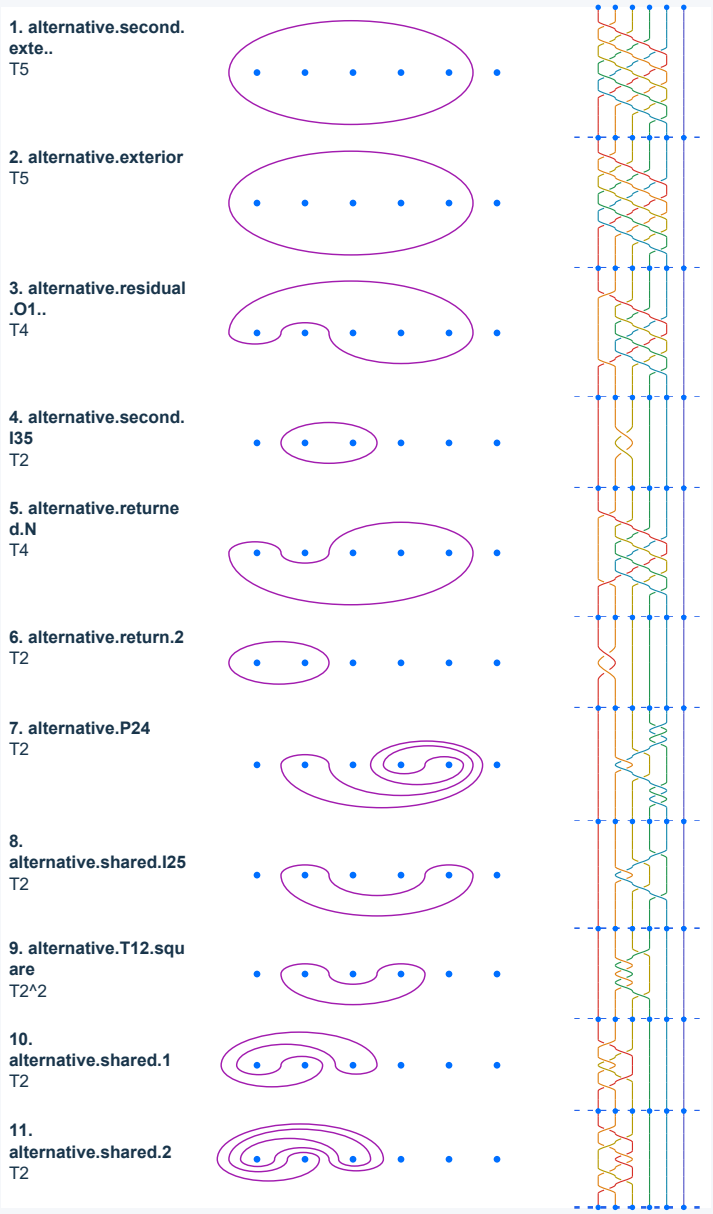
3 - step 063

Cyclic start BEFORE gathering the first daisy

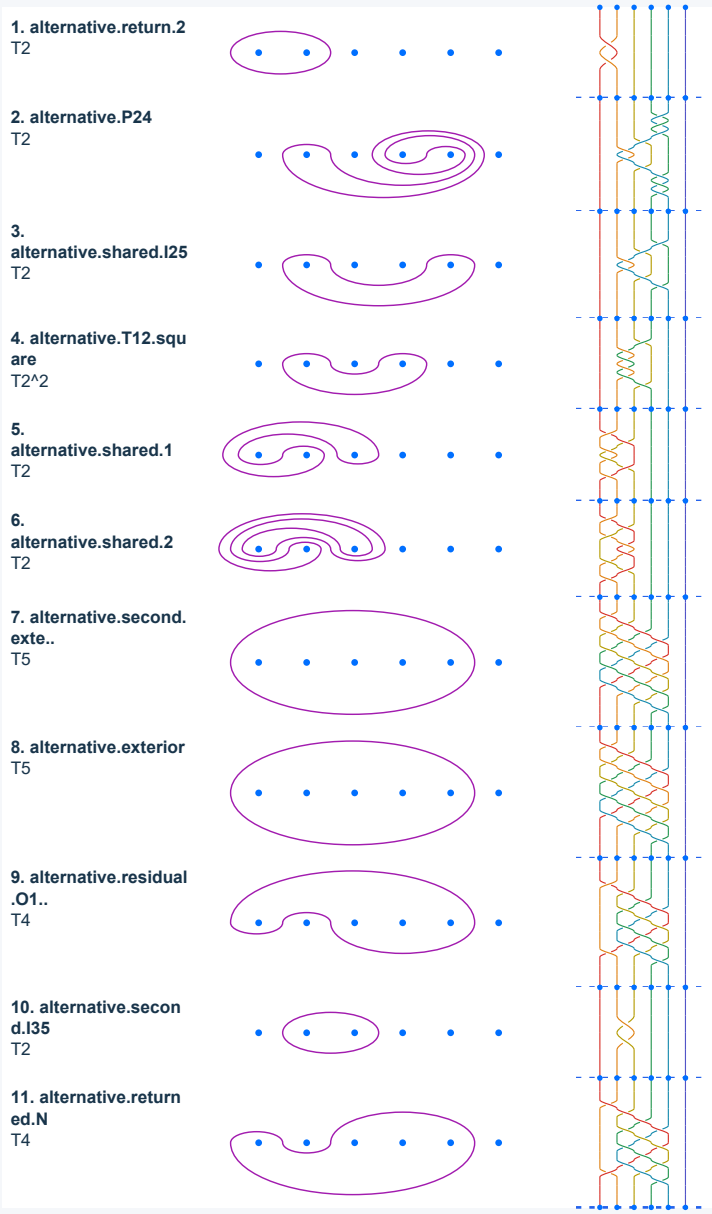
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



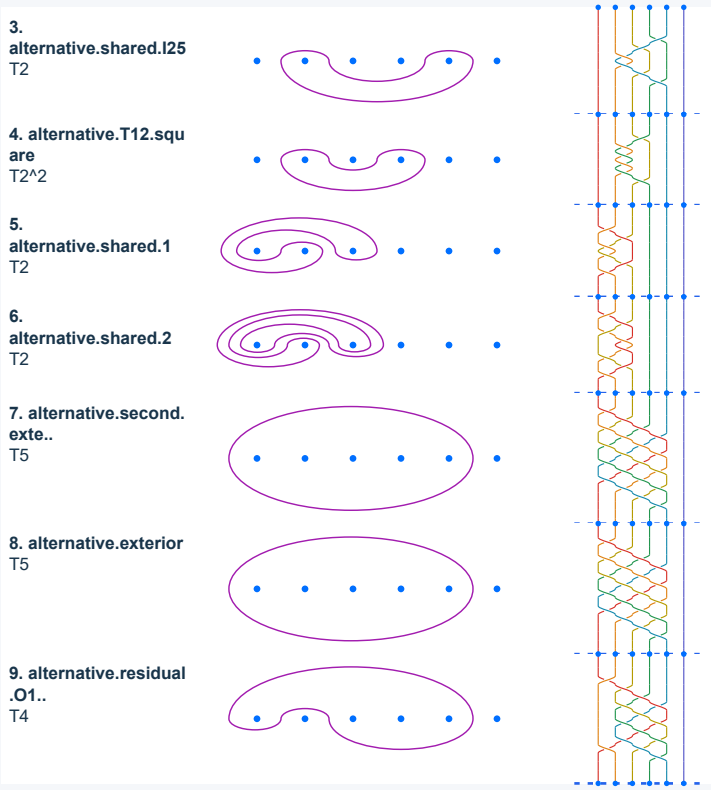
3 - step 064

Carry the SAME outer UP through its first-daisy petals

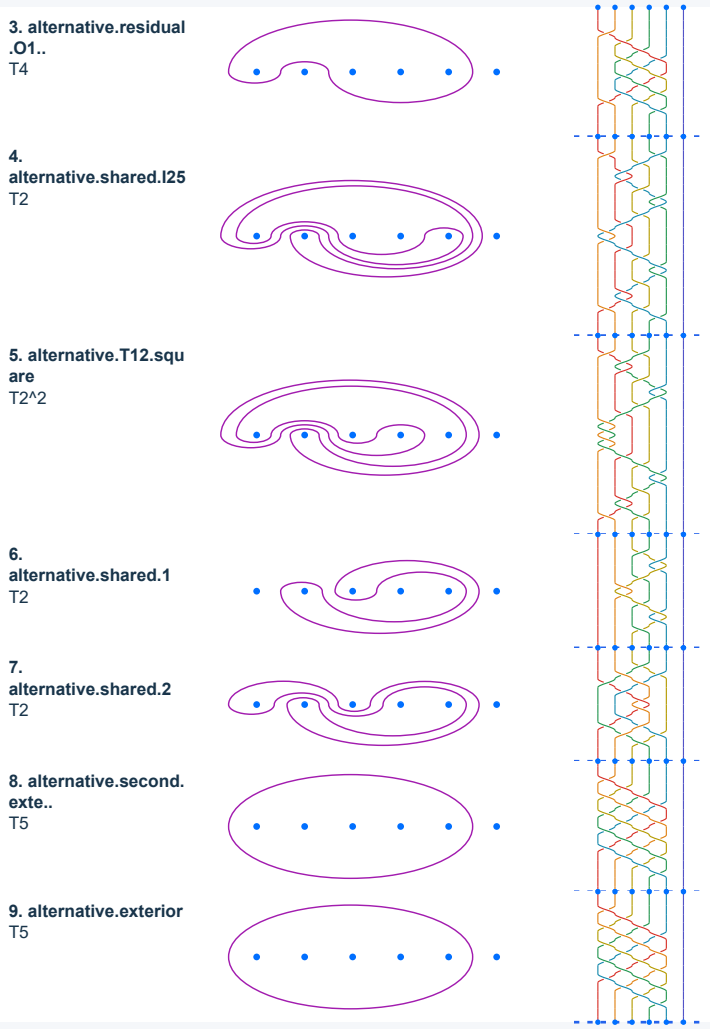
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 2 below. Only the changed interval is shown.

Before



After



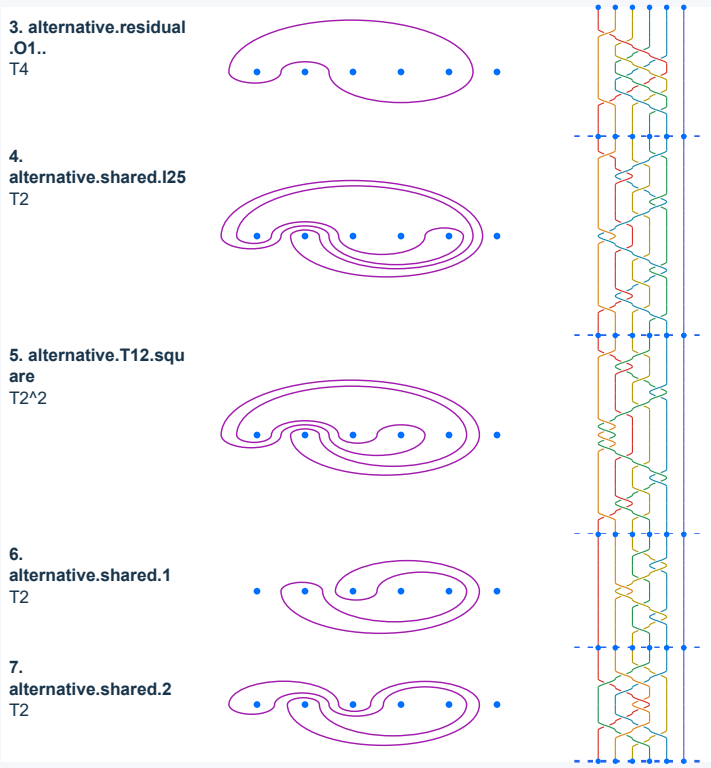
3 - step 065

PC daisy extracts the exterior DIRECTLY from the squared petal

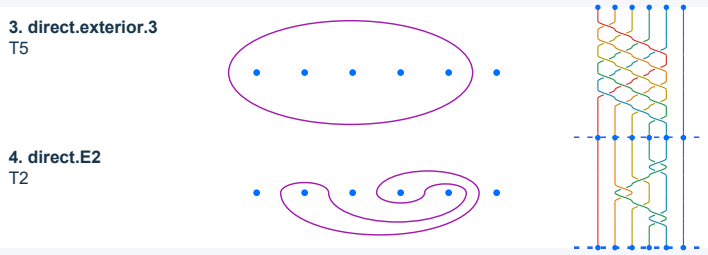
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 4 below. Only the changed interval is shown.

Before



After



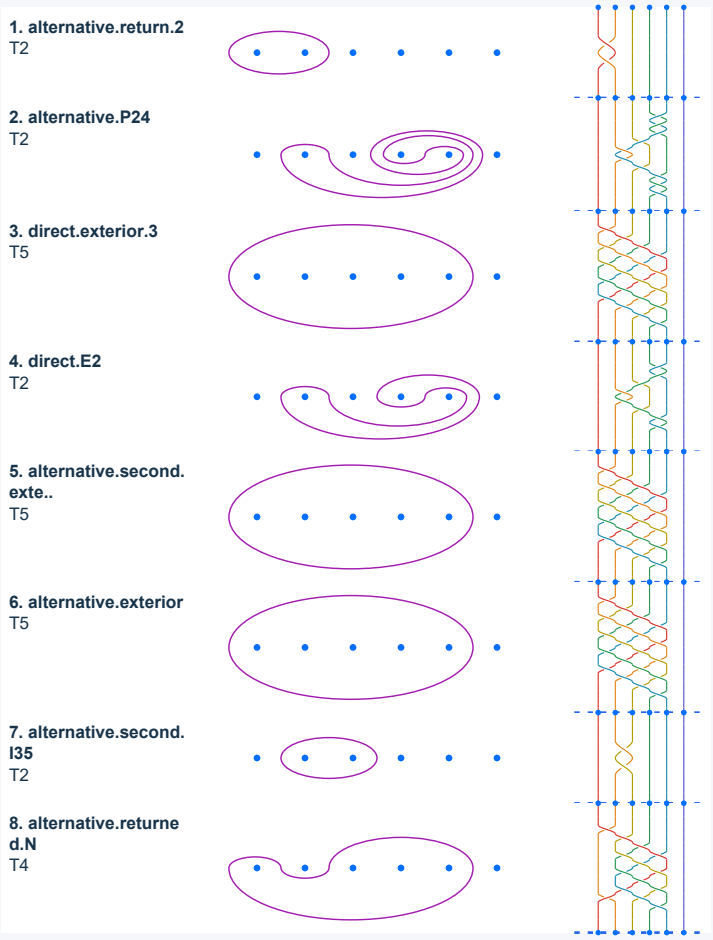
3 - step 066

Cyclic start at the NEW visible exterior

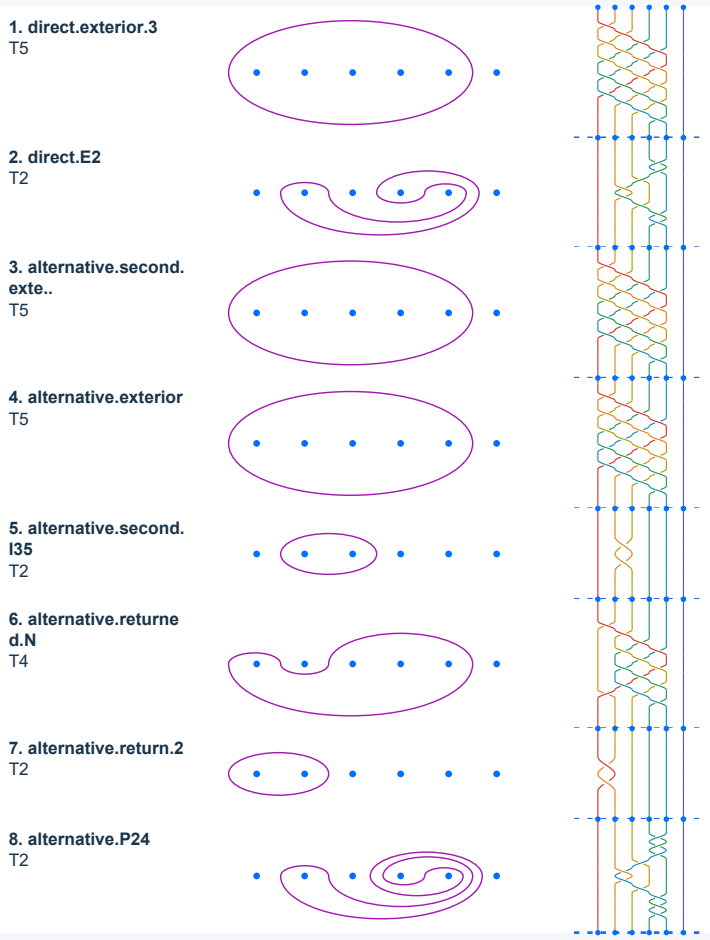
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



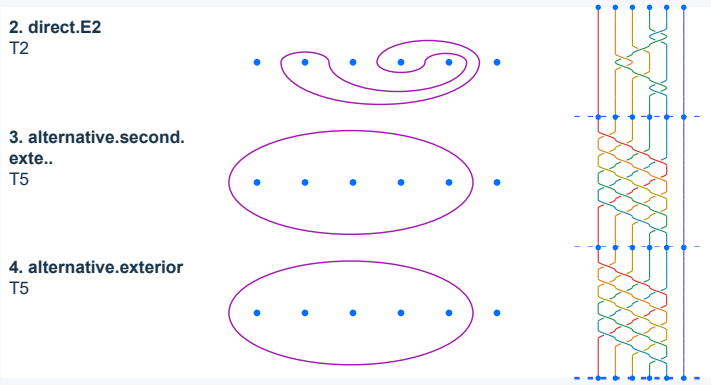
3 - step 067

The extra petal passes DOWN past two central exteriors

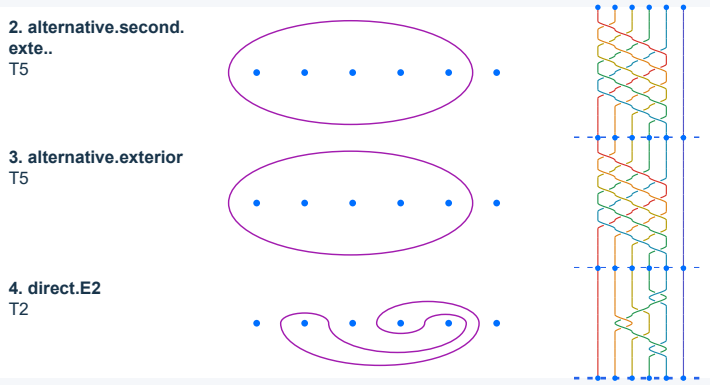
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 4 below. Only the changed interval is shown.

Before



After



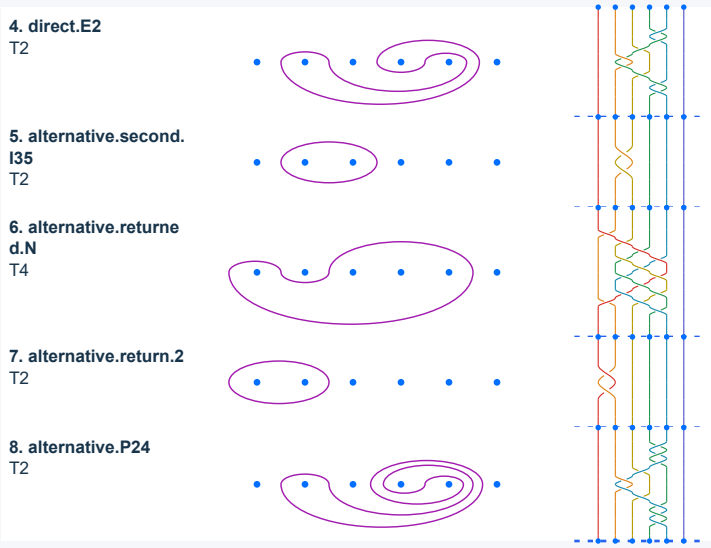
3 - step 068

SECOND actual PC daisy in its reached cyclic order

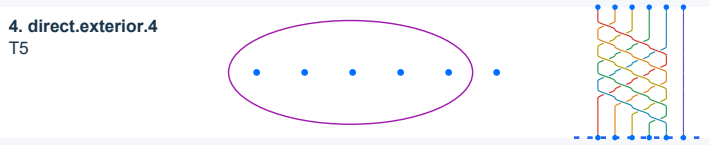
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

3 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



IV. The terminal redraw and peripheral cleanup

Every substitution and sphere redraw is separate. Independent moves share a step number; their individual diagrams remain visible.

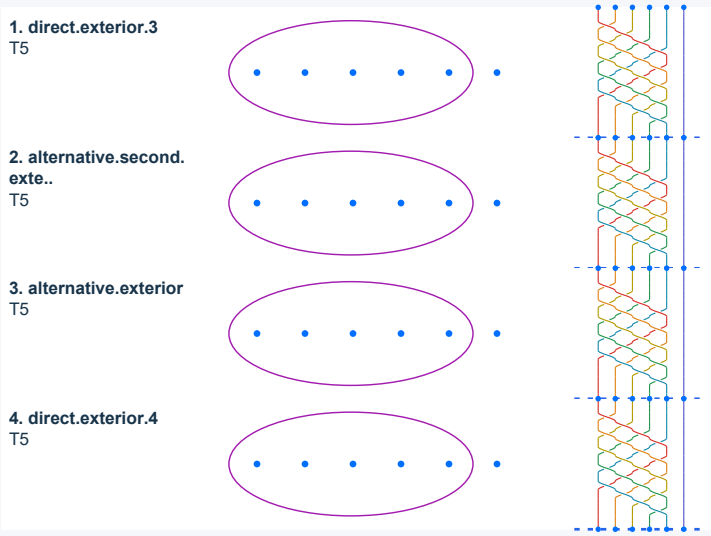
4 - step 069

Combine the FOUR identical visible exterior twists

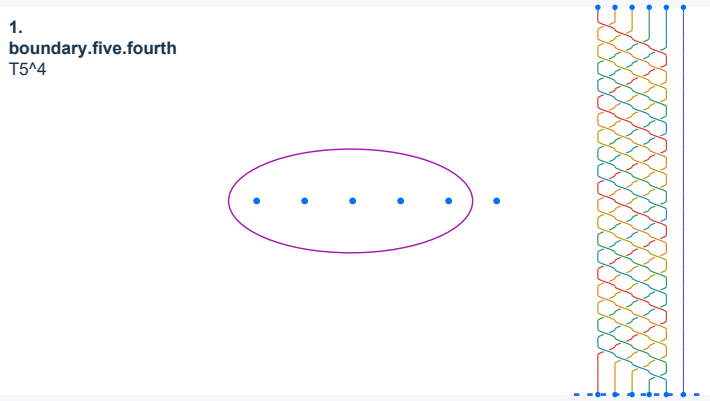
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



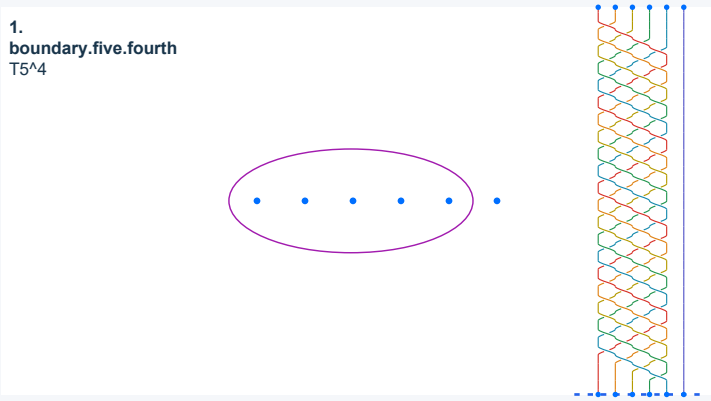
4 - step 070

Redraw the FOURTH-power exterior curve around point6

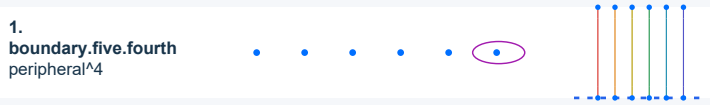
SPHERE REDRAW ONLY: the same curve or arc, with the same factor and power. Framing ledger retained separately.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. REDRAW ONLY: factor identity, position, power and sphere curve are retained.

Before



After



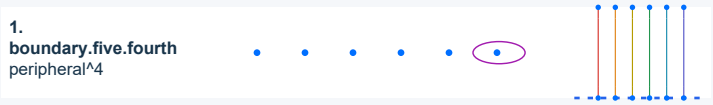
4 - step 071

Remove the already-peripheral FOURTH-power twist

PERIPHERAL CLEANUP: the curve has already been redrawn around one point; its even powered twist is trivial in the chosen lift.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After

Identity / no factors in this interval.

V. Full sphere-redraw discrepancies

Every genuine sphere redraw is listed below. Each has the same geometric curve, an explicit full disk discrepancy, exact marked-sphere action and corrected chosen +I lift. The framing ledger is included in the full word.

SPHERE REDRAW at step 011

Redraw ONE outer curve from its complementary side

Drawing change only; no lantern or other substitution is included. The corrected full discrepancy has exact sphere action and chosen +I lift.

FULL DISK DISCREPANCY including ledger (52 signed letters)

2 3 2 5 -3 -3 4 1 2 3 4 5 1 2 3 4 5 1 1 1 2 3 1 2
3 1 2 3 1 2 3 -5 -5 -1 -1 -5 -4 -3 -2 -1 -5 -4 -3 -2 -1 -4 3 3
-5 -2 -3 -2

SPHERE REDRAW at step 013a

Redraw prefix.P45 in the five-point chart

Drawing change only; no lantern or other substitution is included. The corrected full discrepancy has exact sphere action and chosen +I lift.

FULL DISK DISCREPANCY including ledger (42 signed letters)

5 4 4 1 2 3 4 5 1 2 3 4 5 1 1 -5 -4 -3 -2 -1 -5 -4 -3 -2
-1 -4 -4 -5 -4 2 1 1 2 3 -4 -4 -3 -2 -1 -1 -2 4

LOCAL CURVE DISCREPANCY with its explicit framing correction (42 signed letters)

5 4 4 1 2 3 4 5 1 2 3 4 5 1 1 -5 -4 -3 -2 -1 -5 -4 -3 -2
-1 -4 -4 -5 -4 2 1 1 2 3 -4 -4 -3 -2 -1 -1 -2 4

SPHERE REDRAW at step 013c

Redraw F4 in the five-point chart

Drawing change only; no lantern or other substitution is included. The corrected full discrepancy has exact sphere action and chosen +I lift.

FULL DISK DISCREPANCY including ledger (136 signed letters)

5 4 4 1 2 3 4 5 1 2 3 4 5 1 1 5 -4 -3 -2 -1 -5 -4 -3 -2
-1 -4 3 3 -5 -2 -3 -2 -2 3 4 5 4 5 4 5 4 5 4 5 4 5 -3 2
2 3 2 5 -3 -3 -4 -5 -2 -1 -2 -1 -2 -1 -2 -1 -2 -1 -2 -1 5 4 4 1
2 3 4 5 1 2 3 4 -5 -1 -1 -5 -4 -3 -2 -1 -5 -4 -3 -2 -1 -4 -4 -5
-4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1
-4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1

LOCAL CURVE DISCREPANCY with its explicit framing correction (84 signed letters)

5 4 3 3 -5 -2 -3 -2 -2 3 4 5 4 5 4 5 4 5 4 5 4 5 -3 2
2 3 2 5 -3 -3 -4 -5 -2 -1 -2 -1 -2 -1 -2 -1 -2 -1 -2 -1 -4 -3 -2 -1
-4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1
-4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1

SPHERE REDRAW at step 013g

Redraw F8.1 in the five-point chart

Drawing change only; no lantern or other substitution is included. The corrected full discrepancy has exact sphere action and chosen +I lift.

FULL DISK DISCREPANCY including ledger (234 signed letters)

5 4 4 1 2 3 4 5 1 2 3 4 5 1 1 5 -4 -3 -2 -1 -5 -4 -3 -2
-1 -4 3 3 -5 -2 -3 -2 -2 3 4 5 4 5 4 5 4 5 4 5 4 5 -3 2
3 2 4 4 -3 2 1 -2 3 -4 -4 -2 -3 -2 3 -4 2 3 2 3 2 3 2 3
2 3 2 3 4 -3 2 -4 -2 -3 2 4 -5 -3 -3 4 3 3 5 -4 -2 5 4 2
3 2 3 2 3 -4 -5 3 2 4 2 3 2 5 -3 -3 -4 -5 -4 1 2 3 -4 -3
-2 -3 -2 -3 -2 4 -3 -2 -1 4 5 4 3 3 -5 -2 -3 -2 -4 -2 -3 2 4 -5
-3 -3 -4 3 3 5 -4 -2 3 2 4 -2 3 -4 -3 -2 -3 -2 -3 -2 -3 -2 -3 -2
-3 -2 4 -3 2 3 2 4 4 -3 2 -1 -2 3 -4 -4 -2 -3 -2 3 -5 -4 -5 -4
-5 -4 -5 -4 -5 -4 -5 -4 -3 2 2 3 2 5 -3 -3 4 1 2 3 4 5 1 2
3 4 -5 -1 -1 -5 -4 -3 -2 -1 -5 -4 -3 -2 -1 -4 -4 -5

LOCAL CURVE DISCREPANCY with its explicit framing correction (48 signed letters)

5 4 3 3 -5 -2 -3 -2 -4 -2 -3 5 4 2 3 2 3 2 3 -4 -5 3 2 4
2 3 2 5 -3 -3 -4 -5 -4 1 2 3 -4 -3 -2 -3 -2 -3 -2 4 -3 -2 -1 4

SPHERE REDRAW at step 013h

Redraw F10.1 in the five-point chart

Drawing change only; no lantern or other substitution is included. The corrected full discrepancy has exact sphere action and chosen +I lift.

FULL DISK DISCREPANCY including ledger (268 signed letters)

5 4 4 1 2 3 4 5 1 2 3 4 5 1 1 5 -4 -3 -2 -1 -5 -4 -3 -2
-1 -4 3 3 -5 -2 -3 -2 -2 3 4 5 4 5 4 5 4 5 4 5 -3 2
3 2 4 4 -3 2 1 -2 3 -4 -4 -2 -3 -2 3 -4 2 3 2 3 2 3 2 3
2 3 2 3 4 -3 2 -4 -2 -3 2 4 -5 -3 -3 4 3 3 5 -4 -2 5 4 2
3 2 3 2 3 -4 -5 3 2 4 -2 5 3 4 3 4 3 4 -5 2 2 3 2 5
-3 -3 -4 -5 -4 1 -3 -2 -3 -2 -3 -2 -1 4 5 4 3 3 -5 -2 -3 -2 -4 -2
-3 5 4 -3 -2 -3 -2 -3 -2 -4 -5 2 4 -5 -3 -3 -4 3 3 5 -4 -2 3 2
4 -2 3 -4 -3 -2 -3 -2 -3 -2 -3 -2 -3 -2 -3 -2 4 -3 2 3 2 4 4 -3
2 -1 -2 3 -4 -4 -2 -3 -2 3 -5 -4 -5 -4 -5 -4 -5 -4 -5 -4 -5 -4 -3 2
2 3 2 5 -3 -3 4 1 2 3 4 5 1 2 3 4 -5 -1 -1 -5 -4 -3 -2 -1
-5 -4 -3 -2 -1 -4 -4 -5 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1
-4 -3 -2 -1

LOCAL CURVE DISCREPANCY with its explicit framing correction (54 signed letters)

5 4 3 3 -5 -2 -3 -2 -2 5 3 4 3 4 3 4 -5 2 2 3 2 5 -3 -3
-4 -5 -4 1 -3 -2 -3 -2 -3 -2 -1 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3
-2 -1 -4 -3 -2 -1

SPHERE REDRAW at step 013i

Redraw F8.2 in the five-point chart

Drawing change only; no lantern or other substitution is included. The corrected full discrepancy has exact sphere action and chosen +I lift.

FULL DISK DISCREPANCY including ledger (292 signed letters)

5 4 4 1 2 3 4 5 1 2 3 4 5 1 1 5 -4 -3 -2 -1 -5 -4 -3 -2
-1 -4 3 3 -5 -2 -3 -2 -2 3 4 5 4 5 4 5 4 5 4 5 4 5 -3 2
3 2 4 4 -3 2 1 -2 3 -4 -4 -2 -3 -2 3 -4 2 3 2 3 2 3 2 3
2 3 2 3 4 -3 2 -4 -2 -3 2 4 -5 -3 -3 4 3 3 5 -4 -2 5 4 2
3 2 3 2 3 -4 -5 3 2 4 -2 5 3 4 3 4 3 4 -5 2 -5 -4 2 3
2 3 2 3 4 5 2 3 2 5 -3 -3 -4 -5 -4 -4 -4 -3 -2 -1 -2 -1 -2 -1
3 4 4 4 5 4 3 3 -5 -2 -3 -2 -2 5 -4 -3 -4 -3 -4 -3 -5 2 -4 -2
-3 5 4 -3 -2 -3 -2 -3 -2 -4 -5 2 4 -5 -3 -3 -4 3 3 5 -4 -2 3 2
4 -2 3 -4 -3 -2 -3 -2 -3 -2 -3 -2 -3 -2 -3 -2 -3 -2 4 -3 2 3 2 4 4 -3
2 -1 -2 3 -4 -4 -2 -3 -2 3 -5 -4 -5 -4 -5 -4 -5 -4 -5 -4 -5 -4 -5 -4 -3 2
2 3 2 5 -3 -3 4 1 2 3 4 5 1 2 3 4 -5 -1 -1 -5 -4 -3 -2 -1
-5 -4 -3 -2 -1 -4 -4 -5 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1
-4 -3 -2 -1

LOCAL CURVE DISCREPANCY with its explicit framing correction (58 signed letters)

5 4 3 3 -5 -2 -3 -2 -5 -4 2 3 2 3 2 3 4 5 2 3 2 5 -3 -3
-4 -5 -4 -4 -4 -3 -2 -1 -2 -1 -2 -1 3 4 4 -3 -2 -1 -4 -3 -2 -1 -4 -3
-2 -1 -4 -3 -2 -1 -4 -3 -2 -1

SPHERE REDRAW at step 013j

Redraw F9 in the five-point chart

Drawing change only; no lantern or other substitution is included. The corrected full discrepancy has exact sphere action and chosen +I lift.

FULL DISK DISCREPANCY including ledger (296 signed letters)

5 4 4 1 2 3 4 5 1 2 3 4 5 1 1 5 -4 -3 -2 -1 -5 -4 -3 -2
-1 -4 3 3 -5 -2 -3 -2 -2 3 4 5 4 5 4 5 4 5 4 5 -3 2
3 2 4 4 -3 2 1 -2 3 -4 -4 -2 -3 -2 3 -4 2 3 2 3 2 3 2 3
2 3 2 3 4 -3 2 -4 -2 -3 2 4 -5 -3 -3 4 3 3 5 -4 -2 5 4 2
3 2 3 2 3 -4 -5 3 2 4 -2 5 3 4 3 4 3 4 -5 2 -5 -4 2 3
2 3 3 -2 1 2 -3 2 3 4 5 2 3 2 5 -3 -3 -4 -5 -4 -3 -4 -3 1
2 2 1 -4 -3 -2 -1 2 3 4 -1 -2 -2 -1 3 4 3 4 5 4 3 3 -5 -2
-3 -2 -5 -4 -3 -2 -3 -2 -3 -2 4 5 -2 5 -4 -3 -4 -3 -4 -3 -5 2 -4 -2
-3 5 4 -3 -2 -3 -2 -3 -2 -4 -5 2 4 -5 -3 -3 -4 3 3 5 -4 -2 3 2
4 -2 3 -4 -3 -2 -3 -2 -3 -2 -3 -2 -3 -2 -3 -2 4 -3 2 3 2 4 4 -3
2 -1 -2 3 -4 -4 -2 -3 -2 3 -5 -4 -5 -4 -5 -4 -5 -4 -5 -4 -5 -4 -3 2
2 3 2 5 -3 -3 4 1 2 3 4 5 1 2 3 4 -5 -1 -1 -5 -4 -3 -2 -1
-5 -4 -3 -2 -1 -4 -4 -5

LOCAL CURVE DISCREPANCY with its explicit framing correction (52 signed letters)

5 4 3 3 -5 -2 -3 -2 -5 -4 -3 -2 3 -2 1 2 -3 2 3 4 5 2 3 2
5 -3 -3 -4 -5 -4 -3 -4 -3 1 2 2 1 -4 -3 -2 -1 2 3 4 -1 -2 -2 -1
3 4 3 4

SPHERE REDRAW at step 013k

Redraw F10.2 in the five-point chart

Drawing change only; no lantern or other substitution is included. The corrected full discrepancy has exact sphere action and chosen +I lift.

FULL DISK DISCREPANCY including ledger (296 signed letters)

5 4 4 1 2 3 4 5 1 2 3 4 5 1 1 5 -4 -3 -2 -1 -5 -4 -3 -2
-1 -4 3 3 -5 -2 -3 -2 -2 3 4 5 4 5 4 5 4 5 4 5 4 5 -3 2
3 2 4 4 -3 2 1 -2 3 -4 -4 -2 -3 -2 3 -4 2 3 2 3 2 3 2 3
2 3 2 3 4 -3 2 -4 -2 -3 2 4 -5 -3 -3 4 3 3 5 -4 -2 5 4 2
3 2 3 2 3 -4 -5 3 2 4 -2 5 3 4 3 4 3 4 -5 2 -5 -4 2 3
2 3 3 -2 1 2 -3 2 3 4 5 4 5 4 5 4 5 2 3 2 5 -3 -3 -4
-5 1 1 2 -4 -3 -4 -3 -4 -3 -2 -1 -1 5 4 3 3 -5 -2 -3 -2 -5 -4 -3
-2 3 -2 -1 2 -3 -3 -2 -3 -2 4 5 -2 5 -4 -3 -4 -3 -4 -3 -5 2 -4 -2
-3 5 4 -3 -2 -3 -2 -3 -2 -4 -5 2 4 -5 -3 -3 -4 3 3 5 -4 -2 3 2
4 -2 3 -4 -3 -2 -3 -2 -3 -2 -3 -2 -3 -2 -3 -2 4 -3 2 3 2 4 4 -3
2 -1 -2 3 -4 -4 -2 -3 -2 3 -5 -4 -5 -4 -5 -4 -5 -4 -5 -4 -5 -4 -3 2
2 3 2 5 -3 -3 4 1 2 3 4 5 1 2 3 4 -5 -1 -1 -5 -4 -3 -2 -1
-5 -4 -3 -2 -1 -4 -4 -5

LOCAL CURVE DISCREPANCY with its explicit framing correction (34 signed letters)

5 4 3 3 -5 -2 -3 -2 4 5 4 5 4 5 2 3 2 5 -3 -3 -4 -5 1 1
2 -4 -3 -4 -3 -4 -3 -2 -1 -1

SPHERE REDRAW at step 013l

Redraw F11 in the five-point chart

Drawing change only; no lantern or other substitution is included. The corrected full discrepancy has exact sphere action and chosen +I lift.

FULL DISK DISCREPANCY including ledger (318 signed letters)

5 4 4 1 2 3 4 5 1 2 3 4 5 1 1 5 -4 -3 -2 -1 -5 -4 -3 -2
-1 -4 3 3 -5 -2 -3 -2 -2 3 4 5 4 5 4 5 4 5 4 5 -3 2
3 2 4 4 -3 2 1 -2 3 -4 -4 -2 -3 -2 3 -4 2 3 2 3 2 3 2 3
2 3 2 3 4 -3 2 -4 -2 -3 2 4 -5 -3 -3 4 3 3 5 -4 -2 5 4 2
3 2 3 2 3 -4 -5 3 2 4 -2 5 3 4 3 4 3 4 -5 2 -5 -4 2 3
2 3 3 -2 1 2 -3 2 3 4 5 4 5 4 5 4 -1 -2 3 -4 3 4 -3 2
1 5 2 3 2 5 -3 -3 -4 -5 -4 -4 -4 1 1 1 -4 -3 -2 -1 2 3 4 -1
-1 -1 4 4 4 5 4 3 3 -5 -2 -3 -2 -5 -4 -5 -4 -5 -4 -5 -4 -3 -2 3
-2 -1 2 -3 -3 -2 -3 -2 4 5 -2 5 -4 -3 -4 -3 -4 -3 -5 2 -4 -2 -3 5
4 -3 -2 -3 -2 -3 -2 -4 -5 2 4 -5 -3 -3 -4 3 3 5 -4 -2 3 2 4 -2
3 -4 -3 -2 -3 -2 -3 -2 -3 -2 -3 -2 -3 -2 4 -3 2 3 2 4 4 -3 2 -1
-2 3 -4 -4 -2 -3 -2 3 -5 -4 -5 -4 -5 -4 -5 -4 -5 -4 -5 -4 -3 2 2 3
2 5 -3 -3 4 1 2 3 4 5 1 2 3 4 -5 -1 -1 -5 -4 -3 -2 -1 -5 -4
-3 -2 -1 -4 -4 -5

LOCAL CURVE DISCREPANCY with its explicit framing correction (46 signed letters)

5 4 3 3 -5 -2 -3 -2 -5 -1 -2 3 -4 3 4 -3 2 1 5 2 3 2 5 -3
-3 -4 -5 -4 -4 -4 1 1 1 -4 -3 -2 -1 2 3 4 -1 -1 -1 4 4 4

SPHERE REDRAW at step 013m

Redraw F12 in the five-point chart

Drawing change only; no lantern or other substitution is included. The corrected full discrepancy has exact sphere action and chosen +I lift.

FULL DISK DISCREPANCY including ledger (344 signed letters)

5 4 4 1 2 3 4 5 1 2 3 4 5 1 1 5 -4 -3 -2 -1 -5 -4 -3 -2
-1 -4 3 3 -5 -2 -3 -2 -2 3 4 5 4 5 4 5 4 5 4 5 4 5 -3 2
3 2 4 4 -3 2 1 -2 3 -4 -4 -2 -3 -2 3 -4 2 3 2 3 2 3 2 3
2 3 2 3 4 -3 2 -4 -2 -3 2 4 -5 -3 -3 4 3 3 5 -4 -2 5 4 2
3 2 3 2 3 -4 -5 3 2 4 -2 5 3 4 3 4 3 4 -5 2 -5 -4 2 3
2 3 3 -2 1 2 -3 2 3 4 5 4 5 4 5 4 -1 -2 3 -4 3 4 -3 2
1 5 -1 -2 3 4 3 4 3 4 3 4 3 4 3 4 2 1 2 3 2 5 -3 -3
-4 -5 -4 -4 -4 -3 -2 -1 -2 -1 -2 -1 -2 -1 -2 -1 -2 -1 -2 -1 3 4 4 4 5 4
3 3 -5 -2 -3 -2 -5 -1 -2 3 -4 -3 4 -3 2 1 -4 -5 -4 -5 -4 -5 -4 -3
-2 3 -2 -1 2 -3 -3 -2 -3 -2 4 5 -2 5 -4 -3 -4 -3 -4 -3 -5 2 -4 -2
-3 5 4 -3 -2 -3 -2 -3 -2 -4 -5 2 4 -5 -3 -3 -4 3 3 5 -4 -2 3 2
4 -2 3 -4 -3 -2 -3 -2 -3 -2 -3 -2 -3 -2 -3 -2 -3 -2 4 -3 2 3 2 4 4 -3
2 -1 -2 3 -4 -4 -2 -3 -2 3 -5 -4 -5 -4 -5 -4 -5 -4 -5 -4 -5 -4 -5 -4 -3 2
2 3 2 5 -3 -3 4 1 2 3 4 5 1 2 3 4 -5 -1 -1 -5 -4 -3 -2 -1
-5 -4 -3 -2 -1 -4 -4 -5

LOCAL CURVE DISCREPANCY with its explicit framing correction (52 signed letters)

5 4 3 3 -5 -2 -3 -2 -1 -2 3 4 3 4 3 4 3 4 3 4 3 4 2 1
2 3 2 5 -3 -3 -4 -5 -4 -4 -4 -3 -2 -1 -2 -1 -2 -1 -2 -1 -2 -1 -2 -1
3 4 4 4

SPHERE REDRAW at step 013n

Redraw F13 in the five-point chart

Drawing change only; no lantern or other substitution is included. The corrected full discrepancy has exact sphere action and chosen +I lift.

FULL DISK DISCREPANCY including ledger (412 signed letters)

5 4 4 1 2 3 4 5 1 2 3 4 5 1 1 5 -4 -3 -2 -1 -5 -4 -3 -2
-1 -4 3 3 -5 -2 -3 -2 -2 3 4 5 4 5 4 5 4 5 4 5 4 5 -3 2
3 2 4 4 -3 2 1 -2 3 -4 -4 -2 -3 -2 3 -4 2 3 2 3 2 3 2 3
2 3 2 3 4 -3 2 -4 -2 -3 2 4 -5 -3 -3 4 3 3 5 -4 -2 5 4 2
3 2 3 2 3 -4 -5 3 2 4 -2 5 3 4 3 4 3 4 -5 2 -5 -4 2 3
2 3 3 -2 1 2 -3 2 3 4 5 4 5 4 5 4 -1 -2 3 -4 3 4 -3 2
1 5 -1 -2 3 4 3 4 3 4 3 4 3 4 3 4 2 1 -4 2 3 2 3 2
3 2 3 2 3 2 3 4 2 3 2 5 -3 -3 -4 -5 -4 -4 -3 -2 -1 -2 -1 -2
-1 -2 -1 -2 -1 -2 -1 3 4 4 5 4 3 3 -5 -2 -3 -2 -1 -2 -4 -3 -4 -3
-4 -3 -4 -3 -4 -3 -4 -3 2 1 -5 -1 -2 3 -4 -3 4 -3 2 1 -4 -5 -4 -5
-4 -5 -4 -3 -2 3 -2 -1 2 -3 -3 -2 -3 -2 4 5 -2 5 -4 -3 -4 -3 -4 -3
-5 2 -4 -2 -3 5 4 -3 -2 -3 -2 -3 -2 -4 -5 2 4 -5 -3 -3 -4 3 3 5
-4 -2 3 2 4 -2 3 -4 -3 -2 -3 -2 -3 -2 -3 -2 -3 -2 -3 -2 4 -3 2 3
2 4 4 -3 2 -1 -2 3 -4 -4 -2 -3 -2 3 -5 -4 -5 -4 -5 -4 -5 -4 -5 -4
-5 -4 -3 2 2 3 2 5 -3 -3 4 1 2 3 4 5 1 2 3 4 -5 -1 -1 -5
-4 -3 -2 -1 -5 -4 -3 -2 -1 -4 -4 -5 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1
-4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1
-4 -3 -2 -1

LOCAL CURVE DISCREPANCY with its explicit framing correction (86 signed letters)

5 4 3 3 -5 -2 -3 -2 -4 2 3 2 3 2 3 2 3 2 3 2 3 4 2 3
2 5 -3 -3 -4 -5 -4 -4 -3 -2 -1 -2 -1 -2 -1 -2 -1 -2 -1 -2 -1 -2 -1 3 4 -3
-2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3
-2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1

SPHERE REDRAW at step 014

Redraw the SIX-correction ledger curve around point6

Drawing change only; no lantern or other substitution is included. The corrected full discrepancy has exact sphere action and chosen +I lift.

FULL DISK DISCREPANCY including ledger (120 signed letters)

1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4
1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4
1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4
1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4
1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4

SPHERE REDRAW at step 070

Redraw the FOURTH-power exterior curve around point6

Drawing change only; no lantern or other substitution is included. The corrected full discrepancy has exact sphere action and chosen +I lift.

FULL DISK DISCREPANCY including ledger (80 signed letters)

1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4
1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4
1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4
1 2 3 4 1 2 3 4

Certificates and scope

Preferred graphical route:
https://richardbuckman-math.github.io/surface-diagrams/docs/proofs/six-seven-fifty-blocks/index.html?phase=clearfull

Clarity certificates: clarity-certificate.json. The archived 79-operation source, the longer early prefix, and all their named-pattern witnesses remain unchanged in certificate.json / named_full_route.

101 moves are displayed in 71 certified independent steps. This is a presentation structure, not a mathematical-shortening claim. All source Hurwitz checks, named witnesses, signed PC balances and nonabelian kernels remain referenced.

The first lantern is now an exact disk substitution with its four-point outer retained. The following sphere move only redraws that ONE outer curve as its complementary pair. Framing corrections in the later chart are held in a separate explicit ledger. Peripheral deletion always follows its own curve redraw.

The thirteen genuine sphere redraws retain full discrepancies and corrected lifts. Four other chart redraws are already disk equalities. No standalone odd exterior cap is used. The archive 125-page/177-step PDF remains unchanged.

Priority: favor simpler arcs and curves and less tangled interactions, even when more moves are needed. Parallel steps require actual independence, not a smaller number of navigation labels.

Independent jointly certified six-point / five-arc comparison

